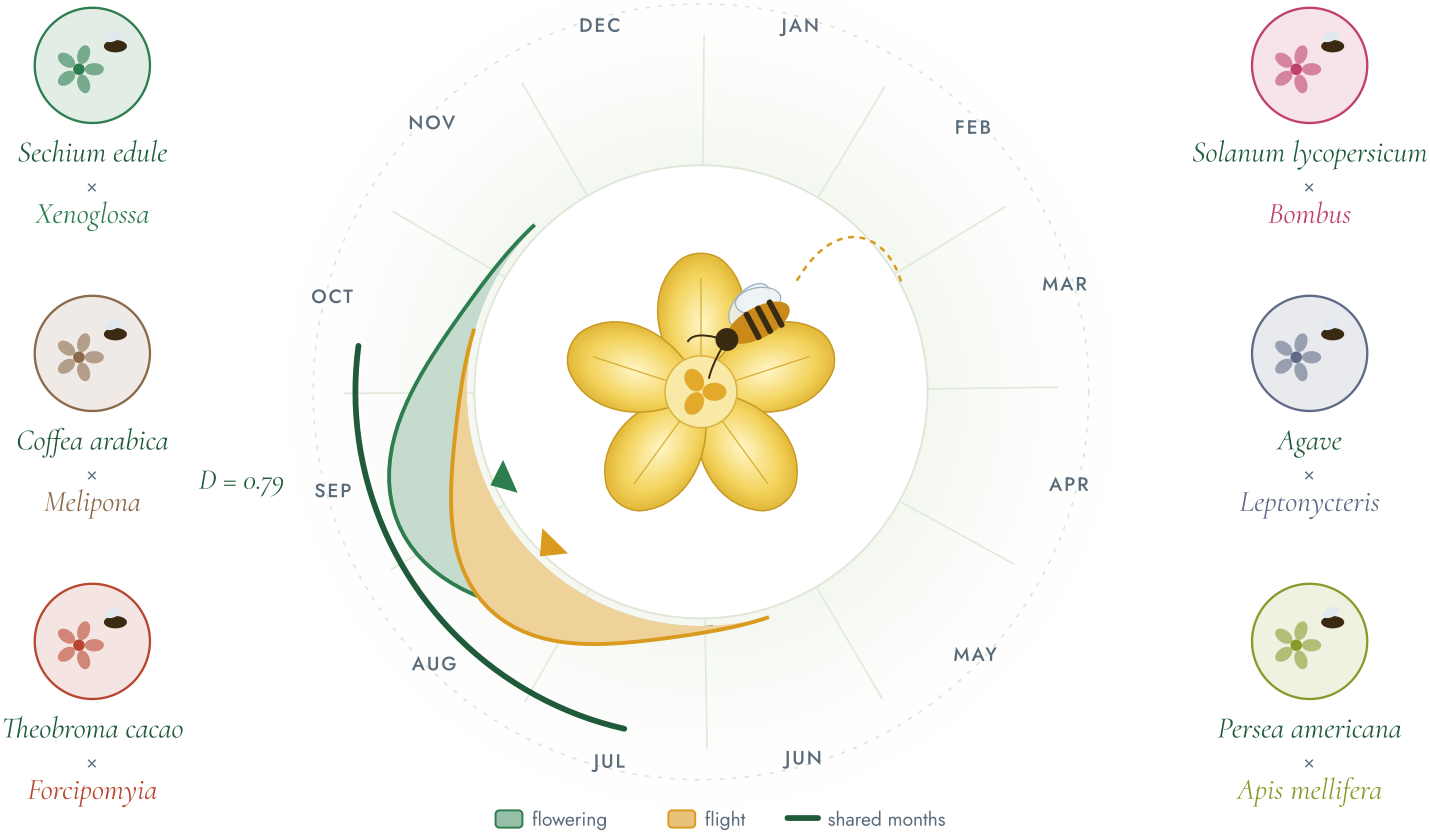
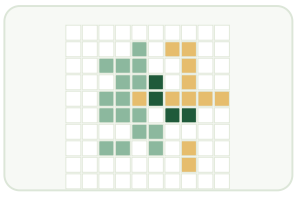
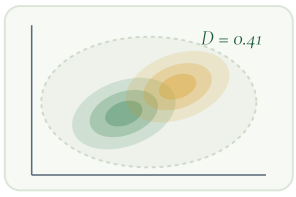
PollinationPro

User's Manual
Manual de usuario

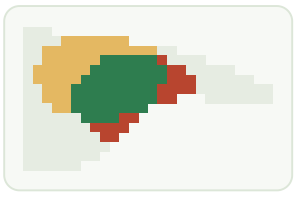
PLANTS, POLLINATORS AND THEIR OVERLAP · PLANTAS, POLINIZADORES Y SU SOLAPAMIENTO




co-occurrence on the grid



niches in environmental space


mismatch map


visit network

Luis Ángel Barrera-Guzmán

FIRST ENGLISH EDITION · 2026 · VERSION 1.2.5

PollinationPro

User's Manual · Plants, pollinators and their overlap

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PROGRAM	PollinationPro, version 1.2.5 (October 2026) · DOI 10.5281/zenodo.23004694
MANUAL	First English edition, 2026, translated from the first Spanish edition. Describes version 1.2.5 of the program.

PROGRAM LICENSE	GNU General Public License, version 3 or later (GPL-3.0-or-later).
MANUAL RIGHTS	© 2026 Luis Ángel Barrera-Guzmán. All rights reserved.

INTERFACE LANGUAGE	Spanish or English, with the ES · EN switch on the top bar. The app opens in Spanish; in this manual the names of buttons, fields and menus are written as they appear on screen in English, for example Clean and apply.

HOW TO CITE THE PROGRAM

Barrera-Guzmán, L.Á. (2026). *PollinationPro: a browser-based platform for analysing plants, pollinators and their overlap with occurrence data* (Version 1.2.5) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23004694

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This is the **concept DOI** on Zenodo: it always resolves to the latest version. The citation of the program is the one given by the Copy the citation button on the app's home page. Online: luisangelbg.github.io/PollinationPro.

PRIVACY

PollinationPro runs entirely inside your browser. Your presence records, your visit observations and your climate layers are read on your computer and **are never sent to any server**. Only when you ask for it does the program query two public services —GBIF, for the names and the records, and a public database of biotic interactions— and whatever they return stays on your machine. Everything else works without the internet.

The illustrations, diagrams and screenshots in this manual are original: they were generated with the program itself or with code written for the manual. All the results shown come from the **practice study** included with the program —"Cucurbits and squash bees"— and from its **practice environmental layers**. Both are **fictitious**: the taxon names are real so that the example reads like a study, but no point, date, visit count or climate value reproduces an observation, and the app says so every time they are loaded. They must not be cited or used to conclude anything about those species.

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How to read this manual

The manual follows the same order as the app: ten numbered blocks that run from top to bottom in the Blocks list on the left. Each chapter explains one block and carries its number —Chapter 5 is Block 5, and its sections, figures and tables are numbered 5.1, 5.2…—. Each one begins with the ecological idea the block needs, goes on to what the screen calculates and how to read it, and ends with the warnings the block can show and what to do about them. This introduction has no number: its sections run I.1, I.2…

A color for each block

The block's color appears on the opening page of its chapter, on the number of each section, on the tab at the edge of the page and in the table of contents, and it is the same as in the strip on the cover and in the app.

Block	Name in the app	What it does
1	Home	Home page, two labs that calculate with the real engine, twenty methods, the theory in plain language and the 60 references
2	Data	The study list, the GBIF records by country, the published interactions, your presence and visit records, and the practice data
3	Cleaning and taxonomy	Fifteen quality rules, cultivated plants decided plant by plant, study area, synonyms, units of analysis and the sentence for the methods
4	Map	A map studio that works offline: its own vector base map, framing and clipping, north arrow, scale, legend, heat or grids, styles and export up to GeoTIFF
5	Co-occurrence	Veech's exact test with a target-group universe, the distance to the nearest visitor against a null model, and range overlap
6	Phenology and elevation	Circular statistics of the dates, phenogram, circular rose, month-by-month overlap corrected for effort, and the elevational distribution
7	Environmental niche	Your layers (or the practice ones), the principal-component space, Schoener's D, the equivalency and similarity tests, and the analysis by variable
8	Potential distribution	Three suitability models with cross-validation, the map of the plant with and without a potential visitor, and the same map with future climate
9	Interaction network	The plant–visitor network of the recorded visits: connectance, nestedness, specialization, modules, null models and sampling completeness
10	Report	The figure catalog up to 900 dpi, the report with the methods written out, and the .zip package that reproduces the study

Types of box

NOTE

Clarifications and practical tips that save time.

IMPORTANT

What can change your conclusions if overlooked.

THEORY IN BRIEF

The idea behind a calculation, without unnecessary formulas.

WORKED EXAMPLE

A case worked step by step with the program's practice data.

DECISION RULE

If the result falls in this case…	**then** this is the recommended reading.

These are guides for interpretation, not laws: what you know about the plant, the visitor and the sampling always comes first.

1.96

KEY FACT

A figure or a fact worth remembering; for example, the standardized effect size beyond which a coincidence departs from chance at roughly $p < 0.05$.

Writing conventions

Written like this	Means
Clean and apply	A button, checkbox or menu of the app, with its text as it appears on screen in English.
Block 4 › Export	A path: open the first item and look for the second inside it.
`decimalLatitude`, `.csv`	A column name, a file name or a value typed as it stands.
Sechium edule, *Xenoglossa*	Scientific names, in italics; the app also writes them this way in tables, legends and figures.
plant · visitor · unit	The two **roles** in the study list, and the **unit of analysis**: a species or an entry of the list (a whole genus, for example), as chosen in Block 3.
SES · D · R^{-} · AUC · TSS · NODF · H_2'	The abbreviations and symbols of the indices; each one is defined the first time it appears, and the glossary (Appendix C) gathers them.
1,856 records · 0.25° · Sep 5	Thousands with a comma and decimals with a point, cells in degrees, dates with the abbreviated month and the day, as in the app (in its own text the manual writes the month in full: September 5). In its own text the manual writes the percent sign without a space (29%); where it quotes the screen, it writes it as the app shows it (29 %).

In

INTRODUCTION · BEFORE THE BLOCKS

Introduction to PollinationPro

What the program is, which questions of pollination ecology it answers with occurrence data, which data it accepts and how to work through it from start to finish. It includes the five ideas that support everything else —coinciding is not pollinating, the record reflects the recorder, the year is a circle, the niche is compared in the environment and the visit is the evidence— and an honest list of what the program does not do.

I.1 What is PollinationPro?

I.2 Questions it answers

I.3 Five ideas behind the manual

I.4 Data it accepts and what it delivers

I.5 What you need and how to open it

I.6 A tour of the interface

I.7 The practice study and the practice layers

I.8 How to work through a study

I.9 What the program does not do

In

I.1 What is PollinationPro?

PollinationPro is a platform for taking a study of plants and their flower visitors **from occurrence records to the paper**: it gathers the GBIF records, the interactions already published and your own observations; it cleans them and harmonizes their taxonomy; it puts them on a map; it measures whether the plant and the visitor coincide in space more than chance predicts, whether they fly and flower in the same months and at the same elevation, whether they share the same climate and where the plant could be left without any potential visitor, today and with future climate; it analyzes the network of recorded visits; and it delivers a report with figures, methods written out and a package that reproduces it.

It works for **any crop** —vegetables, fruit trees, coffee, cocoa, cucurbits, oilseeds—, for its wild relatives and for any native plant, with any group of visitors: bees, bumblebees, stingless bees, flies, butterflies, bats or birds. Everything happens inside the browser: there is nothing to install and no need to know how to program. It is written for botanists, entomologists and thesis students who know that most records of a crop are of planted individuals, that bee taxonomy changes, that records pile up where there are roads, and that a reviewer will strike out the sentence "it is its pollinator because their distributions overlap".

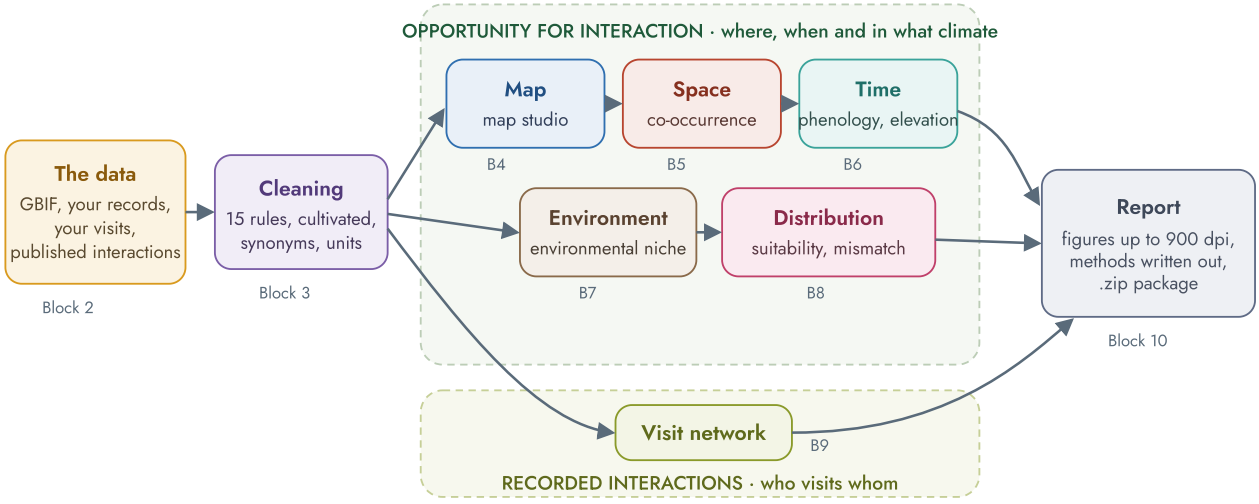


Figure I.1. The path of a study through the blocks. Once cleaned, the presence records feed five readings of the **opportunity for interaction** —where, when and in what climate the plant and the visitor could meet—; the recorded visits feed the network, which is the only reading of **interactions**. The report gathers whatever has been calculated.

I.2 Questions it answers

Each block answers a question that a pollination study with occurrence data really asks. Here they are, in the order in which they are worked through:

Block	The question	What answers it
2	What has been published about my plants and their visitors? How many records do I have, and where from?	Names resolved in GBIF, records by country, published interactions in both directions, and your imported data
3	Which records are errors? Am I studying the crop or its wild populations? What is each taxon called today?	Fifteen rules with their counts, cultivated plants plant by plant, synonyms and the sentence for the methods
4	Where are they? Which states have no records?	The editable map and the tables by state and by country
5	Do they coincide in the territory more than chance predicts, once where sampling took place is accounted for?	Veech's exact test with a target-group universe, sensitivity to cell size, nearest neighbor and hulls
6	Does the visitor fly when the plant flowers, and at the same elevation?	Mean date and concentration of each, Rayleigh, Watson–Williams, Schoener's D month by month, and elevational profiles
7	Do they live in the same climates? Which variable sets them apart?	The environmental space, D and I, equivalency and similarity, and the overlap by variable
8	Where is the plant suitable and no visitor is? What will happen with future climate?	Validated suitability models, the mismatch map and how it changes with future layers
9	How specialized is the relationship? Which visitors are key? How many remain unrecorded?	Connectance, NODF, H_2', d′, modules against null models, strength and completeness
10	How do I hand it in, in a thesis or a paper?	Figures in whatever format is requested, a report with methods written out and references, and a reproducible package

I.3 Five ideas behind the manual

You do not need to be a specialist in pollination ecology to use the app, but five ideas come back again and again. Understanding them turns the results from numbers into arguments.

1. Coinciding is not pollinating

For an insect to pollinate a plant it has to **be where the plant is**, **when the plant flowers**, in **environments both tolerate**; it has to **visit the flower** and, in doing so, **carry and deposit compatible pollen**. Presence records inform the first three links; visit observations, the fourth; the fifth can only be proven in the field. That is why Blocks 5 to 8 speak of **opportunity for interaction** and of **potential pollinators**, and only Block 9 speaks of interactions. A paper with occurrence data is valuable precisely because it narrows down where, when and in what climate the field evidence is worth looking for.

2. The record reflects the recorder

GBIF records pile up near roads, cities, universities and research stations. If the plant and the bee were recorded more where more sampling took place, their records will coincide even if in nature they do not seek each other. Every test in the app that compares an observed value with an expected one does so within the **target group**: the cells, the months or the environmental background where someone recorded something from the study, which are the ones that were really sampled. The first lab of Block 1 shows, with a landscape whose truth is known, what happens when this is not done.

3. The year is a circle

December 31 and January 1 are one day apart, not 364. A species that flowers from November to February has its peak in December–January, not in June as the arithmetic mean of the months would say. That is why dates are read with **circular statistics**: each date is an angle, the mean date is the direction of the average vector, and its length $\bar{R}$, from 0 to 1, says how concentrated the season is.

4. The niche is compared in the environment, not on the map

Two species can live in different places and share a climate, or live close by and use different environments. Block 7 compares them in **environmental space** —the first two principal components of the region's climatic variables—, corrects for what the environment offers and asks whether the niches are identical (equivalency) or more alike than the mere likeness of the available environments would produce (similarity). Block 8 takes that suitability back to the map.

5. The visit is the evidence, and sampling has an end

The network of Block 9 is built not from coincidences but from observed visits: yours or published ones. And a network is always incomplete: how many visitors remain unrecorded is estimated with the rarefaction curve and with Chao1, and **completeness** tells you whether to go back to the field before claiming that a plant has few visitors.

1.4 Data it accepts and what it delivers

WHAT GOES IN

- **A study list**: the scientific names of the plants and the visitors, at any level (species, genus or family), typed by hand or taken from one of the seven example systems. Each name is a unit of analysis.
- **GBIF records**, downloaded from the app by name and by country, only with coordinates, with filters for years, type of record and geospatial issues.
- **Published interactions** from a public database of biotic interactions, searched in both directions (the visitors of each plant and the plants of each visitor).
- **Your presence records** —one row per plant or insect seen, with taxon, latitude and longitude— and **your visit observations** —one row per plant–visitor pair with how many visits—, from `.csv`, `.txt`, `.tsv`, `.xlsx` or text pasted from a spreadsheet.

- **Environmental layers** in GeoTIFF (`.tif`) or ASCII grid (`.asc`), in degrees, for Blocks 7 and 8, and the same variables for a future scenario.
- **A project** saved earlier as `.json`: the list, the records, the interactions and the visits, as they left Block 2.

WHAT COMES OUT

- **On-screen tables** with reading scales: next to almost every number there is a label —*moderate*, *marked*, *acceptable*, *generalist*— with the scale that justifies it one click away.
- **Figures** in every block, each with its ↓ button to export it and its ✎ button to edit its texts, colors and sizes; and the map with its own studio.
- **.csv files** of every important table: raw, cleaned and removed records with the reason, map points, co-occurrence pairs, circular statistics, phenological pairs, environmental values, mismatch classes, network matrix and species.
- **The report**: a single `.html` file that opens in any browser and prints to PDF, with one section per calculated block, the methods written from what was actually run, the references for those methods, the GBIF datasets that must be cited and the calculation record.
- **The package**: a `.zip` with the report, the project, the parameters, the tables and the figures. Whoever receives it can redo the study.

1.5 What you need and how to open it

Online, without downloading anything: open luisangelbg.github.io/PollinationPro in the browser; it is also in the catalog of the LABG Suite (luisangelbg.github.io). Everything is calculated on your computer just as in the local copy: your data are not uploaded anywhere.

On your computer: a computer with a recent browser —Edge, Chrome, Firefox or Safari from the last two or three years— and the program folder. There is no installation: copy the folder and open index.html with a double click. It works from a USB stick and, except for the GBIF download and the search for published interactions, without the internet.

1. Copy the whole PollinationPro folder to your computer. Inside it there must be index.html and the folders js, css, data (the outlines of the countries and of the states of Mexico) and tests.
2. Open index.html with a double click. The first screen is Block 1, the home page. On Windows, Open PollinationPro.bat also opens it as a small local server (port 9750), useful for opening it from another machine on the network.
3. Choose the language with ES · EN and the light or dark theme with the sun button. Both are remembered for next time.
4. If you want to check that the engine calculates correctly in your browser, open tests/index.html: there are 191 tests against published values, exact arithmetic, formulas solved by hand, independent implementations and simulations under the null hypothesis. They must all come out green.

In

I THE APP REMEMBERS THE STUDY

The list, the records, the interactions and the visits are kept in the browser of that computer (in its local database), so that when you open the app again it shows "The previous session was recovered". The options of each block, the map style and the changes made to each figure are also remembered. This is only a convenience: the real backup is the `.json` project of Block 2, which opens on any other machine. The environmental layers are not kept: they are chosen again in Block 7.

I.6 A tour of the interface

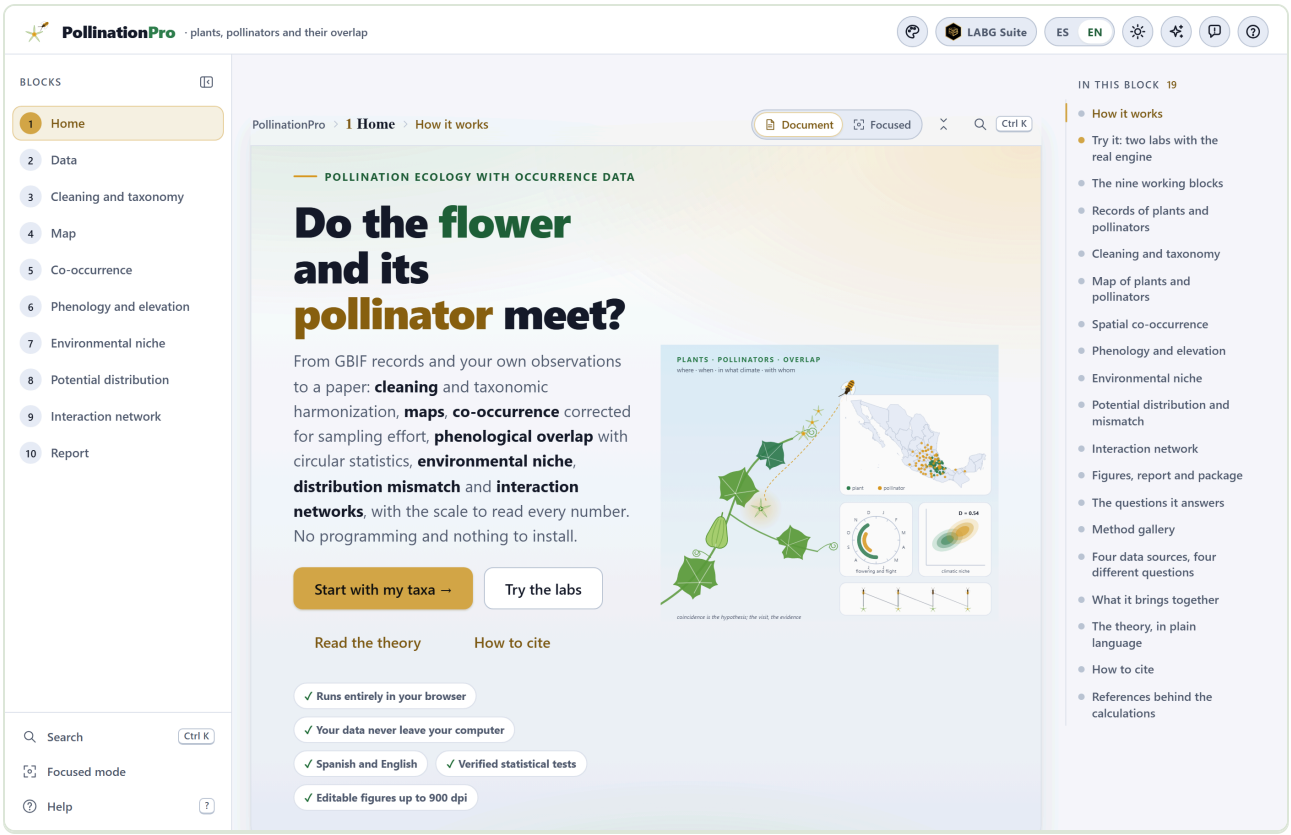


Figure I.2. The opening screen. At the top, the fixed bar with the brand, the palette button (the figure studio), the language switch and the theme switch; on the left, the Blocks list with the ten blocks; on the right, the In this block index with its sections; and, in the center, the home page of Block 1 with the question the program answers: *do the flower and its pollinator meet?*

Element	What it is for
The Blocks list and the In this block index	On the left, the ten blocks, always visible, with a ✓ on those that are finished; on the right, the sections of the open block. You can jump to any block at any time; a block that needs something that does not exist yet says so and offers the button to go and sort it out.
The cards	Each block is divided into numbered cards —"1 · The study list", "2 · Bring the GBIF records"…— that are read from top to bottom: first what you choose, then what is calculated.

Element	What it is for
The result tiles	A row of large figures with their label and a line of context underneath. It is the summary of the card.
The ? sign	Next to a title, a figure or a column, it opens a help sheet: what it measures, how to read it, the scale that says whether the value is low or high and the most common mistake, with its references. There are 50 sheets; each block ends with its 📖 Interpretation guide for this block, which gathers its own.
The scale labels	Next to many numbers there is a word in color —*moderate*, *marked*, *acceptable*, *generalist*—. It comes from the scale of the corresponding help sheet, applied to the value you have on screen.
The ↓ and ✎ buttons of each figure	In the corner of every figure. ↓ exports it (PNG, JPG or SVG, 150 to 900 dpi, light or dark). ✎ opens its six-tab editor: title, axes, series, legend, any text (rewritten, resized, recolored or hidden) and annotations (Section 10.3).
The figure studio	The palette button on the top bar. It changes the palette (nine, among them Okabe–Ito for color-blind readers and a grayscale one), the font, the scales and the background of **all** the figures at once, on screen and in what is exported.
Continue to … →	At the foot of each block. It takes you to the next one.
The messages	Colored bands under a card: blue or green for information, amber for a warning, red for an error that prevents going on. Each chapter ends with those of its block.

I.7 The practice study and the practice layers

Block 2 includes two practice studies that the program itself generates with a fixed seed, without querying the internet: **"Cucurbits and squash bees"** —five species of *Sechium* and the genus *Cucurbita*, with *Xenoglossa*, *Peponapis* and the honey bee, in Mexico— and **"Coffee and stingless bees"** —*Coffea arabica* with the honey bee and four stingless bees, in southern Mexico and Central America—. Each one comes with presence records and a table of visits. They are **fictitious**: the points are drawn around invented centers, the dates around invented peaks and the visits with Poisson counts; the names are real so that the example reads like a study. And they carry **errors planted on purpose** —duplicates, a (0, 0), swapped coordinates, a longitude without its sign, the centroid of Mexico, rounded coordinates, a future year, cultivated plants— so that Block 3 has something to find.

Block 7 also includes four **practice environmental layers** for Mexico and Central America at 0.1°: an invented relief and three climatic variables derived from it with simple rules (cooler with altitude and towards the north, rainier towards the southeast and on the Gulf slope), with the names of the bioclimatic variables BIO1, BIO4 and BIO12, and a warmer and drier **practice future** for Block 8. They look like a climate, but they are not one.

In the practice study	Records	Its role in the manual
Sechium edule	424	The main plant: chayote, with 29% cultivated records
S. compositum, *S. chinantlense*, *S. hintonii*, *S. mexicanum*	71 · 46 · 40 · 35	The wild relatives, with small ranges

In the practice study	Records	Its role in the manual
Cucurbita (genus)	605	The squashes, with 36% cultivated: the plant on which the squash bees depend
Xenoglossa and *Peponapis*	106 · 216	The squash bees, which Block 3 brings together under *Xenoglossa*
Apis mellifera	705	The honey bee: a generalist with a long season, for contrast
Visits at five sites	354 rows	953 visits by 11 visitors to 6 plants: the network of Block 9

! THE PRACTICE DATA ARE NOT CITED

The results in this manual —that chayote and *Xenoglossa* coincide at random in space, that they flower and fly 26 days apart, that the practice future reduces the suitable area by 62%— are properties of the invented data, not of the species. They are for learning to read the screen. Your real study starts with Resolve the names in GBIF.

i THE FIGURES IN THIS MANUAL

All the results in the text were obtained by running the app with the practice study **"Cucurbits and squash bees"**, cleaned with the default values of Block 3, and —in Blocks 7 and 8— with the program's **practice layers**; they were copied from the screen. If you repeat the steps, you should see the same numbers: the practice data are generated with a fixed seed. Only the randomization tests (the null models of Blocks 5, 7 and 9) may vary in the last decimal of a p-value from one run to the next.

I.8 How to work through a study

The order of the blocks is the order in which a pollination paper with occurrence data is written. You do not have to do all of it: a study of distribution and phenology ends at Block 6; one of a visit network needs Blocks 2 and 9. Whatever a block needs from another it calculates on its own when you open it.

1. **Block 2.** Build the study list, resolve the names in GBIF, choose the countries, count and download. Search for the published interactions and import your records and your visits. Save the project.
2. **Block 3.** Review the fifteen rules, decide plant by plant what to do with the cultivated ones, set the study area, confirm the synonyms and the unit of analysis, and apply. Copy the sentence for the methods.
3. **Block 4.** Look at the map: a point in the sea or in the wrong country stands out at once and sends you back to Block 3. Build the figure for the paper and export it.
4. **Blocks 5 and 6.** Read the matrix of pairs and the sensitivity to cell size; then the phenogram, the phenological pairs and elevation. Note which pairs coincide in space and in time.

5. **Blocks 7 and 8.** Load your layers, calculate the environmental space and the tests; fit the suitability models, check their AUC and TSS, read the mismatch and, if you have future layers, how it changes.
6. **Block 9.** With your visits and the published ones, read the network against the null models, the key visitors and the completeness of each plant.
7. **Block 10.** Go through the state of the study, export the figures, check the report in the preview and download the package.

> **! WHAT IS UPSTREAM RULES WHAT IS DOWNSTREAM**
>
> If you change the list or the records in Block 2, the cleaning is redone. If you change a rule, a synonym or the unit of analysis in Block 3 and apply again, everything below is recalculated when you open it —the map, the pairs, the calendars, the environmental space—. The report of Block 10 is written with whatever exists at that moment: that is why it is best generated at the end, and generated again if anything was touched.

1.9 What the program does not do

- **It does not prove that a visitor pollinates.** It measures opportunity for interaction and analyzes recorded visits. Pollen load, deposition after a single visit and fruit set are proven in the field, and the program says so in every interpretation guide.
- **It does not include real climate layers.** Climate databases do not allow their files to be redistributed: they are downloaded once from their website and chosen in Block 7. The practice layers are invented and are only for learning.
- **It does not replace the official GBIF download for the final paper.** The download from the app is for exploring; GBIF asks for every download to be cited with its DOI, which is obtained by repeating it on its portal with the same filters. The report lists the datasets that contributed records and reminds you of this.
- **It is not a full species distribution modeling program.** The three models of Block 8 are simple and transparent, designed to compare the plant with its visitors by the same method; a study centered on the distribution of one species deserves a specialized program and more variables.
- **It does not correct taxonomy on its own.** It resolves the names against the GBIF taxonomic backbone and applies the synonym rules that you confirm; the decision about which name is correct still belongs to the specialist.
- **It does not decide for you.** The decision rules and the scales are guides for reading a number, not verdicts. No warning blocks the work: it points the issue out, explains what it means and leaves the decision to whoever knows the species.

1

BLOCK 1 · HOME

Home: labs, methods and theory

The home page is not a brochure: its two labs generate virtual landscapes and calendars and run on them the same tests the app will run on your records. Since the truth is known in a virtual landscape, you can see when a test is right and when it is wrong. Here you also find the map of the blocks, the questions they answer, the twenty methods with their source publication, what each data source can claim, the theory in plain language and the references.

1.1 What you see when you open the app

1.2 How it works: the seven steps of a paper

1.3 Lab 1: co-occurrence and sampling bias

1.4 Lab 2: phenological overlap

1.5 The blocks, the questions and the method gallery

1.6 Four data sources, four questions

1.7 The theory, in plain language

1.8 How to cite, and the references

1.1 What you see when you open the app

The first screen (Figure I.2) asks a single question: *do the flower and its pollinator meet?* Below it, a paragraph sums up the path —from GBIF records and your own observations to a paper— and four buttons lead to the four places where people usually start.

Button	Where it leads
Start with my taxa →	To Block 2, to type the study list.
Try the labs	Down to the two labs on this same page (Sections 1.3 and 1.4).
Read the theory	Down to "The theory, in plain language", and opens its first item (Section 1.7).
How to cite	Down to the citation of the program (Section 1.8).

The illustration on the right sums up the app in one picture: a cucurbit vine with flower and fruit, a bee arriving, the map of Mexico with plant and pollinator records, a year clock with flowering and flight, a niche diagram with its D and, at the bottom, a visit network. Its caption says it all: *coincidence is the hypothesis; the visit, the evidence*. The five badges under the buttons sum up what the program guarantees: it runs entirely in the browser, the data never leave the computer, it is in Spanish and English, its tests are verified, and its figures can be edited and exported up to 900 dpi.

1.2 How it works: the seven steps of a paper

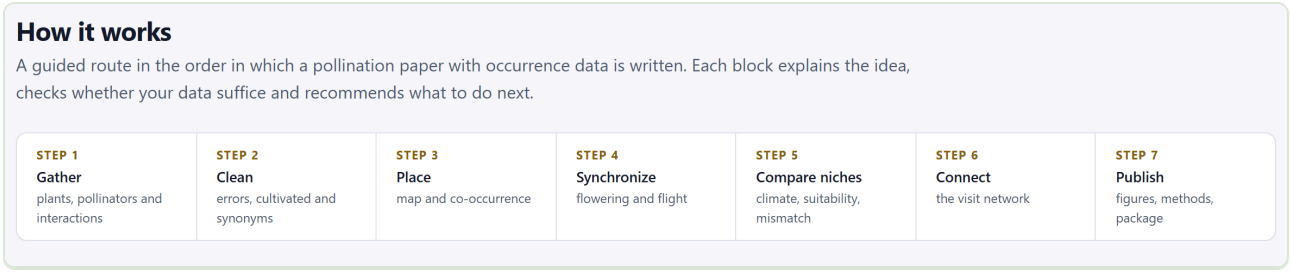


Figure 1.1. The guided route, in the order in which a pollination paper with occurrence data is written. Each step groups one or two blocks.

Step	What is done	Blocks
1 · Gather	plants, pollinators and interactions	2
2 · Clean	errors, cultivated and synonyms	3
3 · Place	map and co-occurrence	4 and 5
4 · Synchronize	flowering and flight	6
5 · Compare niches	climate, suitability, mismatch	7 and 8
6 · Connect	the visit network	9
7 · Publish	figures, methods, package	10

B1

1.3 Lab 1: co-occurrence and sampling bias

The first lab answers the question that is most often got wrong in studies with occurrence data: *do the plant and the bee coincide more than expected?* It generates a landscape of 36 × 24 cells with an environmental gradient. The plant and the bee occupy the cells that suit them according to their optimum and their niche breadth. Then someone samples, closer to a road the higher the bias, and only the sampled cells yield records. Veech's exact test (Chapter 5) is run on those records with three different universes, and compared with **the truth**: the same test with the real presences, which in the field are never known.

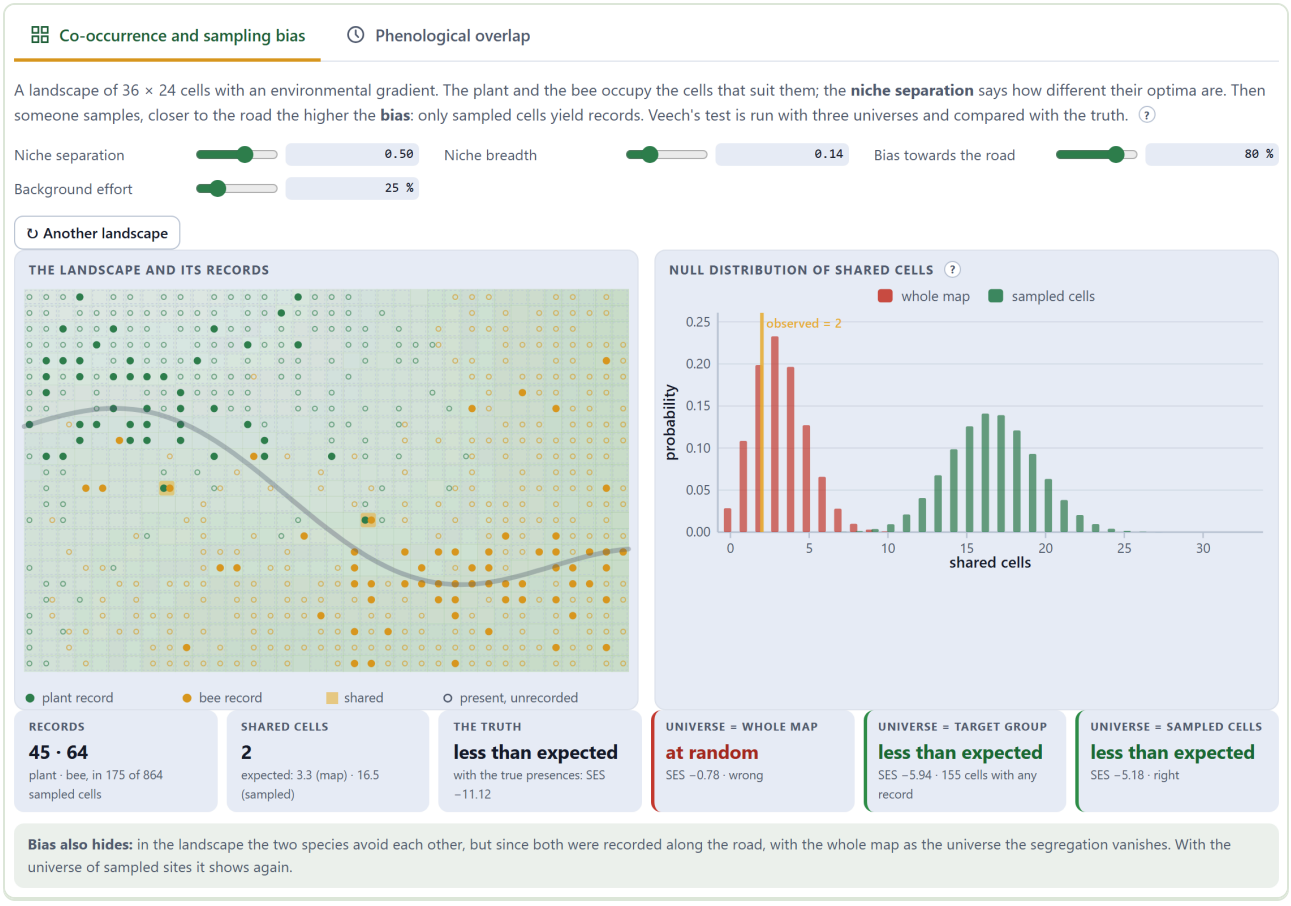


Figure 1.2. The lab with its starting values: niche separation 0.50, breadth 0.14, bias towards the road 80% and background effort 25%. Filled dots are records; open circles, presences nobody recorded; the gray band, the road.

Control	What it changes
Niche separation	How different the environmental optima of the plant and the bee are, from 0 (the same) to 0.8. With 0 they share the landscape; with 0.8 they avoid each other.
Niche breadth	How much of the gradient they tolerate. Large breadths make almost everything coincide.
Bias towards the road	How strongly sampling concentrates along the road, from 0 to 100%.

Control	What it changes
Background effort	The probability of sampling a cell far from the road.
↻ Another landscape	Draws another landscape with the same controls. With the same controls and the same landscape, the result is always the same.

The tiles underneath are the lesson. **Records** and **Shared cells** give what was observed —45 plant records and 64 bee records, 2 shared cells in Figure 1.2— and how many would be expected with two of the universes. **The truth** gives what happens in the landscape: the two species avoid each other ("less than expected", SES −11.12). The next three tiles are the same test with three universes: the **whole map** calls it "at random" and gets it wrong (in red); the **target group** —the 155 cells where the sampler recorded the plant, the bee or another taxon of the same group— and the **sampled cells** recover the segregation (in green). The paragraph below sums it up: *"Bias also hides: in the landscape the two species avoid each other, but since both were recorded along the road, with the whole map as the universe the segregation vanishes."*

▸ WORKED EXAMPLE · BIAS ALSO INVENTS

Set the niche separation to 0.20, the breadth to 0.25 and the background effort to 5%, with the bias at 80%. In the landscape, the plant and the bee are distributed at random with respect to each other (SES −0.37). But their records pile up along the road, and with the whole map as the universe the test declares a huge positive association —54 shared cells against 9.3 expected, SES 16.38— that does not exist. With the target group (SES −1.68) and with the sampled cells (SES −0.86) it comes out "at random" again. This is *the bias trap*, and it is the most common mistake in papers with occurrence data.

Now raise the background effort to 100% and lower the bias to 0: the whole landscape is sampled and the whole-map universe is right. The target group, on the other hand, leans towards "less than expected" (SES −2.36), because it leaves out the cells where nobody recorded anything and, with them, the shared absences. The app explains it in the same paragraph: the target group corrects bias, and when there is no bias it costs a little accuracy.

✓ DECISION RULE · WHICH UNIVERSE TO USE

If your records come from GBIF or from collections	**then** use the target group (the default option in Block 5): your records carry the road bias.
If your records come from systematic sampling (a grid or transects walked evenly)	**then** the universe is the cells you walked; "all land" is only correct if you sampled all the land.

The ? sheet on *Sampling bias* explains it with references (Phillips et al. 2009; Kramer-Schadt et al. 2013).

1.4 Lab 2: phenological overlap

The second lab puts the circular statistics of Block 6 to work. A plant flowers and a bee flies around two peak dates, with two spreads; the app draws records from both calendars, calculates the mean date and the concentration of each one, tests whether they differ and measures how much they overlap month by month.

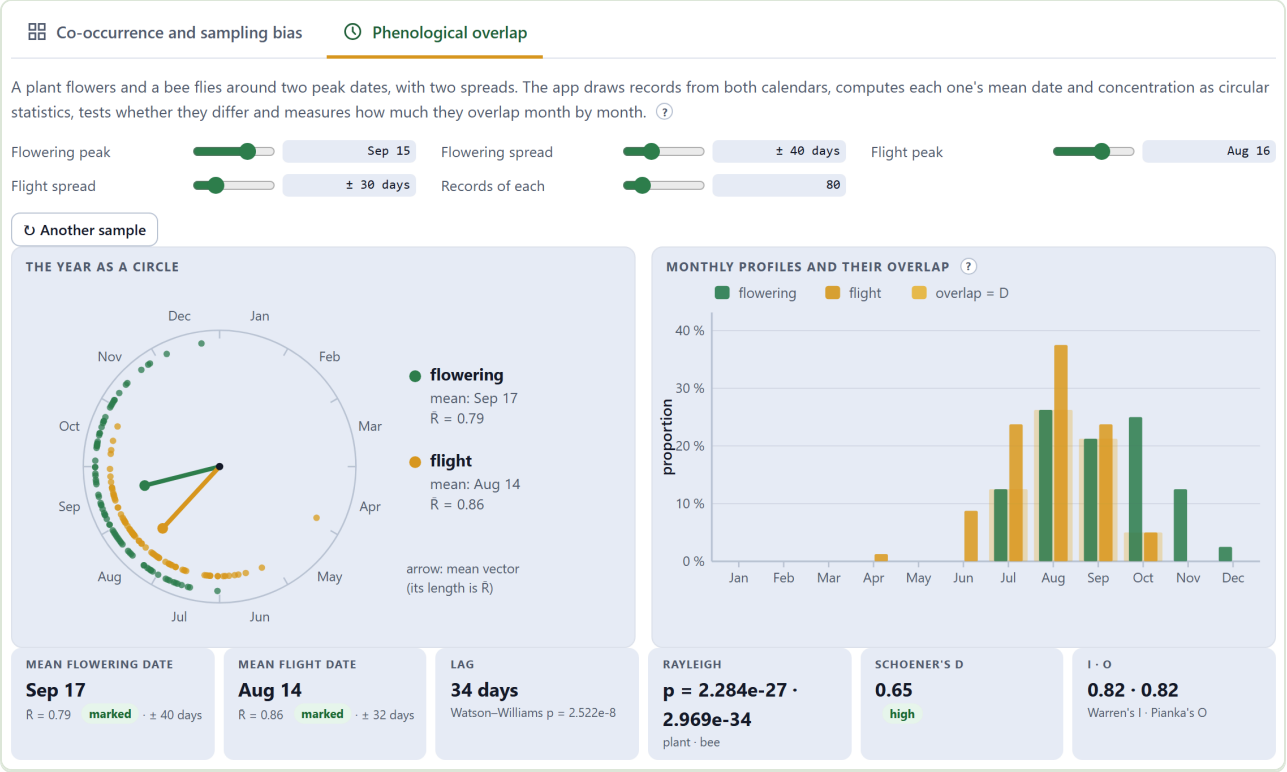


Figure 1.3. Flowering peaking on September 15 (± 40 days) and flight peaking on August 16 (± 30 days), 80 records of each. On the left, each record is a dot on the circle of the year and the arrow is the mean vector; on the right, the monthly profiles and their overlap.

Tile	What it says	In Figure 1.3
Mean date of flowering and of flight	The direction of the mean vector, with $\bar{R}$—from 0, dates spread over the whole year, to 1, all on the same day—and its scale label, and the circular standard deviation in days.	Sep 17 · $\bar{R}$ 0.79 *marked* · ± 40 days; Aug 14 · $\bar{R}$ 0.86 · ± 32 days
Lag	The days between the two mean dates and the Watson–Williams test that they are the same.	34 days · $p = 2.5 \times 10^{-8}$
Rayleigh	Whether each calendar is seasonal (small p) or uniform over the year.	$p \approx 10^{-27}$ and 10^{-34}
Schoener's D	The fraction of the two monthly profiles that is shared, from 0 to 1.	0.65 *high*
I · O	Warren's I and Pianka's O, two overlap indices that are more lenient with small differences.	0.82 · 0.82

THEORY IN BRIEF · WHY D CAN BE HIGH AND THE LAG SIGNIFICANT

Watson–Williams asks whether the two **mean** dates are equal; with 80 records on each side, a lag of one month is highly significant. D asks what fraction of the **whole calendars** is shared, and two two-month seasons one month apart still overlap a great deal. Both things are true at once: the bee flies before the flowering peak, but most of the flowering happens with bees flying. A paper reports both. Move the **flight peak** towards March and you will see D drop to zero long before the lag reaches six months.

1.5 The blocks, the questions and the method gallery

Below the labs, the home page offers three indexes of the app. **The nine working blocks** (Figure 1.4) are cards that take you to the block with one click; each one says what it does, with a drawing that sums up its main figure.



Figure 1.4. The cards of Blocks 2 to 10. The amber label gives the dimension each block looks at: data, quality, map, space, time, environment, distribution, network and report.

The **method gallery** gathers the twenty methods of the app with their source publication and the block where they are used. The buttons at the top filter it by family: Data and quality, Space, Time, Environment and Networks and diversity.

B1

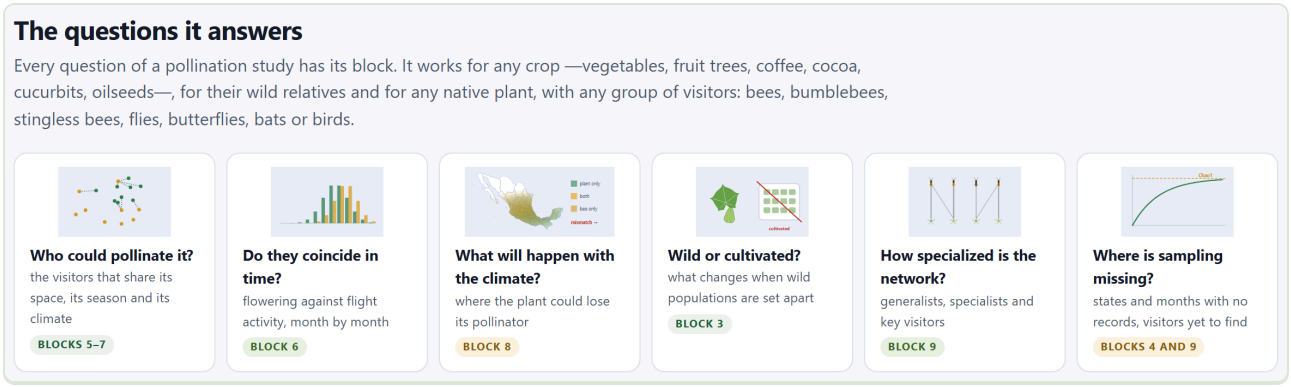


Figure 1.5. The questions it answers, with the block that answers each one: who could pollinate it, whether they coincide in time, what will happen with the climate, wild or cultivated, how specialized the network is, and where sampling is missing.

Family	Methods
Data and quality	GBIF download by taxon and country · taxonomic harmonization · record cleaning · wild versus cultivated
Space	Probabilistic co-occurrence (Veech 2013) · similarity by cells (Jaccard, Sørensen, Simpson and cover) · nearest-neighbor distance against a target-group null · overlap of convex hulls
Time	Circular statistics (mean date, $\bar{R}$, Rayleigh, Watson–Williams) · month-by-month phenological overlap corrected for effort · elevational overlap with kernel densities
Environment	PCA of the environment (Broennimann et al. 2012) · Schoener's D and Warren's I · niche equivalency and similarity (Warren et al. 2008) · environmental suitability (envelope, Mahalanobis and background logistic) · mismatch map
Networks and diversity	Bipartite network · NODF nestedness (Almeida-Neto et al. 2008) · H_2' and d' specialization (Blüthgen et al. 2006) · visitor richness with rarefaction, Chao1 and coverage

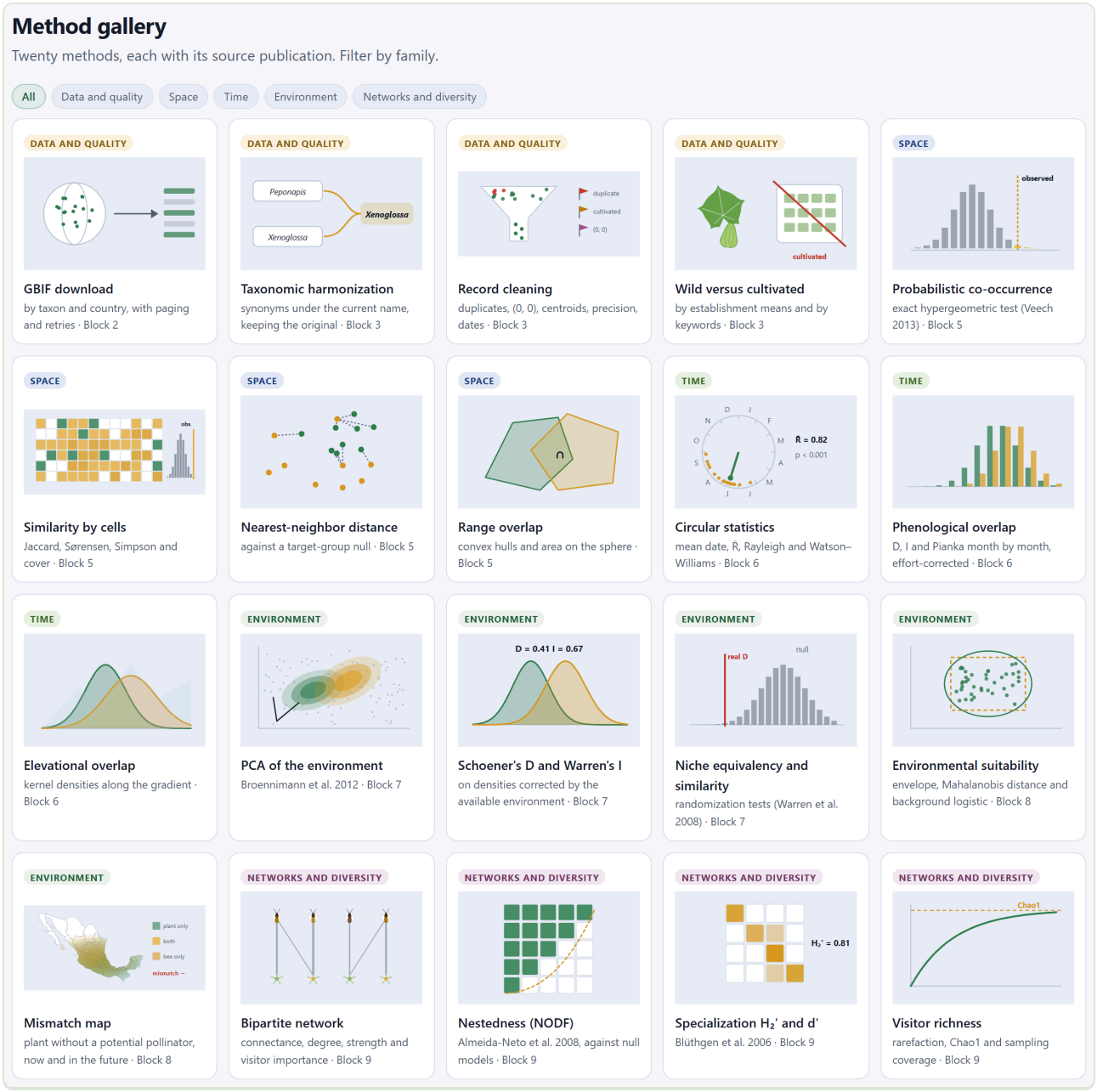


Figure 1.6. The method gallery. Each card gives the method, what it consists of in one line, and the block where it is used.

1.6 Four data sources, four questions

A table on the home page says what each type of data can claim and what it cannot. It is the best defense against the sentence a reviewer strikes out: each source answers a different question, and a good paper combines at least two.

B1

Four data sources, four different questions

What each kind of data can assert and what it cannot. A good paper combines at least two.

	GBIF records	Your presence records	Your visit observations	Interaction databases
What they say	the taxon was at that place on that date	the same, with your own quality control	this visitor touched this flower	someone published that A visits or pollinates B
What they do not say	where it is absent; whether it interacts	where it is absent	whether it carried or deposited pollen	where, when or how often, almost never
Main bias	uneven effort: cities and roads	your field sites	hours and sites observed	what was studied and digitized
What they are for	opportunity for interaction	opportunity and verification	the network and specialization	context and comparison
In the app	✓ Blocks 2–8	✓ Blocks 2–8	✓ Block 9	✓ Blocks 2 and 9

What it brings together

One place
data, cleaning, space, time, niche and network chained: what leaves one block enters the next

Cited methods
every test carries its original publication, and the report cites only those you used

Validated
every formula checked against exact arithmetic, closed forms or simulation on the test page

Scales to read by
every number carries beside it the scale that says whether it is low or high, and the commonest mistake

Editable figures
palette, font and background of the whole app in one panel, and every figure with its own ✎ editor of texts, colors and sizes; export up to 900 dpi

Nothing to install
opens with a double click; only the GBIF download needs the internet and your data never leave your computer

Figure 1.7. The four sources and, below them, the six things the app brings together in one place: the chaining of the blocks, the cited methods, the validation, the scales to read by, the editable figures and working without installing anything.

1.7 The theory, in plain language

Five collapsible items explain what needs to be understood before downloading the first record. Each one shows on the right the block where it applies.

Item	What it explains
Coinciding is not pollinating: the chain of evidence *(start here)*	The five links, and why Blocks 5 to 8 speak of "potential pollinators" and Block 9 of recorded interactions.
Sampling bias and the target group *(Block 5)*	Why records coincide where more sampling took place, the target-group solution and the hypergeometric distribution.
The year is a circle *(Block 6)*	The date as an angle, the mean vector and $\bar{R}$, Rayleigh, Watson–Williams and the three overlap indices.
The niche in environmental space *(Blocks 7 and 8)*	The method of Broennimann et al. (2012), the tests of Warren et al. (2008) and the idea of mismatch.
Crops, wild relatives and their pollinators *(what it is for)*	Six crop systems with their visitors and their typical question, why planted plants should be separated from wild ones, and the example of *Peponapis*, a subgenus of *Xenoglossa* since Freitas et al. (2023).

$$P(j \text{ shared}) = C(N_1, j) \cdot C(N - N_1, N_2 - j) / C(N, N_2) \quad E[j] = N_1 \cdot N_2 / N$$

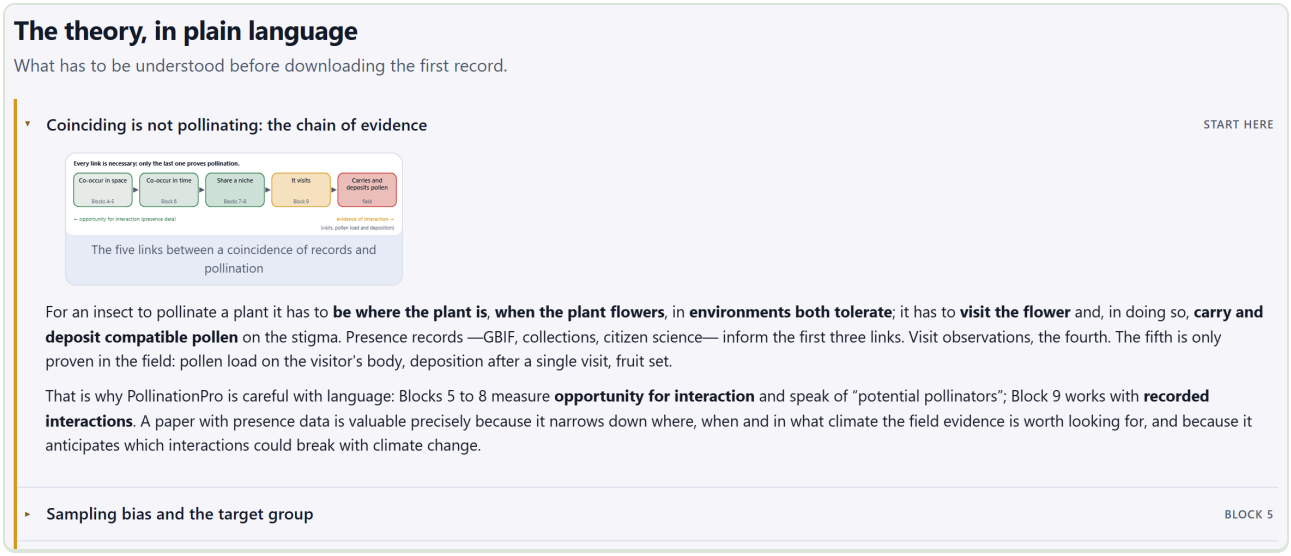


Figure 1.8. The first item, "Coinciding is not pollinating", with the diagram of the five links: co-occur in space (Blocks 4–5), in time (Block 6), share a niche (Blocks 7–8), the visit (Block 9) and the carrying and deposition of pollen (field).

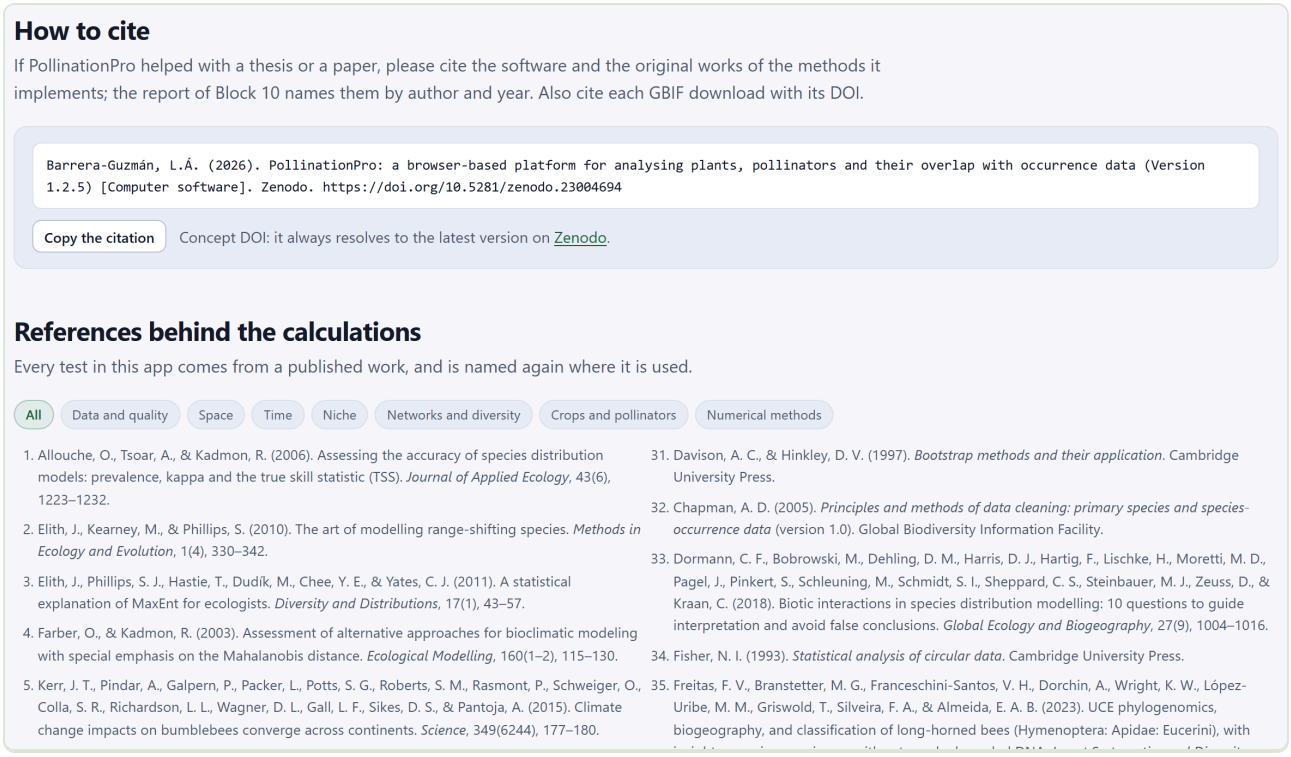
The formula above is that of Veech's test: if the plant occupies N1 of N cells and the visitor N2, and they were placed at random with respect to each other, the number of shared cells would follow the hypergeometric distribution. It is the only formula in the manual worth remembering, because the reading of the whole of Chapter 5 depends on it.

1.8 How to cite, and the references

Copy the citation copies it to the clipboard in the language of the interface. The citation of the program is not enough: each method is cited to its original work —the report of Block 10 names them by author and year, and only those you used—, and each GBIF download is cited with its DOI. The list of references can be filtered into seven families: data and quality, space, time, niche, networks and diversity, crops and pollinators, and numerical methods. Appendix E reproduces it in full.

> **I** THE HOME-PAGE FIGURES CAN BE EXPORTED TOO
>
> The figures of the two labs have their ↓ and ✎ buttons like any other. They are useful for a class or a presentation: the co-occurrence lab, with high and low bias, is the quickest way to explain to a committee why an analysis corrects for sampling effort.

B1



How to cite

If PollinationPro helped with a thesis or a paper, please cite the software and the original works of the methods it implements; the report of Block 10 names them by author and year. Also cite each GBIF download with its DOI.

```
Barrera-Guzmán, L.Á. (2026). PollinationPro: a browser-based platform for analysing plants, pollinators and their overlap with occurrence data (Version 1.2.5) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23004694
```

Copy the citation Concept DOI: it always resolves to the latest version on Zenodo.

References behind the calculations

Every test in this app comes from a published work, and is named again where it is used.

All | Data and quality | Space | Time | Niche | Networks and diversity | Crops and pollinators | Numerical methods

1. Allouche, O., Tsoar, A., & Kadmon, R. (2006). Assessing the accuracy of species distribution models: prevalence, kappa and the true skill statistic (TSS). *Journal of Applied Ecology*, 43(6), 1223–1232.
2. Elith, J., Kearney, M., & Phillips, S. (2010). The art of modelling range-shifting species. *Methods in Ecology and Evolution*, 1(4), 330–342.
3. Elith, J., Phillips, S. J., Hastie, T., Dudík, M., Chee, Y. E., & Yates, C. J. (2011). A statistical explanation of MaxEnt for ecologists. *Diversity and Distributions*, 17(1), 43–57.
4. Farber, O., & Kadmon, R. (2003). Assessment of alternative approaches for bioclimatic modeling with special emphasis on the Mahalanobis distance. *Ecological Modelling*, 160(1–2), 115–130.
5. Kerr, J. T., Pindar, A., Galpern, P., Packer, L., Potts, S. G., Roberts, S. M., Rasmont, P., Schweiger, O., Colla, S. R., Richardson, L. L., Wagner, D. L., Gall, L. F., Sikes, D. S., & Pantoja, A. (2015). Climate change impacts on bumblebees converge across continents. *Science*, 349(6244), 177–180.
31. Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap methods and their application*. Cambridge University Press.
32. Chapman, A. D. (2005). *Principles and methods of data cleaning: primary species and species-occurrence data* (version 1.0). Global Biodiversity Information Facility.
33. Dormann, C. F., Bobrowski, M., Dehling, D. M., Harris, D. J., Hartig, F., Lischke, H., Moretti, M. D., Pagel, J., Pinkert, S., Schleuning, M., Schmidt, S. I., Sheppard, C. S., Steinbauer, M. J., Zeuss, D., & Kraan, C. (2018). Biotic interactions in species distribution modelling: 10 questions to guide interpretation and avoid false conclusions. *Global Ecology and Biogeography*, 27(9), 1004–1016.
34. Fisher, N. I. (1993). *Statistical analysis of circular data*. Cambridge University Press.
35. Freitas, F. V., Branstetter, M. G., Franceschini-Santos, V. H., Dorchin, A., Wright, K. W., López-Uribe, M. M., Griswold, T., Silveira, F. A., & Almeida, E. A. B. (2023). UCE phylogenomics, biogeography, and classification of long-horned bees (Hymenoptera: Apidae: Eucerini), with

Figure 1.9. The citation of the program, with the version you have open, and the list of the 60 references behind the calculations, which can be filtered by family.

2

BLOCK 2 · DATA

Records of plants and visitors

Which plants and which visitors enter the study, and all their records: those from GBIF, the interactions already published and your own. Nothing is analyzed yet: here things are gathered, counted and checked. It is the only block that uses the internet, and only when you ask it to.

2.1 Where the data come from

2.2 The study list

2.3 The GBIF records

2.4 The interactions already published

2.5 Your own data

2.6 The practice data

2.7 What there is, and the project

2.8 The warnings in this block

2.1 Where the data come from

A pollination study with occurrence data combines four sources that say different things (Section 1.6). Block 2 brings them together in five cards: the study list, the GBIF records, the published interactions, your own data and a summary of what there is.

Source	What it contributes	Where it is used
GBIF	Presence records from collections, herbaria, citizen observations and projects, with coordinates, date, type of record and source dataset.	Blocks 3 to 8
Biotic-interaction database	Published plant–visitor pairs ("visits flowers of", "pollinates"), almost never with a place or a date.	Block 9
Your presence records	The same as GBIF, with your own quality control: plants and insects you saw, with their point.	Blocks 3 to 8
Your visit observations	Plant–visitor pairs you observed, with how many visits and, if you can, site and date.	Block 9

2.2 The study list

Everything starts with the names. They are typed at any level —species (*Coffea arabica*), genus (*Bombus*) or family— in the **Plants** column or in the **Visitors** column, and each name is an **entry** of the list: a genus gathers the records of all its species. Several names are separated with commas or semicolons. Or you can start from one of the seven **example systems**, which are starting points, not recommendations: Mexican cucurbits and squash bees; coffee and bees; cocoa and pollinating midges (*Forcipomyia*); Solanaceae and bumblebees; avocado and its visitors; agaves and nectar-feeding bats; and vanilla and orchid bees.

Resolve the names in GBIF looks up each name in the GBIF taxonomic backbone. When there are **homonyms** —the same name in two kingdoms, or a fossil name alongside the living one— the app picks the one from the kingdom that matches the role (plants or animals) with the most records, and leaves a selector to change it. In Figure 2.1, *Cucurbita* L. has 94,500 records and a homonym with 180; among the chayote species, *Sechium edule* has 4,278 and the wild ones between 53 and 145, and among the visitors the honey bee exceeds 1.3 million. Each name is shown with its number of records worldwide; after counting with the chosen countries (Section 2.3), the number becomes that of those countries.

What a name can say	What it means and what to do
accepted	The name is accepted in the GBIF backbone. This is the normal case.
synonym or *doubtful*	GBIF considers it a synonym or doubtful. The records are requested by the key of the chosen name; check whether you should type the accepted name instead.
GBIF does not recognize the name: check the spelling	There was no match. It is almost always a typo or a very recent name.
practice data (not resolved in GBIF)	The entry comes from the fictitious data (Section 2.6): it has no GBIF key and does not need one.

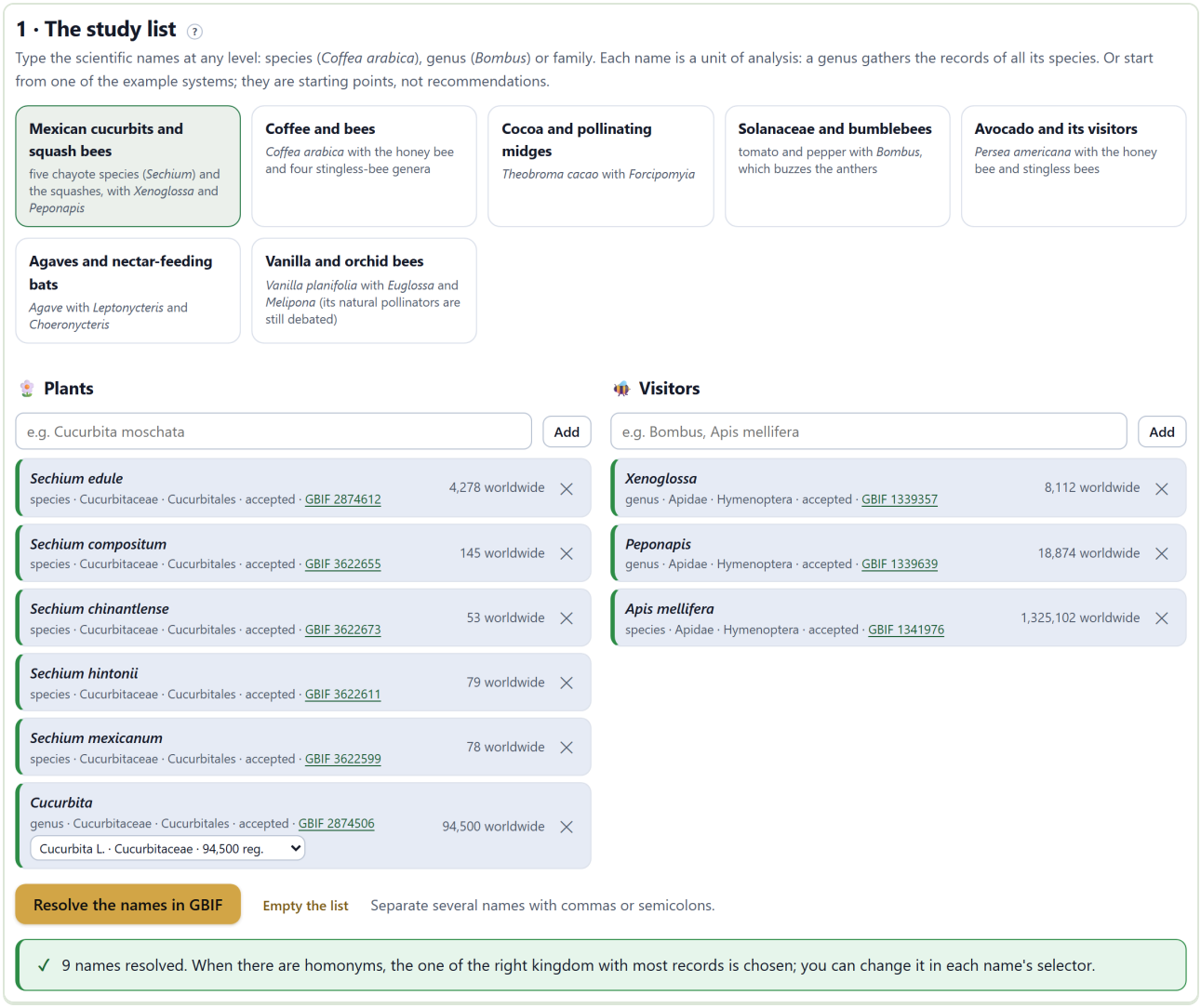


Figure 2.1. The system of the Mexican cucurbits resolved in GBIF (real query of October 31, 2026; your numbers will differ). Each entry shows its rank, family, order, taxonomic status and the GBIF key with its link; *Cucurbita* also shows a selector, because GBIF has more than one name with that spelling.

> **CHOOSING AN EXAMPLE SYSTEM STARTS A NEW STUDY**
>
> When you click one of the seven systems, the list, the records and the published interactions are replaced by those of the example, and the countries are set to its own. If you already had a study, save it first as a project (Section 2.7). To remove a single name use its ×: it takes its records with it, and the app says so. Empty the list deletes the whole study.

B2

2.3 The GBIF records

The second card requests the records of each resolved name, **only with coordinates** and filtered by country on the server. Choosing the countries here is much better than downloading the world and filtering afterwards: the count already comes filtered and the download is shorter.

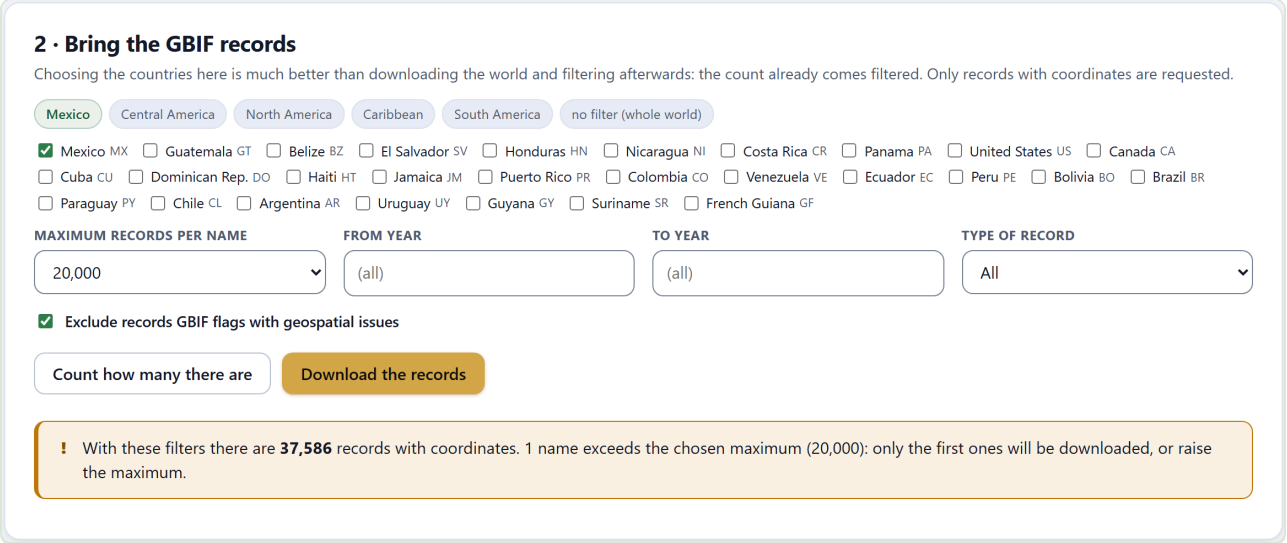


Figure 2.2. The download filters and the result of Count how many there are for the list of the cucurbits in Mexico (real query of October 31, 2026): 37,586 records with coordinates, and one name (*Apis mellifera*) that exceeds the maximum of 20,000.

Control	What it does
Regions and countries	The buttons Mexico, Central America, North America, Caribbean and South America check or uncheck their countries; the checkboxes are ticked one by one. no filter (whole world) unchecks them all.
Maximum records per name	5,000, 20,000, 50,000 or 99,000. A name with more is downloaded up to the maximum and marked as cut.
From year and **To year**	Left empty, all years are requested.
Type of record	All, collection specimens only, observations only, or specimens and observations without fossils or records of unknown type.
Exclude records GBIF flags with geospatial issues	Checked by default: GBIF has already detected impossible, swapped or out-of-country coordinates in them. Block 3 also checks what GBIF does not flag.
Count how many there are	Asks how many records there are with those filters, without downloading. It is worth always doing this first.
Download the records · Stop	Downloads name by name, in pages of 300 with retries, with a progress bar per name. Stop interrupts the download; whatever has already arrived is kept.

When it finishes, the app says how many records arrived and reminds you that GBIF asks for every download to be cited with its DOI. Downloading a name again replaces only its GBIF records, never yours: if a name failed or was stopped, simply repeat the download.

THE APP'S DOWNLOAD IS FOR EXPLORING; THE PAPER'S IS REPEATED ON GBIF

The app requests the records from the public GBIF interface, which does not issue a DOI. For the final paper, repeat the download on the GBIF portal with the same filters —they are kept in the calculation record of Block 10— and cite the DOI it gives you. The report also lists the datasets that contributed cleaned records, because GBIF asks for each one to be credited.

DECISION RULE · HOW MANY RECORDS TO DOWNLOAD

If a generalist visitor (the honey bee) exceeds the maximum	**then** there is no need to raise it: 20,000 well-spread records are enough for its map, its calendar and its niche. Note it in the methods.
If the plant or the visitor you care about exceeds the maximum	**then** raise it to 50,000 or 99,000, or narrow the years: a cut leaves out records at random from the order in which GBIF delivers them.
If a name has fewer than 20 records in the area	**then** its tests will be weak: consider merging it with its genus (a single entry) or widening the area.

2.4 The interactions already published

Search published interactions queries a public database of biotic interactions, which gathers what has been published in papers, databases and collections, in both directions: the visitors to the flowers of each plant in the list, and the plants whose flowers each visitor visits. The result is a table of pairs with the visitor's group (bee, wasp, hoverfly, fly, butterfly or moth, beetle, bat, bird), how many records support it and whether at least one of them calls it **pollination** and not just a visit.

Up to 4,000 records are requested per name. The table shows up to 300 pairs sorted by number of records; ↓ Interactions (.csv), at the foot, downloads all the records with their type, their source (when the database gives it) and their coordinates (when they have them). When the database returns the citation of the original study, a **Sources** column also appears with how many studies support each pair.

WHAT THE TABLE TEACHES BEFORE THE ANALYSIS

The full table already offers three useful readings. The first rows are of squashes with squash bees (*Peponapis*); *Sechium edule* appears further down, outside Figure 2.3, only with the honey bee (2 records, one of which claims pollination) and with *Trigona corvina*, and with no squash bee: the chayote–*Xenoglossa* link that the practice study explores is undocumented. Some plants appear under names that are no longer used (*Sicyos edulis*, an old synonym of chayote, with the honey bee and three other visitors): Block 9 can apply the synonyms of Block 3 to them. And the same pair appears at species and at genus level (*Cucurbita pepo* and *Cucurbita*): Block 9 lets you group plants and visitors by genus.

B2

3 · Interactions already published

Biotic-interaction databases gather what has been published in papers and collections: "such visitor visits the flowers of such plant" or "pollinates it". The search goes both ways: the visitors of each plant in the list and the plants each visitor visits. They give context and feed the network of Block 9; they almost never carry a place or a date.

Search published interactions

✓ 2,390 published interaction records.

Plant	Visitor	Group	Records	Pollinates?	Sources
Cucurbita IN LIST	*Peponapis pruinosa* IN LIST	bee	174	visit only	87
Cucurbita pepo IN LIST	*Peponapis pruinosa* IN LIST	bee	150	at least one says so	75
Deinandra fasciculata	*Xenoglossa pomonae* IN LIST	bee	80	visit only	80
Cucurbita moschata IN LIST	*Peponapis pruinosa* IN LIST	bee	37	visit only	36
Cucurbita pepo IN LIST	*Apis mellifera* IN LIST	bee	33	at least one says so	27
Cucurbita IN LIST	*Peponapis limitaris* IN LIST	bee	26	visit only	1
Cucurbita IN LIST	*Bombus impatiens*	bee	26	visit only	25
Cucurbita pepo IN LIST	*Bombus impatiens*	bee	25	at least one says so	23
Cucurbita maxima IN LIST	*Peponapis pruinosa* IN LIST	bee	24	visit only	12
Cucurbita IN LIST	*Peponapis crassidentata* IN LIST	bee	20	visit only	1
Cucurbita IN LIST	*Xenoglossa gabbii* IN LIST	bee	20	visit only	1
Cucurbita IN LIST	*Apis mellifera* IN LIST	bee	19	at least one says so	14

Figure 2.3. The first rows of 2,390 published interaction records for the list of the cucurbits (real query of October 31, 2026). At the top are the squashes (*Cucurbita*) with *Peponapis pruinosa*. The in list label marks the names that belong to an entry of your list; *Deinandra fasciculata*, a composite visited by *Xenoglossa pomonae*, does not carry it because it arrived through the visitor search.

2.5 Your own data

The fourth card imports two kinds of table, each with its own tab. **Presence records**: one row per plant or insect seen, with taxon, latitude and longitude and, if you have them, date, elevation, locality and whether it is wild or cultivated. **Visit observations**: one row per plant–visitor pair observed, with how many visits and, if you can, site and date. It accepts `.csv`, `.txt`, `.tsv` and `.xlsx` —in UTF-8 or Windows-1252, with any separator— or the text that is pasted when you copy cells from a spreadsheet. ↓ Download an empty template gives a table with the headings the app recognizes.

Table	Fields it recognizes (by their heading, in Spanish or English)
Presence records	**taxon***, **latitude***, **longitude***, date (or year, month and day separately), elevation, uncertainty, country, state, locality, type of record, wild/cultivated, reproductive condition, role (plant/visitor), notes, collector or observer. Darwin Core headings (`scientificName`, `decimalLatitude`, `eventDate`…) are recognized too.
Visit observations	**plant***, **visitor***, number of visits (1 if missing), date (or year, month and day), site, latitude, longitude, type of interaction, notes.

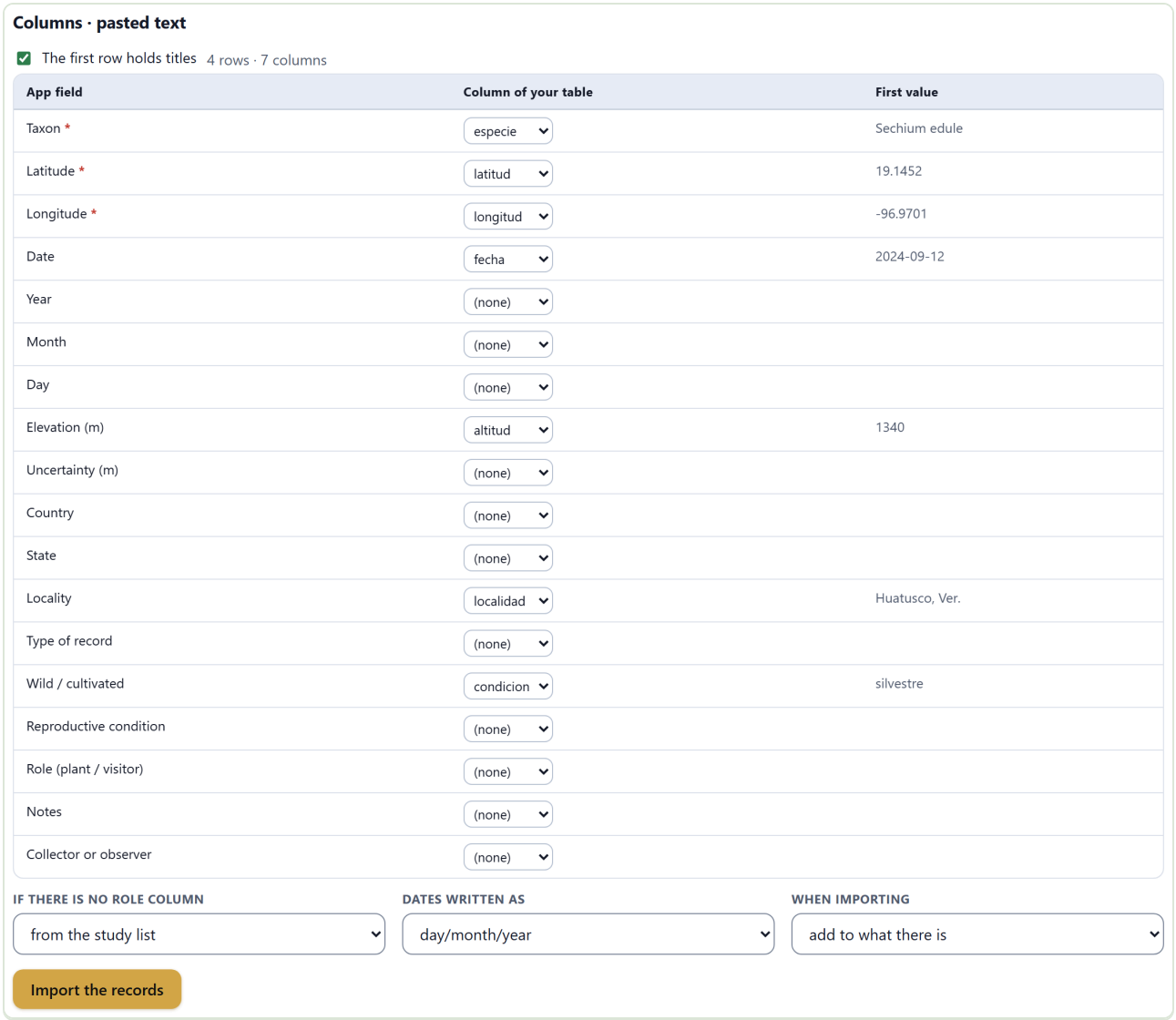


Figure 2.4. A table of four records pasted from a spreadsheet. The app recognized the columns `especie`, `latitud`, `longitud`, `fecha`, `altitud`, `localidad` and `condicion` on its own; the first value of each one lets you check this at a glance. The fields with an asterisk are required.

Below the assignment there are three options. **If there is no role column**: the role of each record is taken from the study list (the normal case), or all are marked as plants or all as visitors. **Dates written as**: day/month/year or month/day/year, for dates with slashes; those of the form `2024-09-12` are not ambiguous. **When importing**: add to what there is, or replace your previous data of that kind (never the GBIF data).

The records of a name that is not in the list do not enter the analyses until they belong to an entry. In the table of **Names not in the list** you choose, for each one, whether to add it as a plant, as a visitor or to leave it out, and Add to the list applies it.

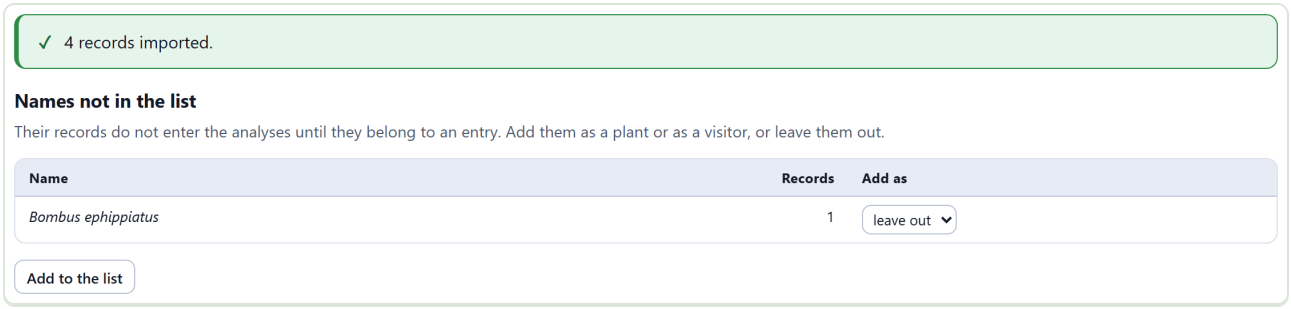


Figure 2.5. After importing, the app reports the names that no entry of the list claims. *Xenoglossa gabbii* went in on its own because the list has the entry *Xenoglossa*; *Bombus ephippiatus* did not, and waits for you to decide.

2.6 The practice data

Under the list, the collapsible No internet, or just practicing? Fictional data generated by the program loads one of the two practice studies (Section I.7) without querying the internet: Cucurbits and squash bees or Coffee and stingless bees. With them you can reproduce this manual.

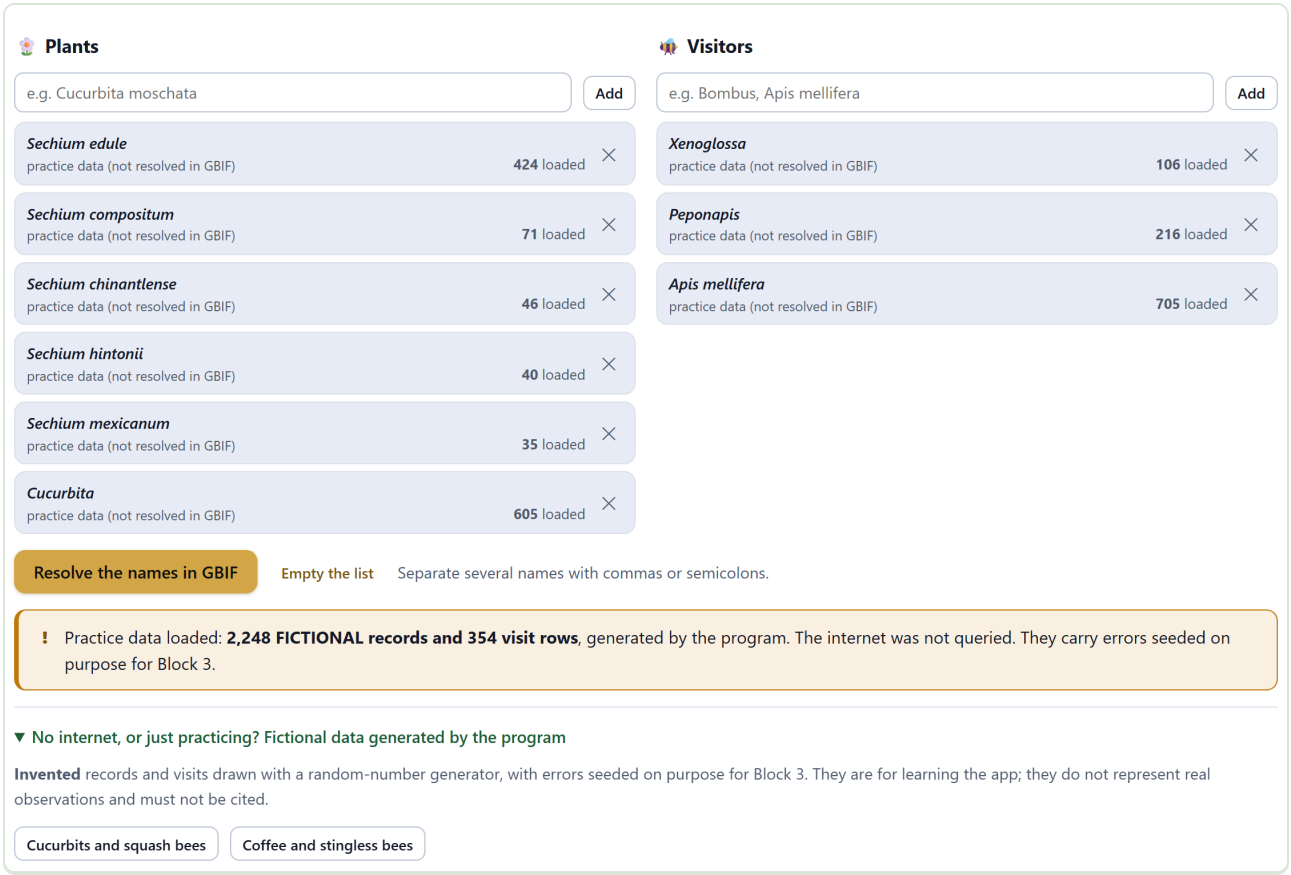


Figure 2.6. The cucurbit practice study: six plants and three visitors marked as "practice data (not resolved in GBIF)", with their records loaded. The message says it plainly: 2,248 records and 354 rows of visits, all **fictitious**, with errors planted on purpose for Block 3.

2.7 What there is, and the project

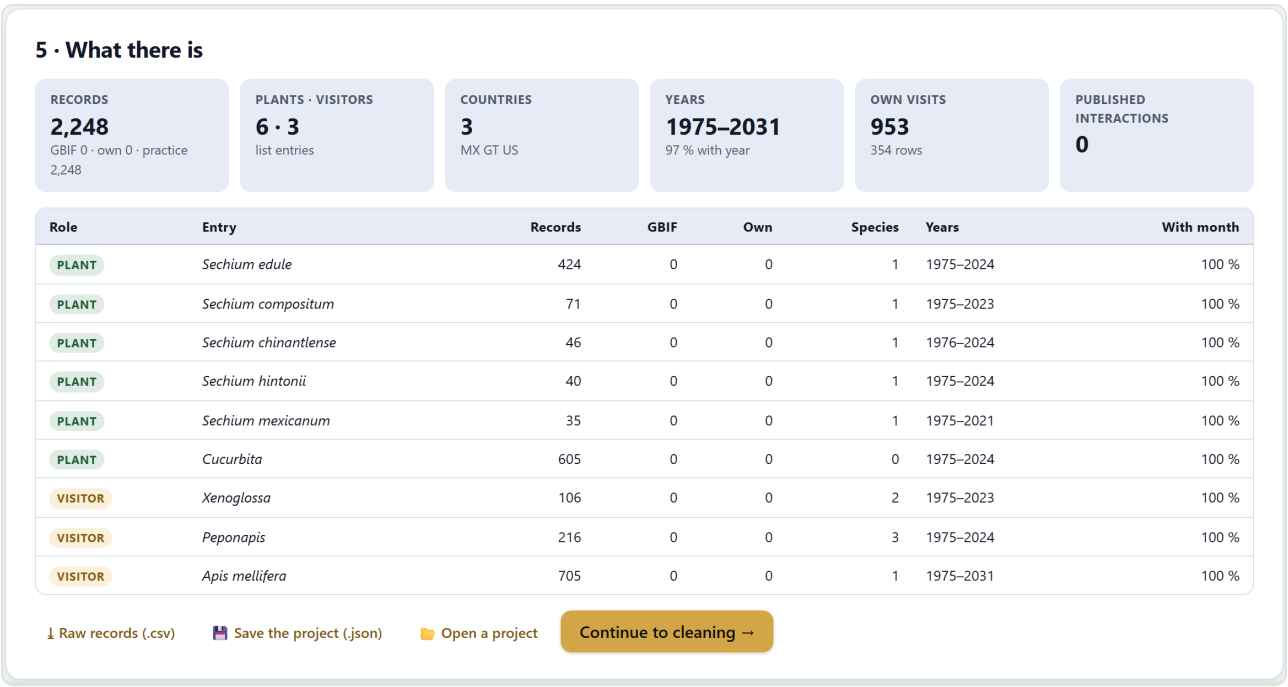


Figure 2.7. The summary of the practice study. The table gives, for each entry, the role, the records (from GBIF and your own), how many species it gathers, the range of years and the fraction with a month.

Tile	What it says	In the practice study
Records	The total and its origin: GBIF, own, practice.	2,248 · practice 2,248
Plants · visitors	The entries of the list with each role.	6 · 3
Countries	In how many countries the records fall, with their codes.	3 · MX GT US
Years	The range and the fraction of records with a year. A future year is the first sign of an error.	1975–2031 · 97 % with year
Own visits	The imported visits and how many rows describe them.	953 in 354 rows
Published interactions	The records brought from the interaction database.	0

The buttons at the foot: ⤓ Raw records (.csv) downloads all the records as they arrived; 💾 Save the project (.json) saves the list, the records, the visits, the interactions and the download filters in a single file; 📂 Open a project recovers it on any computer; Continue to cleaning → takes you to Block 3.

> **I** SAVE THE PROJECT WHEN THE DOWNLOAD IS FINISHED
>
> The browser already remembers the study (Section I.5), but the `.json` project is the only thing that survives a change of computer or the clearing of the browser's data. And it is what lets you repeat the analysis with exactly the same records months later, when GBIF will already hold others.

B2

2.8 The warnings in this block

Message	What it means and what to do
First resolve the names in GBIF (section 1)	Counting and downloading need the GBIF key of each name.
Could not query GBIF… Check your connection, or practice with the fictional data below	No internet, or the service is down. The practice data work offline.
N names exceed the chosen maximum	Only the first records of those names will be downloaded. Raise the maximum or narrow the years (rule in Section 2.3).
Download finished… Some names failed or were stopped	Repeat the download: only the GBIF records of those names are replaced. A name that was cut keeps what it managed to download and is marked as such ("cut: N kept").
N published interaction records; N queries failed	Some of the queries to the interaction database did not respond. Repeat the search.
Still to assign: …	A required field (taxon, latitude and longitude; or plant and visitor) has no column. Choose it in the assignment.
N records imported. N rows without a taxon or readable coordinates were skipped (the first: row N)	Those rows were not imported. Check the original table.
That file is not a PollinationPro project	The `.json` does not come from this app's save button.
The previous session was recovered…	Not a warning: the browser remembered the study from last time.

3

BLOCK 3 · CLEANING AND TAXONOMY

Cleaning and taxonomy

Records arriving from hundreds of collections carry errors that, left uncleaned, turn into results. Here you decide what is removed, what is only flagged and how the units of analysis are named: fifteen quality rules, cultivated plants decided plant by plant, the study area, synonyms and the unit. The report comes out written for the methods section.

3.1 Why clean

3.2 The fifteen rules

3.3 Cultivated plants and study area

3.4 Synonyms and units of analysis

3.5 The report

3.6 Reviewing and downloading records

3.7 The warnings in this block

3.1 Why clean

A single record at the geographic center of Mexico —the place written down when only the country is known— can create a co-occurrence between two species that have never been seen together. A backyard chayote in Mexico City stretches the climatic niche of the species toward places where it never lived in the wild. And a bee recorded as *Peponapis* in one collection and as *Xenoglossa* in another splits into two units that do not recognize each other. Cleaning is not a formality: it is the first scientific decision of the study.

B3

The app applies the checks on which the record-cleaning literature agrees (Chapman 2005; Zizka et al. 2020) and keeps every decision in view, with its count. Nothing is applied until you press `Clean and apply`, and everything can be redone: the raw records of Block 2 are not touched.

3.2 The fifteen rules

For each rule you choose whether it **removes** the record, only **flags** it (it stays in the analyses, with the flag in the table) or is **not applied**. The column on the right says how many records it reaches with the current parameters, and how many it would remove.



Figure 3.1. The fifteen rules with their default actions and what they reach in the practice study, and the four parameters below: distance to a centroid, maximum uncertainty, minimum decimals and year of the old record.

Rule	What it detects	Default	Practice
No valid coordinates	Latitude or longitude empty or beyond ±90 / ±180.	remove	0
(0, 0) coordinates	The default value of many databases: a point in the Gulf of Guinea.	remove	1
Latitude equal to longitude	Nearly always a data-entry error.	remove	0
Swapped or unsigned coordinates	The point does not fall in the country the record states, but does after changing the sign or swapping latitude and longitude. This rule does not remove: it **corrects**.	correct	2
Country or state centroid	Within the chosen distance of the center of a country or of a Mexican state, with two definitions of "center" (Section 3.2.1).	remove	9
Outside the stated country	The point falls in another country and no correction explains it, or it falls in the sea.	flag only	1
In the sea or outside the outlines	It falls in no country. The world outlines are coarse: check before removing coastal records.	flag only	1
Imprecise coordinate	Declared uncertainty larger than the threshold (10 km).	remove	1
Rounded coordinate	Latitude and longitude with fewer decimals than the minimum (2, about 1.1 km). A whole degree is ~110 km.	flag only	2
Duplicates	Same taxon, same coordinates and same date: one is kept.	remove	18
Impossible date	Year after the current one, or month or day out of range.	remove	1
Old record	Before the chosen year (1950): useful for history, doubtful for the current climate.	flag only	0
Fossils and living collections	A botanical-garden specimen or a fossil does not say where the wild population lives.	remove	0
Cultivated plants	Plants only: "managed" establishment means, or words such as cultivated, garden, backyard, nursery, greenhouse, plantation, milpa, coffee farm (Section 3.3).	remove	340
Outside the study area	The point does not fall in the countries —or Mexican states— chosen as the area (Section 3.3).	remove	32

The seven errors planted in the practice study all come out: the (0, 0), the swapped coordinate and the unsigned longitude (both corrected), the center of Mexico, the rounded coordinate, the year 2031 and the record with 25 km of uncertainty, in addition to the 18 duplicates. The centroid rule reaches nine records and not only the planted one: with points drawn at random around invented centers, eight fell within 5 km of the center of some state. The same happens with real data, and that is why the distance is a parameter.

3.2.1 Two definitions of "center"

The place written down when only the country or the state is known is not always the same. Some databases use the **area centroid** of the polygon; others, the **center of the bounding box**. The "center of Mexico" that circulates as a placeholder, at 23.6345° N and 102.5528° W, is the second one, and it lies tens of kilometers from the first. The app computes both for every country in the world and for each of the 32 Mexican states from its own outlines, and the rule reaches a record if it falls near either of them.

DECISION RULE · REMOVE OR FLAG ONLY

If the error means the point does not represent where the organism was (0, 0; centroid; fossil; living collection; impossible date)	**then** remove it: leaving it shifts the maps, the cells and the niche.
If the record is doubtful but may be correct (rounded, old, coastal)	**then** flag it and review it in the table of Section 3.6 before deciding. If your analysis cell is 0.25° or larger, a coordinate with two decimals changes nothing.
If one rule reaches many records of a single entry	**then** look at the table "By list entry": it may be a collection with a systematic problem, and it is worth knowing before removing it altogether.

3.3 Cultivated plants and study area

Most records of a crop are of planted individuals, whose distribution reflects agriculture and not the natural niche. What to do with them depends on the question: if you study the crop as it is grown (coffee in its farms), keep its cultivated records; if you study the wild populations or the wild relatives, remove them. That is why each plant is decided **separately**, and the app proposes keeping them when more than half of that plant's records are cultivated.

A record is considered cultivated if its establishment means is "managed", "cultivated", "introduced" or "planted", or if its locality or its notes say cultivated, sown, planted, orchard, vegetable garden, garden, backyard, nursery, greenhouse, market, plot, fruit orchard, milpa, coffee farm, plantation, ornamental or escaped from cultivation, in Spanish or English. The column **Living collection** counts separately the specimens from botanical gardens and field genebanks, which are handled by the rule "Fossils and living collections".

CULTIVATED RECORDS WITHOUT A LABEL ARE NOT DETECTED

The rule only sees what the record says. A garden chayote recorded without notes passes as wild. With widely grown crops, check on the map of Block 4 the points in cities and in farming areas, and consider narrowing the study area to the known natural distribution.

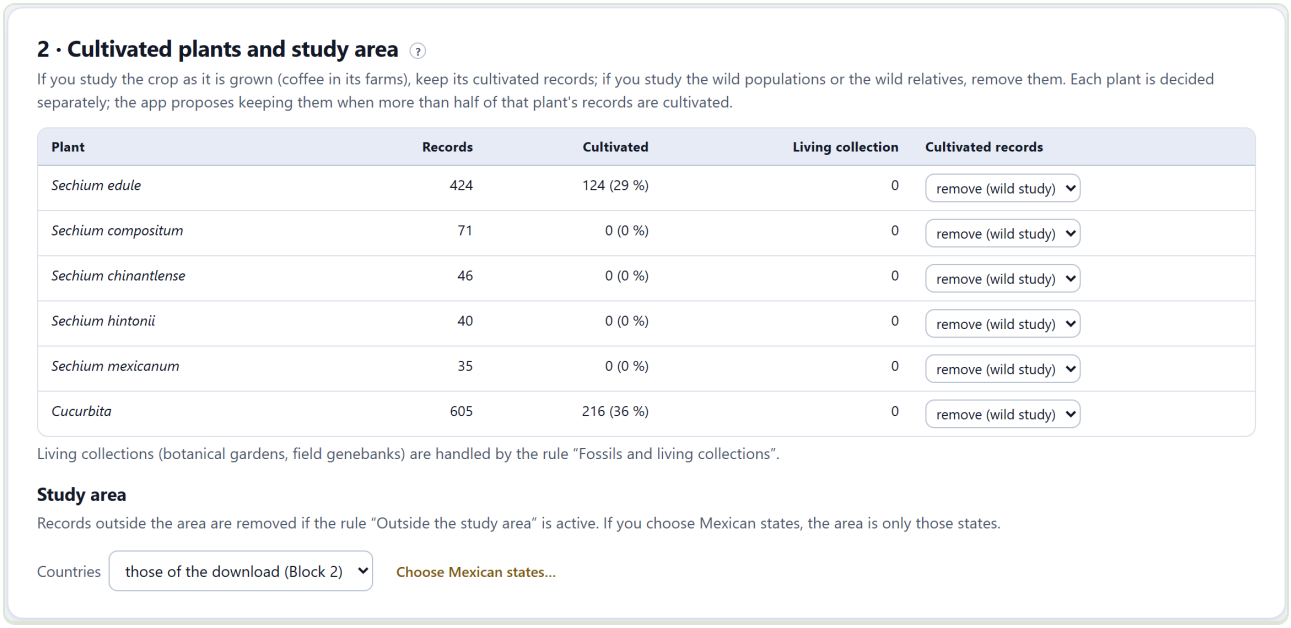


Figure 3.2. The plants of the practice study. *Sechium edule* has 124 cultivated records (29%) and *Cucurbita* 216 (36%); the four wild relatives, none. With fewer than half cultivated, the app proposes removing them: the study is about the wild populations.

The study area. Records outside the area are removed if the rule "Outside the study area" is active. The Countries selector offers *those of the download (Block 2)* —the usual choice— or *all (no area)*. Choose Mexican states... opens the list of the 32 states: if you choose some, the area is only those. In the practice study the area is Mexico, and the rule removes 27 records, almost all from the Guatemalan side of the southern border.

3.4 Synonyms and units of analysis

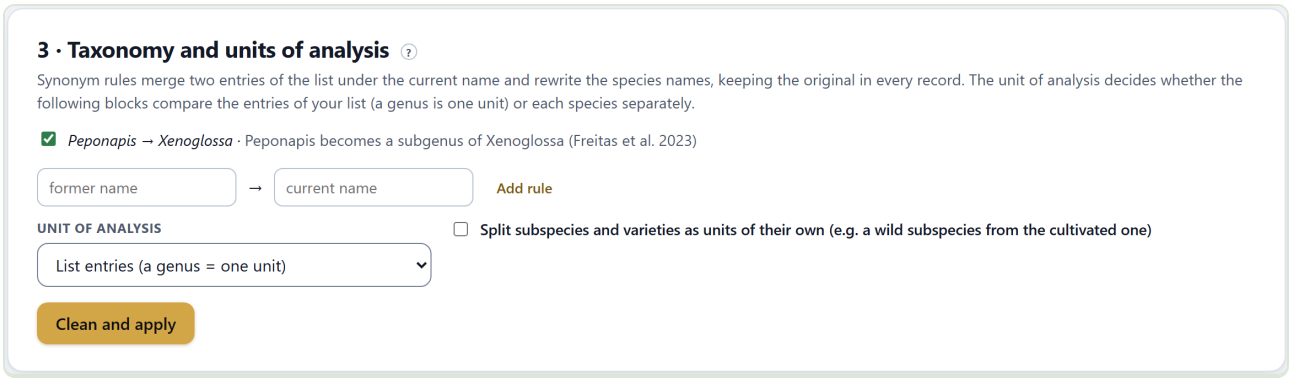


Figure 3.3. The taxonomy of the practice study. The app proposed the rule *Peponapis* → *Xenoglossa* on its own because both are on the list, with the work that supports it.

Synonym rules bring two entries of the list together under the current name and rewrite the species names —*Peponapis crassidentata* becomes *Xenoglossa crassidentata*—, keeping the original name in every record. The app comes with the rules it knows and proposes them when both entries are on the list; the one for the squash bees follows Freitas et al. (2023), who place *Peponapis* as a subgenus of *Xenoglossa*. Each rule is turned on or off with its checkbox, and you can add others by typing the **former name** and the **current name** and pressing Add rule.

Unit of analysis	What the following blocks compare
List entries (a genus = one unit)	Each entry is a unit: *Cucurbita* gathers all its species and *Xenoglossa*, after the synonym, gathers five. It is the right choice when the question is about a functional group (the squash bees) or when the species have few records.
Species (each species apart)	Each species is a unit, even if the list has a genus. It is the right choice when the question is about particular species and each one has enough records.
Split subspecies and varieties as units of their own (e.g. a wild subspecies from the cultivated one)	With species as the unit, it also splits subspecies and varieties (for example, a wild subspecies from the cultivated one).

When you are done, Clean and apply runs the rules, applies the synonyms, assigns the units and opens the report.

3.5 The report



Figure 3.4. The report of the practice study: of 2,248 raw records, 1,856 are kept (82.6%), 392 are removed, 2 coordinates are corrected and 8 units of analysis remain.

The box **For the methods** is a paragraph ready to copy. It says how many records were kept, which rules removed or corrected how many, at what distance and with which two definitions the centroids were evaluated, what maximum uncertainty was used and which synonym rules were applied, with their reference. The report of Block 10 takes it up again with the citations of the cleaning methods.

WORKED EXAMPLE · THE PARAGRAPH OF THE PRACTICE STUDY

"Of 2,248 records, 1,856 were kept (82.6 %). Removed or corrected: (0, 0) coordinates (1); swapped or unsigned coordinates (2); country or state centroid (9); imprecise coordinate (1); duplicates (18); impossible date (1); cultivated plants (335); outside the study area (27). Centroids were evaluated at 5 km with two definitions (area centroid and center of the bounding box), the maximum uncertainty was 10 km and the synonym rules Peponapis → Xenoglossa (Freitas et al. 2023) were applied."

By rule

Rule	Action	Reached	Removed
No valid coordinates	remove	0	0
(0, 0) coordinates	remove	1	1
Latitude equal to longitude	remove	0	0
Swapped or unsigned coordinates	correct	2	0
Country or state centroid	remove	9	9
Outside the stated country	flag only	1	0
In the sea or outside the outlines	flag only	1	0
Imprecise coordinate	remove	1	1
Rounded coordinate	flag only	2	0
Duplicates	remove	18	18
Impossible date	remove	1	1
Old record	flag only	0	0
Fossils and living collections	remove	0	0
Cultivated plants	remove	340	335
Outside the study area	remove	32	27

By list entry

Role	Entry	Before	After	Main reason for removal
PLANT	*Sechium edule*	424	296	Cultivated plants (121)
PLANT	*Sechium compositum*	71	62	Outside the study area (7)
PLANT	*Sechium chinantlense*	46	45	Duplicates (1)
PLANT	*Sechium hintonii*	40	40	—
PLANT	*Sechium mexicanum*	35	35	—
PLANT	*Cucurbita*	605	377	Cultivated plants (214)
VISITOR	*Xenoglossa*	106	104	Country or state centroid (1)
VISITOR	*Peponapis → Xenoglossa*	216	212	Outside the study area (3)
VISITOR	*Apis mellifera*	705	685	Outside the study area (10)

Units of analysis

Role	Unit	Records	Species within	Original names merged
PLANT	*Cucurbita*	377	—	—
PLANT	*Sechium edule*	296	1	—
PLANT	*Sechium compositum*	62	1	—
PLANT	*Sechium chinantlense*	45	1	—
PLANT	*Sechium hintonii*	40	1	—
PLANT	*Sechium mexicanum*	35	1	—
VISITOR	*Apis mellifera*	685	1	—
VISITOR	*Xenoglossa*	316	5	*Peponapis crassidentata, Peponapis azteca, Peponapis pruinosa*

Figure 3.5. The three tables of the report. The units table shows that *Xenoglossa* ended up with 316 records of five species and gathers the original names *Peponapis crassidentata*, *P. azteca* and *P. pruinosa*.

Table	What it is for
By rule	The action of each rule, how many records it reached and how many it removed. A rule can reach more than it removes because another rule had already removed them: *Cultivated plants* reached 340 and removed 335.
By list entry	The records before and after and the main reason for removal. In the practice study, chayote goes from 424 to 296 (mostly cultivated, 121) and *Cucurbita* from 605 to 377 (cultivated, 214). The *Peponapis* row says "*Peponapis* → *Xenoglossa*".
Units of analysis	The units that Blocks 4 to 8 will see, with their cleaned records, how many species they gather and which original names were merged into each one.

B3

DECISION RULE · HOW MUCH CAN BE LOST

If cleaning keeps more than 80% of the records	**then** it is the usual result with well-filtered GBIF data; report it and move on.
If it keeps between 50 and 80%	**then** look at which rule removes the most: if it is cultivated plants, it is a decision of the question, not a quality problem.
If it keeps less than half, or a unit is left with fewer than 20 records	**then** review the parameters (distance to the centroid, uncertainty, area) and consider bringing species together in their genus.

These are guides: a rare species with 15 good records is worth more than a common one with 1,500 doubtful ones.

3.6 Reviewing and downloading records

The Show selector filters the records table: all the removed ones, the corrected coordinates or those of a given rule, with their count. Each row gives the unit, the latitude and longitude (already corrected), the country the record states and the country where it falls, the Mexican state, the date, the record type, the flags of the rules —with ✕ if it was removed— and the locality or the notes.

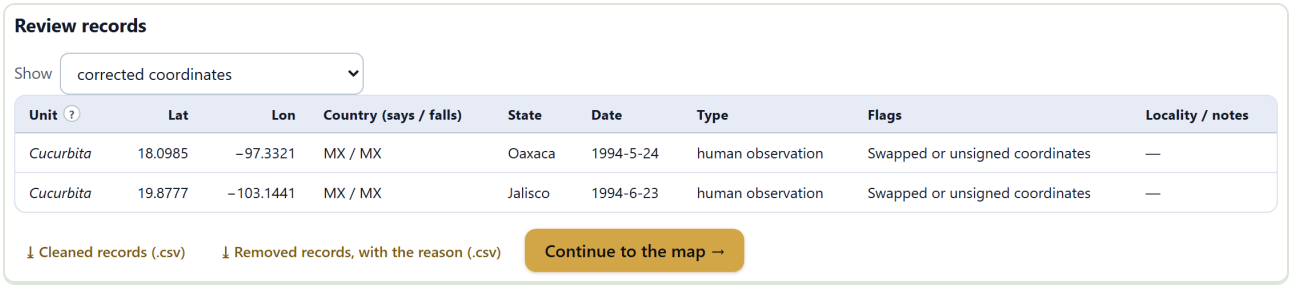


Figure 3.6. The two corrected coordinates of the practice study, already in place: one in Oaxaca and the other in Jalisco. Below, the two downloads and the step to the map.

↓ Cleaned records (.csv) (`clean_records.csv`) downloads those that enter the analysis, with their unit, the original and the harmonized name, and the Mexican state; ↓ Removed records, with the reason (.csv) (`removed_records.csv`) downloads the rest with the rule that removed them. Both are the supplementary material a reviewer expects to see.

3.7 The warnings in this block

Message	What it means and what to do
Records are needed first	Block 2 is empty. Download from GBIF, upload your records or load the practice ones.
A rule reaches many coastal records ("In the sea or outside the outlines")	The world outlines are coarse along the coasts. Leave it on *flag only* and check those points on the map of Block 4 before removing them.
A unit is left with very few records	The following blocks will analyze it anyway, but their tests will have little power and some will leave it "unclassified". Bring it together in its genus or widen the area.
I changed the data of Block 2 after cleaning	There is no need to come back here: the cleaning is redone by itself, with the same rules, as soon as a block needs it. If what you change is a rule, a synonym or the unit, press Clean and apply.

4

BLOCK 4 · MAP

The map and its studio

The cleaned records over the territory, with no internet connection. The map is the first quality check —a point in the sea or in the wrong country catches the eye— and the first figure of the paper. Everything is edited from a studio with eight tabs: background and boundaries, framing and clip, texts and fonts, north arrow, scale bar and legend, points, heat or grids, colors, ready-made looks and export at journal size, up to GeoTIFF.

4.1 The map as a check and as a figure

4.2 The map studio

4.3 Base map, framing and clip

4.4 Texts, north arrow, scale and legend

4.5 Data and colors

4.6 Ready-made looks and export

4.7 Records by territory

4.1 The map as a check and as a figure

Before measuring co-occurrences you have to see where the points are. A record in the sea, a cloud of points at the center of a state, a mountain species that shows up on the coast, a desert bee in the rainforest: all of that is seen on the map in seconds and never in a table. If something like that appears, go back to Block 3 and adjust a rule.

The same map is the first figure of the paper. That is why Block 4 is not a viewer but a complete editing studio, with a **vector base map of the program's own**: the outlines of the world's countries and of the 32 Mexican states (Natural Earth, public domain) travel with the app. No tiles are downloaded from any service, so it works without internet, it is exported without asking anyone for permission and it looks the same on any computer.

4.2 The map studio

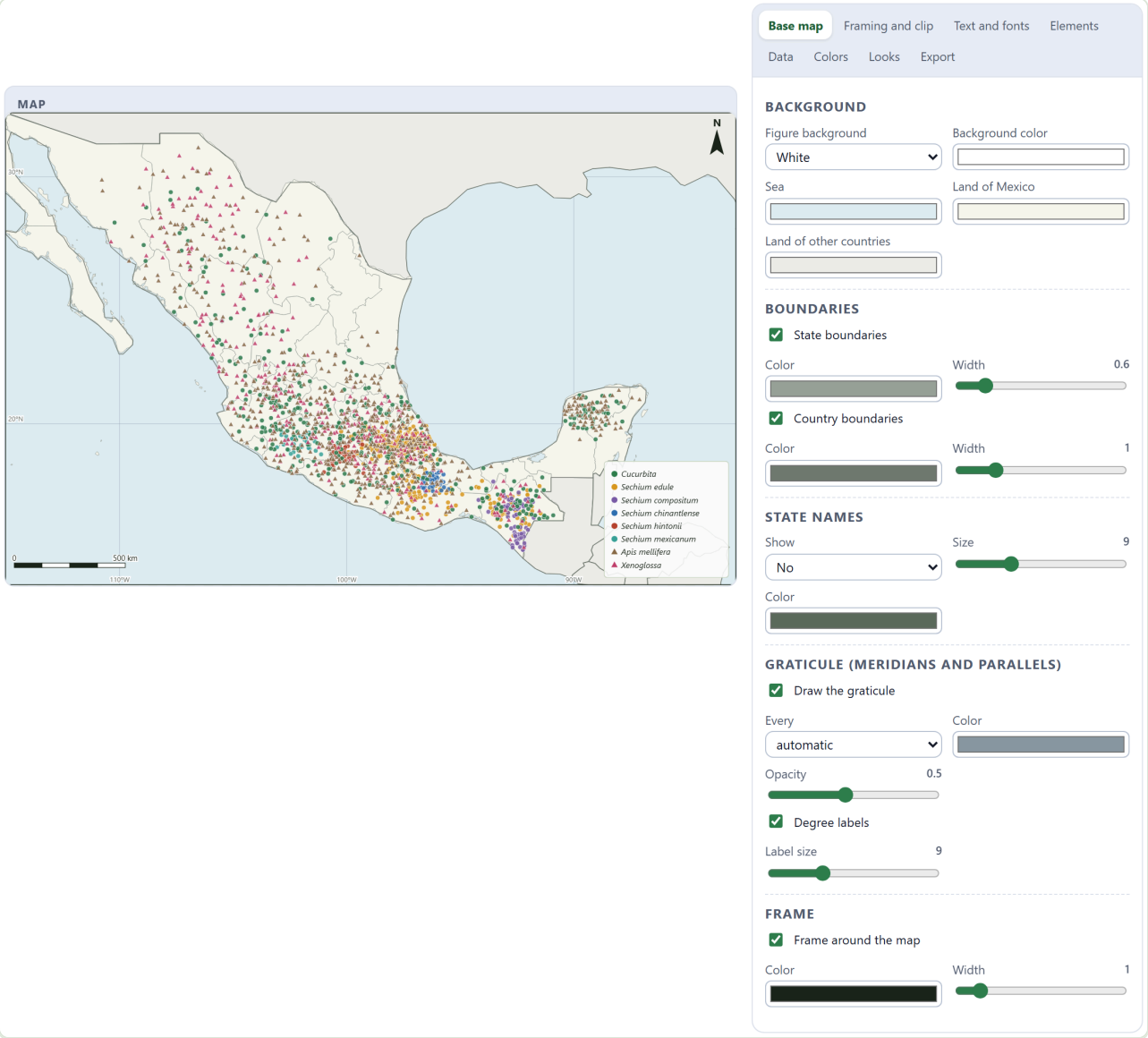


Figure 4.1. The studio with its starting values: extent fitted to the records, plants as circles and visitors as triangles, one color per unit, north arrow, scale bar in alternating blocks, graticule with degrees and light legend. On the right, the Base map tab.

The map on the left is redrawn with every change on the right. The eight tabs group the controls; everything you choose is remembered in that browser, so the map is the same the next time, and a complete style can be saved to a file (Section 4.6).

Tab	What is edited
Base map	Figure background, sea, land of Mexico and of other countries; state and country boundaries; state names; graticule; frame.
Framing and clip	The extent of the map, the margin, the legend outside the map and the clip to Mexico or to chosen states.
Text and fonts	Title, subtitle, source or credit, legend title, the typeface family and the sizes and styles.
Elements	North arrow (eight designs), scale bar (eight formats) and legend.
Data	What is drawn (points, heat, grids, hulls), the symbols and each unit.
Colors	The palette of the units, the density ramp and the color of each unit.
Looks	Six ready-made looks, saving and loading a style, resetting everything.
Export	Format, paper size and resolution, and the download.

4.3 Base map, framing and clip

Base map. The **figure background** can be white, that of the app theme, a custom color or transparent. You choose the colors of the sea, of the land of Mexico and of the land of other countries (telling them apart helps to read a map of Mexico); whether the state and country boundaries are drawn, with their color and width; whether the **state names** are written —no, abbreviation or full name—, with their size and color; the **graticule** of meridians and parallels, how many degrees apart (automatic, 1°, 2°, 5°, 10° or 20°), its color, its opacity and its degree labels; and the **frame** around the map.

Framing. The **extent** can be fitted to the records (with a margin in percent), cover the whole of Mexico, the chosen states, the world or your own coordinates (north, south, west and east). Legend outside the map, on the right moves the legend out of the map area so that it does not cover points.

Clip. Clip the map to Mexico or to the chosen states draws only what falls inside that silhouette. Outside the clip there is the background color or nothing (transparent, useful for placing the map over another figure). Draw the outline of the clip traces the outer silhouette —not the internal borders— with the color and width you choose. When you choose *chosen states* in the extent or in the clip, the list of the 32 states appears so you can tick them.

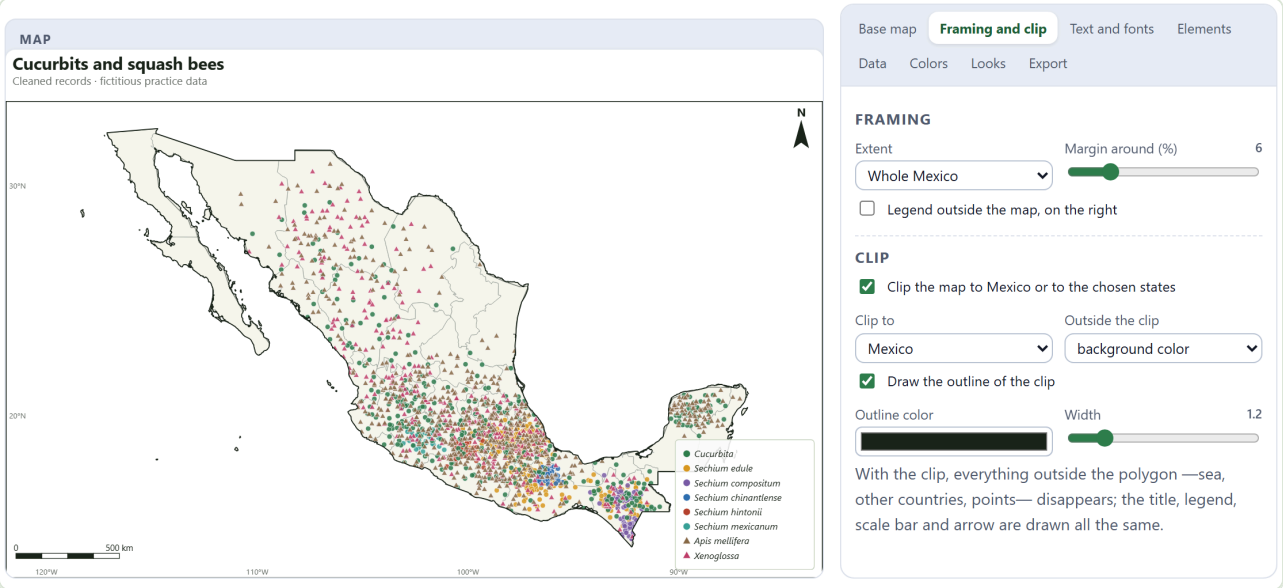


Figure 4.2. The whole of Mexico, clipped to its silhouette, with title and subtitle. Outside the clip the sea, the other countries and the points that fell outside disappear; the north arrow, the scale bar, the graticule and the legend are drawn all the same.

4.4 Texts, north arrow, scale and legend

Text and fonts. Four texts —title, subtitle, source or credit, legend title— and six generic typeface families (system sans, humanist sans, serif, classic serif, condensed sans and monospace), which exist on any computer and therefore look the same on the reviewer's. You choose the text color, the size of the title (with bold, italics and alignment), of the subtitle and of the credit, and the corner of the credit. Unit names are written in italics in the legend, as scientific names require.

Element	Options
North arrow	None, arrow, split arrow, needle, triangle, circle, 4-point rose, 8-point rose and minimal; position, size and color.
Scale bar	Alternating blocks, stepped (two rows), solid bar, line with ends, ruler, kilometers and miles, text only and ratio (1:n); unit (kilometers or miles), position, divisions, thickness and text size. The ratio is computed for the paper size chosen in Export and changes if the paper changes.
Legend	Whether to draw it, position, box (light, dark, outline only or none), text size and one to three columns.



Figure 4.3. The Elements tab with an eight-point rose, the ruler-style scale bar and the legend in two columns. Each element can be placed in any of the four corners.

4.5 Data and colors

The Data tab decides what is drawn. Each **representation** answers a different question:

Representation	What it is for
Points	The standard distribution figure: one symbol per record, with color and shape by unit or by role.
Heat map (density)	Where the records are concentrated, with adjustable radius and intensity. It shows sampling effort as much as abundance.
Grid: records per cell	How many records there are in each cell (0.1° to 2°), with the density ramp.
Grid: where plant and visitor meet	The cells with plant and visitor, plants only or visitors only: the visual version of the test of Block 5.
Convex hulls with points	The minimum area enclosing the points of each unit, and their overlap.

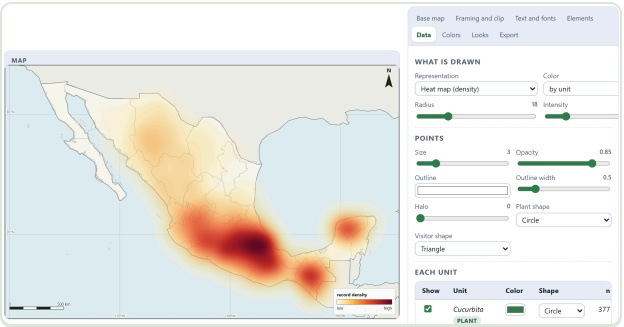


Figure 4.4. Heat map with the "Heat" ramp.

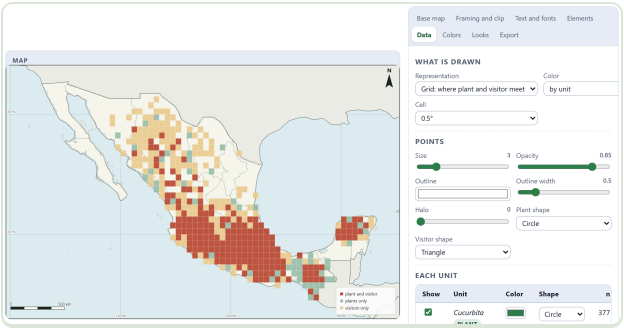


Figure 4.5. Where plant and visitor meet, in 0.5° cells.

In Figure 4.5 the shared cells (in rust red) are concentrated in central and southern Mexico, where more sampling was done; the visitor-only cells (light amber) dominate the north, where *Xenoglossa* and the honey bee are recorded without wild cucurbits. It is a visual reading; Block 5 says whether that co-occurrence is more or less than chance predicts.

The symbols. Size, opacity, outline color and width, halo (a light border that separates the points from the background) and the shape of the plants and of the visitors: circle, square, triangle, diamond, cross or star. With *by role* color, all the plants go in one color and all the visitors in another.

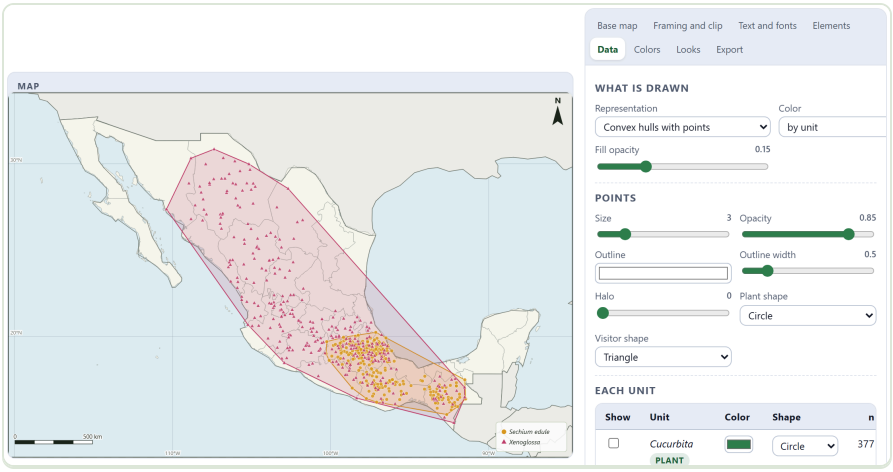


Figure 4.6. Convex hulls with only two units visible: chayote and *Xenoglossa*. The **Each unit** table lets you hide units, change their color and shape, and says how many records each one has.

Each unit. The table of the Data tab —repeated in Colors— has one row per unit with the **Show** checkbox, its color, its shape and its number of records. The buttons Show all, Plants only and Visitors only change the visibility at a stroke. The territory tables of Section 4.7 and the points download follow what is shown.

Colors. Six palettes for the units —PollinationPro, Okabe–Ito (color-blind safe), Earth, Vivid, Soft and Grays— and six ramps for density —Greens, Amber, Violets, Heat, Cool and Grays—. A color changed by hand in a unit is kept when you change palette; Back to the palette colors undoes it.

4.6 Ready-made looks and export

A **look** applies the background, the colors, the typography and the elements at once; the texts (title, credit) and each unit's own colors are kept.

Look	What it is like	What for
Publication	White, thin frame, graticule and light legend.	The figure of a paper.
Black and white	Grays and different shapes per unit.	Journals that charge for color.
Dark presentation	Dark background, large points with halo.	Slides.
Field guide	Earth tones, serif, compass rose, state abbreviations.	Books, outreach posters.

Look	What it is like	What for
Poster	Large texts and points, boxed title.	Conference posters.
Data only	No sea or graticule, faint boundaries.	Composite figures, insets.

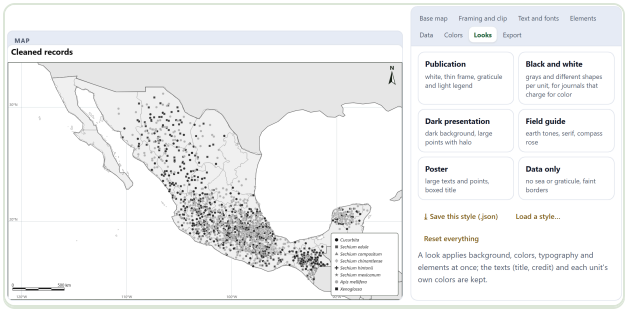


Figure 4.7. The *Black and white* look.

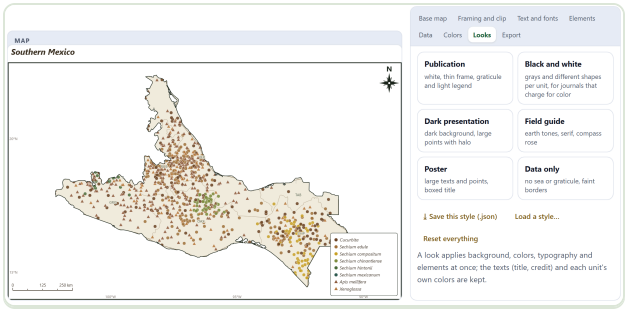


Figure 4.8. *Field guide*, clipped to six southern states.

B4

↓ Save this style (.json) saves all the map options —except the data— to a file (`map_style.json`); Load a style... applies it to another study or on another computer. That way all the map figures of a thesis come out alike. Reset everything goes back to the starting values.

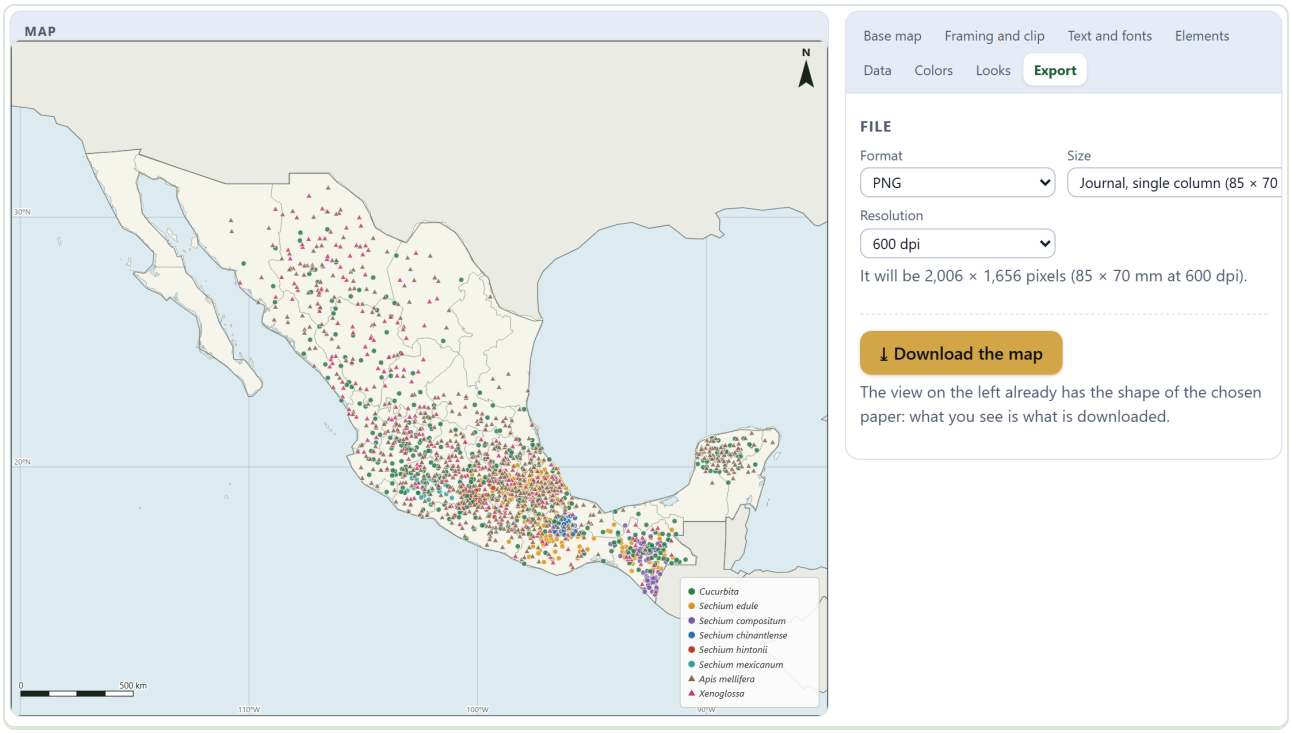


Figure 4.9. The export to a journal column (85 × 70 mm) at 600 dpi: "It will be 2,006 × 1,656 pixels". The view on the left took the shape of the paper: what you see is what is downloaded.

Option	Values
Format	PNG, JPEG, WebP, SVG (vector, editable), TIFF and GeoTIFF (georeferenced).
Size	Screen (960 × 620 px), A4 landscape and portrait, letter landscape, journal single column (85 × 70 mm), 1.5 columns (120 × 90 mm) and double column (175 × 120 mm), 16:9 slide and custom (in millimeters).
Resolution	150, 300, 600 or 900 dpi. Below, you can read the size in pixels that will come out.

THE GEOTIFF OPENS IN PLACE

The GeoTIFF carries the georeferencing keys of a longitude–latitude grid on WGS 84 (EPSG 4326): in a geographic information system it falls exactly on your layers. For an exact fit over the whole image, remove the title and keep the legend inside the map; the app reminds you of this when downloading.

DECISION RULE · WHICH FORMAT TO HAND IN

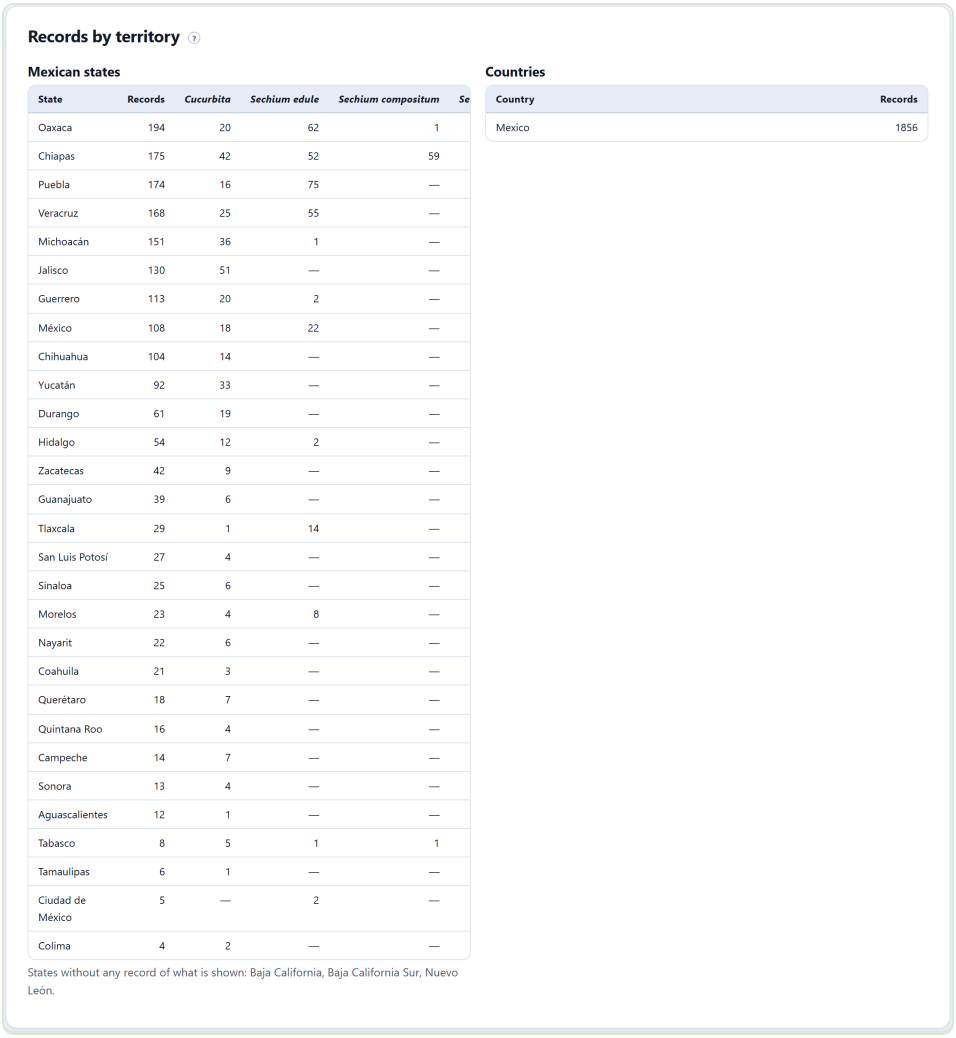
If the journal asks for the final figure	**then** TIFF or PNG at 600 dpi at the column width it asks for (single, one and a half or double).
If you are going to touch up the figure in another program	**then** SVG: every text and every point remains editable.
If the figure goes into a geographic information system or onto another map	**then** GeoTIFF, without a title.

4.7 Records by territory

The tables follow what the map shows: if units are hidden, they are recomputed. The states table gives the total and one column per unit (up to ten); the sentence below names the states without any record of what is shown. It is the first answer to "where is sampling missing?": a state without records of a plant that should be there is a sampling gap or a real distribution limit, and it is worth knowing which before interpreting the following blocks. `↓ Points shown (.csv)`, under the studio, downloads the visible points with their unit and coordinates (`map_points.csv`).

WHAT THE MAP DOES NOT SAY

A map of presence records is not a distribution map: it shows where someone recorded the species, not where it lives. Gaps may be absences or lack of sampling, and concentrations may be abundance or effort. The caption of the paper's figure should say so ("cleaned records of..., GBIF and own observations"), and Block 8 is the one that estimates the potential distribution.



Records by territory

Mexican states

State	Records	*Cucurbita*	*Sechium edule*	*Sechium compositum*	*Se*
Oaxaca	194	20	62	1	
Chiapas	175	42	52	59	
Puebla	174	16	75	—	
Veracruz	168	25	55	—	
Michoacán	151	36	1	—	
Jalisco	130	51	—	—	
Guerrero	113	20	2	—	
México	108	18	22	—	
Chihuahua	104	14	—	—	
Yucatán	92	33	—	—	
Durango	61	19	—	—	
Hidalgo	54	12	2	—	
Zacatecas	42	9	—	—	
Guanajuato	39	6	—	—	
Tlaxcala	29	1	14	—	
San Luis Potosí	27	4	—	—	
Sinaloa	25	6	—	—	
Morelos	23	4	8	—	
Nayarit	22	6	—	—	
Coahuila	21	3	—	—	
Querétaro	18	7	—	—	
Quintana Roo	16	4	—	—	
Campeche	14	7	—	—	
Sonora	13	4	—	—	
Aguascalientes	12	1	—	—	
Tabasco	8	5	1	1	
Tamaulipas	6	1	—	—	
Ciudad de México	5	—	2	—	
Colima	4	2	—	—	

States without any record of what is shown: Baja California, Baja California Sur, Nuevo León.

Countries

Country	Records
Mexico	1856

Figure 4.10. The records of what is shown on the map, by Mexican state and by country. Oaxaca (194), Chiapas (175), Puebla (174) and Veracruz (168) hold the most records; Baja California, Baja California Sur and Nuevo León have none.

5

BLOCK 5 · CO-OCCURRENCE

Spatial co-occurrence

Do the plant and the visitor coincide in the territory more than chance predicts, once where sampling happened is discounted? Three independent readings: the cells they share, with Veech's exact test and its sensitivity to scale; the distance from each plant to the nearest visitor, against a null model that keeps the geography of sampling; and the overlap of their ranges.

5.1 More than chance predicts?

5.2 Every pair

5.3 The chosen pair and the scale

5.4 The distance to the nearest visitor

5.5 The overlap of ranges

5.6 How to read the three tests together

5.7 The warnings in this block

5.1 More than chance predicts?

That the plant and the visitor have records in the same places says nothing unless you know how many co-occurrences would be expected by chance. The test of this block compares the observed with the expected, and the answer depends on two decisions: **among which sites** chance is spread (the universe) and **at what scale** you look (the cell size).

THEORY IN BRIEF · VEECH'S EXACT TEST

The territory is cut into cells. If the universe has N cells, the plant occupies N1 and the visitor N2, and they were placed at random with respect to each other, the number of shared cells j would follow the **hypergeometric** distribution. The app computes the probability of observing as many or more (positive association) and as many or fewer (negative association), with no simulations or normality assumptions (Veech 2013). The **standardized effect size** (SES) says how many standard deviations the observed lies from the expected; |SES| > 1.96 corresponds roughly to p < 0.05 (Gotelli 2000).

$$P(j) = C(N_1, j) \cdot C(N - N_1, N_2 - j) / C(N, N_2) \quad E[j] = N_1 \cdot N_2 / N \quad SES = (j - E[j]) / \sqrt{Var[j]}$$

The default **universe** is the **target group**: the cells where any unit of the study was recorded, which are the ones that were really sampled (Phillips et al. 2009). The alternative, all the land in the extent, also spreads chance over cells nobody visited; the lab of Block 1 (Section 1.3) shows the error that produces.

B5

5.2 Every pair

The first card tests every plant against every visitor with three controls: the **cell** size (0.1°, 0.25°, 0.5°, 1° or 2°, that is, about 11, 28, 55, 110 or 220 km), the **universe** and the Benjamini–Hochberg **adjustment for multiple comparisons**.

Tile	What it says
Universe	How many cells enter the chance model, with the size and type of universe.
Pairs tested	Plants × visitors.
More than expected · Less · At random	The verdicts, with adjusted p if the adjustment is on.
Unclassified	The pairs for which chance could never be rejected: those that expect less than one shared cell, or for which not even the most extreme result would reach α = 0.05. With so few data no conclusion is possible, and the app does not pretend otherwise.

Clicking a square of the matrix chooses that pair for the following cards. The drop-down Table of every pair sorts them by SES, from aggregation to segregation:



Figure 5.1. The twelve pairs of the practice study in 0.25° cells with the target group: 918 universe cells, no pair more than expected, four less than expected and eight at random. In the matrix, each square carries the SES; green is aggregation, violet segregation, the intensity is |SES| and the dotted border marks the significant ones. The pair with the thick border is the chosen one.

Besides the test, each pair carries two descriptive indices: **Jaccard**'s (shared cells over the cells of either of the two) and the **cover** (what fraction of the plant's cells has the visitor). ↓ Pairs (.csv) downloads the complete table (`cooccurrence_pairs.csv`), with the Sørensen and Simpson indices and both tails of p.

> **THEORY IN BRIEF · WHY ADJUST**
>
> With 6 plants and 2 visitors, 12 pairs are tested; with 30 pairs, at $\alpha = 0.05$, one or two "significant" ones would be expected even if there were no association at all. The Benjamini and Hochberg (1995) adjustment controls the expected proportion of false positives among the pairs declared associated. Use the adjusted p to decide and report both; if you are only interested in one pair chosen beforehand by the hypothesis, its unadjusted p is the one that applies.

B5

▼ Table of every pair

Plant	Visitor	Plant cells	Visitor cells	Shared	Expected	SES	p (two-sided)	adjusted p (BH)	Verdict	Jaccard	Cover
Sechium edule	*Xenoglossa*	158	267	51	46.0	0.97	0.38	0.571	at random	0.136	32 %
Sechium hintonii	*Apis mellifera*	24	505	15	13.2	0.75	0.594	0.713	at random	0.029	63 %
Sechium hintonii	*Xenoglossa*	24	267	7	7.0	0.01	1	1	at random	0.025	29 %
Sechium chinantlense	*Xenoglossa*	16	267	4	4.7	−0.36	0.963	1	at random	0.014	25 %
Sechium mexicanum	*Xenoglossa*	21	267	4	6.1	−1.02	0.443	0.591	at random	0.014	19 %
Sechium mexicanum	*Apis mellifera*	21	505	9	11.6	−1.13	0.362	0.571	at random	0.017	43 %
Sechium chinantlense	*Apis mellifera*	16	505	6	8.8	−1.42	0.244	0.487	at random	0.012	38 %
Sechium compositum	*Xenoglossa*	36	267	5	10.5	−2.05	0.052	0.124	at random	0.017	14 %
Sechium edule	*Apis mellifera*	158	505	71	86.9	−2.80	0.007	0.021	less than expected	0.120	45 %
Sechium compositum	*Apis mellifera*	36	505	0	19.8	−6.77	2.707e-13	1.624e-12	less than expected	0.000	0 %
Cucurbita	*Xenoglossa*	315	267	47	91.6	−6.83	2.647e-12	1.059e-11	less than expected	0.088	15 %
Cucurbita	*Apis mellifera*	315	505	118	173.3	−7.72	1.693e-14	2.032e-13	less than expected	0.168	37 %

Figure 5.2. The pairs table. Chayote with *Xenoglossa* heads the list with 51 shared cells against 46.0 expected (SES 0.97, at random); *Cucurbita* with the honey bee closes it with 118 against 173.3 (SES −7.72, less than expected).

5.3 The chosen pair and the scale

Tile	What it says	Chayote × *Xenoglossa*
Shared cells	The observed ones, out of those of the plant and those of the visitor.	51 of 158 and 267
Expected by chance	$N_1 \cdot N_2 / N$, with the size of the universe.	46.0 · N = 918
SES	The standardized effect size, with its scale label.	0.97 · *chance*
p (more) · p (less)	The probability of as many or more shared cells (test of positive association) and of as many or fewer (negative).	0.19 · 0.857
Cover	What fraction of the plant's cells has the visitor.	32 %

The **sensitivity** figure repeats the test with the five cell sizes. In the practice study the verdict changes: less than expected at 0.1°, at random at 0.25°, more than expected at 0.5°, at random at 1° and 2°. The app writes it below and advises: *"Report it that way: the association depends on the resolution it is looked at."* When all five agree, it says the result is robust to the scale.

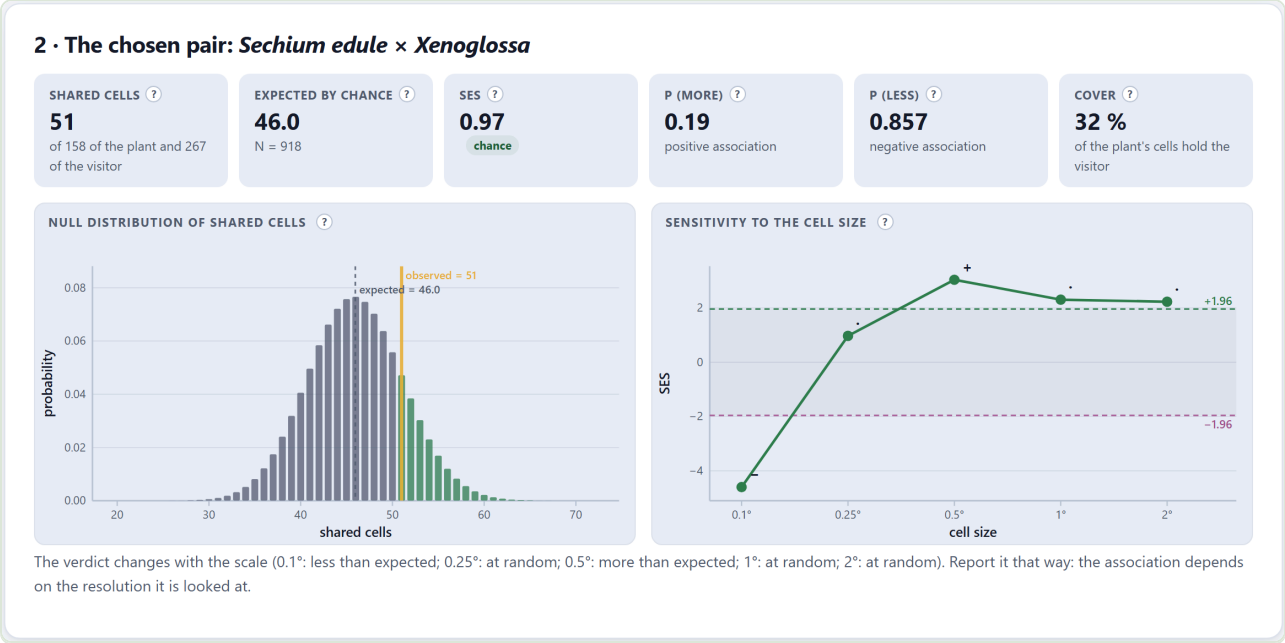


Figure 5.3. Chayote and *Xenoglossa*. On the left, the hypergeometric distribution of shared cells under chance, with the observed (51) and the expected (46.0); the colored bars are the tail running outward from the observed. On the right, the SES with five cell sizes, the ±1.96 band and the verdict of each one (+, − or ·).

B5

DECISION RULE · WHICH CELL SIZE

If the question is whether the visitor can reach the flower	**then** use small cells (0.1–0.25°): a solitary bee forages hundreds of meters or a few kilometers from its nest; a bumblebee, several kilometers.
If the question is whether they share a region (biogeography, history)	**then** 0.5–1° cells.
If the verdict changes with the cell	**then** do not pick the one that suits you: report the sensitivity figure and say at what scale there is association and at which there is not.

With very fine cells and few records almost nothing coincides and the pairs are left unclassified; with huge cells almost everything coincides. Neither extreme is informative.

5.4 The distance to the nearest visitor

The third card uses no grid. For each point of the plant it measures the distance to the nearest record of the visitor and takes the median. The **null model** replaces the visitor's records with as many sites drawn without replacement from all the records of the study —the target group—, which keeps the geography of sampling and breaks only the link between the two taxa; it is repeated 99 to 999 times (Davison and Hinkley 1997). Beforehand, the points are thinned to one per kilometer so that a heavily recorded site does not weigh a hundredfold.

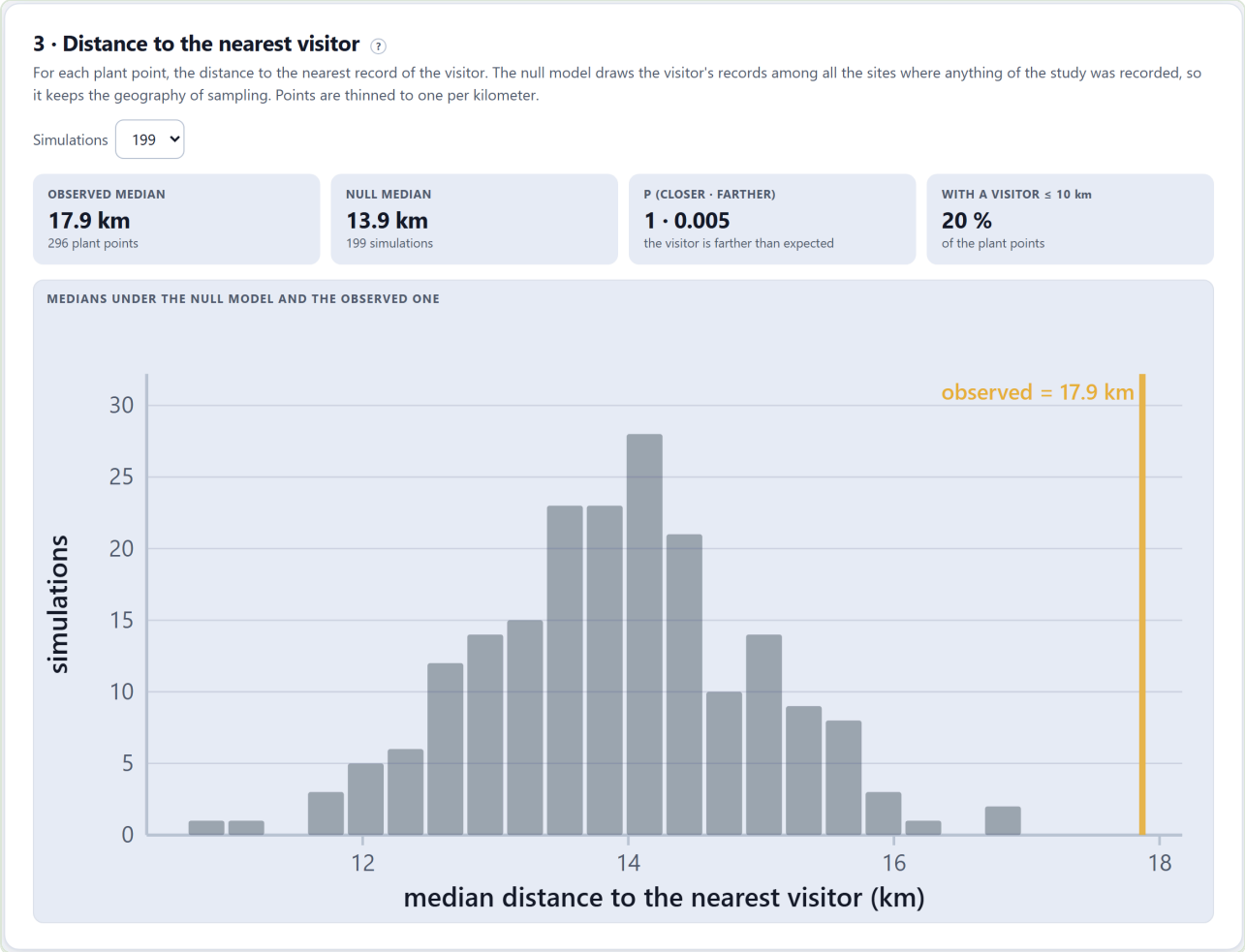


Figure 5.4. Chayote and *Xenoglossa*: the observed median, 17.9 km, lies outside the 199 medians of the null model (null median 13.9 km). p (farther) = 0.005: the visitor is farther from the plant than sampling would explain.

Tile	What it says
Observed median	The typical distance from a plant point to the nearest visitor, with how many plant points enter.
Null median	The median of the simulated medians.
p (closer · farther)	The fraction of simulations with a median equal or smaller (the visitor is closer than expected) and equal or larger (farther). With 199 simulations, the smallest possible p is 0.005.
With a visitor ≤ 10 km	What fraction of the plant points has a record of the visitor within 10 km: a figure easy to communicate.

TWO TESTS, TWO ANSWERS: IT IS NOT A CONTRADICTION

At 0.25°, the cell test says "at random" for chayote and *Xenoglossa*; the distance test says "farther than expected". They measure different things: the first counts whether they share 28 km cells; the second, how close each chayote is to the nearest visitor. In the practice data, *Xenoglossa* spreads over the center and the north, where there are many records of other units, and chayote is concentrated in the center and the southeast: within the cells they share, the points are not particularly close together. Report both and say what each one measures.

5.5 The overlap of ranges

The fourth card draws the **convex hull** of each unit —the smallest polygon without indentations that encloses all its points (Andrew 1979)— and computes its area on the sphere (Chamberlain and Duquette 2007), that of their intersection (Sutherland and Hodgman 1974) and what fraction of the plant's area is covered by the visitor's.



Figure 5.5. The area of chayote (301,078 km²) falls almost entirely (99%) within that of *Xenoglossa* (1,444,639 km²).

It is the coarsest reading of the three: a convex hull includes everything lying between the extreme points —in Figure 5.5, a good part of the Gulf of Mexico—, so it exaggerates the area of species with scattered records. It serves to say whether the visitor covers, at least in extent, the area of the plant; not to say whether they coincide within it.

B5

5.6 How to read the three tests together

Test	Question	In the practice study (chayote × *Xenoglossa*)
Cells (Veech)	Do they share more cells than by chance?	At random at 0.25°; it depends on the scale.
Nearest neighbor	Is the visitor closer to the plant than by chance?	Farther than expected.
Hulls	Does the visitor cover the area of the plant?	Yes, 99%.

WORKED EXAMPLE · HOW TO WRITE IT UP

"Xenoglossa occupies practically the whole area of chayote (99% of its convex hull), but its records do not coincide with those of the plant more than expected by chance within the sampled cells (0.25°: 51 shared cells against 46.0 expected, SES = 0.97), and the result depends on the scale. The distance from each plant record to the nearest visitor record was larger than under a target-group null model (median 17.9 versus 13.9 km; p = 0.005). Xenoglossa is therefore a potential pollinator whose spatial co-occurrence with chayote must be confirmed with visitation observations."

Remember that these numbers come from the practice data: they serve to learn how to write it up, not to conclude anything about chayote.

B5

COINCIDING IS NOT INTERACTING

Even with the three tests in favor, what the block demonstrates is **opportunity for interaction**. The sentence a reviewer will strike out is "X is a pollinator of Y because their distributions overlap"; the correct one, "X and Y overlap in space, which makes X a potential pollinator to be confirmed with observations of visits and pollen loads" (Blanchet et al. 2020; Dormann et al. 2018).

5.7 The warnings in this block

Message	What it means and what to do
Plants and visitors are needed	The list needs at least one plant and one visitor with cleaned records.
The grid is too fine for the "all land" universe in this extent	With 0.1° cells over a large area, the all-land universe has too many cells. Use larger cells or the target group.
The verdict changes with the scale…	It is not an error: it is a result. Report it with the sensitivity figure.
Many "unclassified" pairs	There are few records for the chosen cell. Use larger cells or bring species together in their genus (Block 3).

6

BLOCK 6 · PHENOLOGY AND ELEVATION

Phenology and elevation

Does the visitor fly when the plant flowers, and at the same elevation? Record dates are read as circular statistics —the year is a circle—: mean date, concentration, seasonality and difference between calendars. Monthly profiles are compared with the same overlap indices as the niche, with the option of correcting for the sampling effort of each month. And elevation is compared with densities along the gradient.

6.1 The year is a circle

6.2 The calendar of every unit

6.3 The circular statistics of each unit

6.4 Plant and visitor

6.5 Elevation

6.6 The warnings in this block

6.1 The year is a circle

December 31 and January 1 are one day apart. Treating dates as numbers from 1 to 365 gives a plant that flowers from November to February its “mean” in June. Circular statistics turn each date into an angle and solve the problem (Batschelet 1981; Fisher 1993; Morellato et al. 2010).

THEORY IN BRIEF · MEAN DATE, $\bar{R}$ AND THE TWO TESTS

Day d is equivalent to the angle $\theta = 2\pi(d - ½)/365$. The **mean date** is the direction of the average vector of all the records, and its length $\bar{R}$, from 0 to 1, measures concentration: 1 if all fall on the same day, 0 if they are spread evenly over the year. The **circular standard deviation**, in days, is its more intuitive version. The **Rayleigh** test says whether the distribution is seasonal (small p) or uniform; the **Watson–Williams** test, whether two calendars have the same mean date (Zar 2010), and it requires that they not be too spread out ($\kappa \geq 1$).

$$\bar{C} = \Sigma\cos\theta_i / n \quad \bar{S} = \Sigma\sin\theta_i / n \quad \bar{R} = \sqrt{(\bar{C}^2 + \bar{S}^2)} \quad \bar{\theta} = \operatorname{atan2}(\bar{S}, \bar{C}) \quad Z = n\cdot\bar{R}^2$$

A dated record does not necessarily say that the plant was in flower: a herbarium specimen may be vegetative or in fruit. That is why the checkbox Plants: only records with flowers uses only the records whose reproductive condition says flower or anthesis. And the number of records per month also reflects when people go to the field: the checkbox Correct by the monthly effort of the target group divides each profile by the records of the whole study in that month (Section 6.2).

6.2 The calendar of every unit

B6

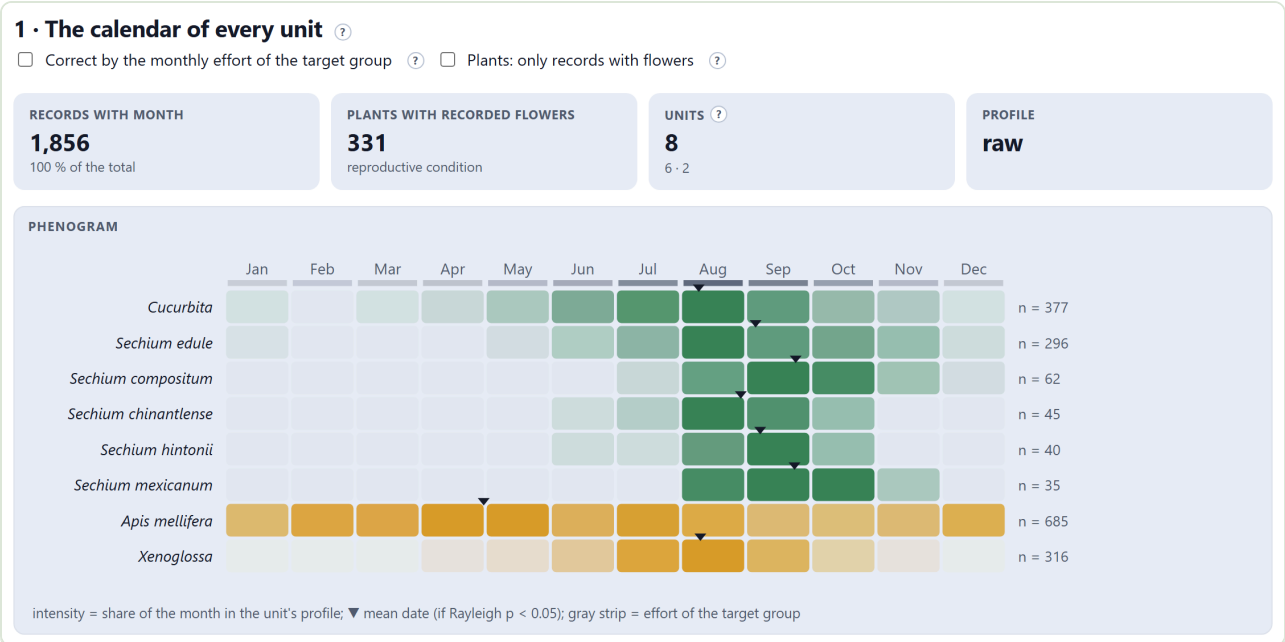


Figure 6.1. The phenogram of the practice study. Each row is a unit and each square a month; the color intensity is the fraction of that unit's records in that month, and the triangle marks the mean date when the Rayleigh test is significant. The gray strip at the top is the effort of the whole study.

The tiles say how many records carry a month (all 1,856 cleaned ones, 100%), how many plants have flowers recorded in their reproductive condition (331), how many units enter (6 plants and 2 visitors) and whether the profile is raw or corrected. The phenogram shows the essentials at a glance: the five *Sechium* species and the squashes flower from July to October; *Xenoglossa* flies from June to September; the honey bee is recorded all year round.



Figure 6.2. The same phenogram corrected by the monthly effort. The honey bee, which was recorded all year round, appears concentrated in winter: in those months the study has few records, and its records weigh more when divided by the effort.

! THE EFFORT CORRECTION CAN ALSO MISLEAD

Correcting divides the fraction of each month by the effort of that month, and the months with very few records in the whole study (fewer than five) are set to zero so as not to divide by almost nothing. Even so, with few records the correction amplifies the noise, as in Figure 6.2. The correction changes the **monthly profiles** and the overlap indices; the mean date and $\bar{R}$ are always computed with the dates of the records. Report the raw and the corrected result, and say which one you use to draw conclusions.

6.3 The circular statistics of each unit



▼ Circular statistics by unit

Role	Unit	Dated	Mean date	R̄	Circular SD (days)	Rayleigh p	Two peaks?
PLANT	*Cucurbita*	377 / 377	Aug 10	0.65 marked	54	2.372e-78	—
PLANT	*Sechium edule*	296 / 296	Sep 5	0.72 marked	47	1.447e-78	—
PLANT	*Sechium compositum*	62 / 62	Sep 24	0.85 marked	33	7.675e-26	—
PLANT	*Sechium chinantlense*	45 / 45	Aug 30	0.89 marked	28	1.307e-21	—
PLANT	*Sechium hintonii*	40 / 40	Sep 8	0.91 marked	26	2.878e-20	—
PLANT	*Sechium mexicanum*	35 / 35	Sep 24	0.89 marked	29	1.569e-16	—
VISITOR	*Apis mellifera*	685 / 685	May 2	0.12 diffuse	120	5.403e-5	—
VISITOR	*Xenoglossa*	316 / 316	Aug 11	0.80 marked	39	3.470e-109	—

Figure 6.3. The circular statistics of the eight units. The plants have marked flowering seasons ($\bar{R}$ from 0.65 to 0.91); *Xenoglossa* flies in a concentrated season ($\bar{R}$ 0.80, mean date August 11); the honey bee is recorded almost all year ($\bar{R}$ 0.12, *diffuse*).

Column	What it says
Dated	How many of the unit's records enter, out of how many it has.
Mean date	The direction of the mean vector, as day and month.
$\bar{R}$	The concentration, with its label: *diffuse*, *moderate* or *marked*.
Circular SD (days)	The spread around the mean date.
Rayleigh p	Whether there is seasonality. With hundreds of records it is nearly always very small: look at $\bar{R}$.
Two peaks?	"possible" when the second harmonic is strong ($\bar{R}_2 > 0.3$ and larger than $\bar{R}$ + 0.15): two flowering seasons a year, or two populations with different calendars. With two peaks, the mean date falls between them and describes neither.

DECISION RULE · HOW SEASONAL IT IS

If $\bar{R} < 0.3$	**then** the unit is recorded almost all year or has two seasons (*diffuse*): its mean date says little and it should not be compared with Watson–Williams.
If $\bar{R}$ is between 0.3 and 0.6	**then** there is a preferred season, but a broad one (*moderate*).
If $\bar{R} > 0.6$	**then** there is a defined season (*marked*) and the mean date summarizes it well.

The ? sheet for $\bar{R}$ carries this scale. They are reading conventions, not laws.

6.4 Plant and visitor

The second card compares one plant with one visitor, chosen in its two selectors.

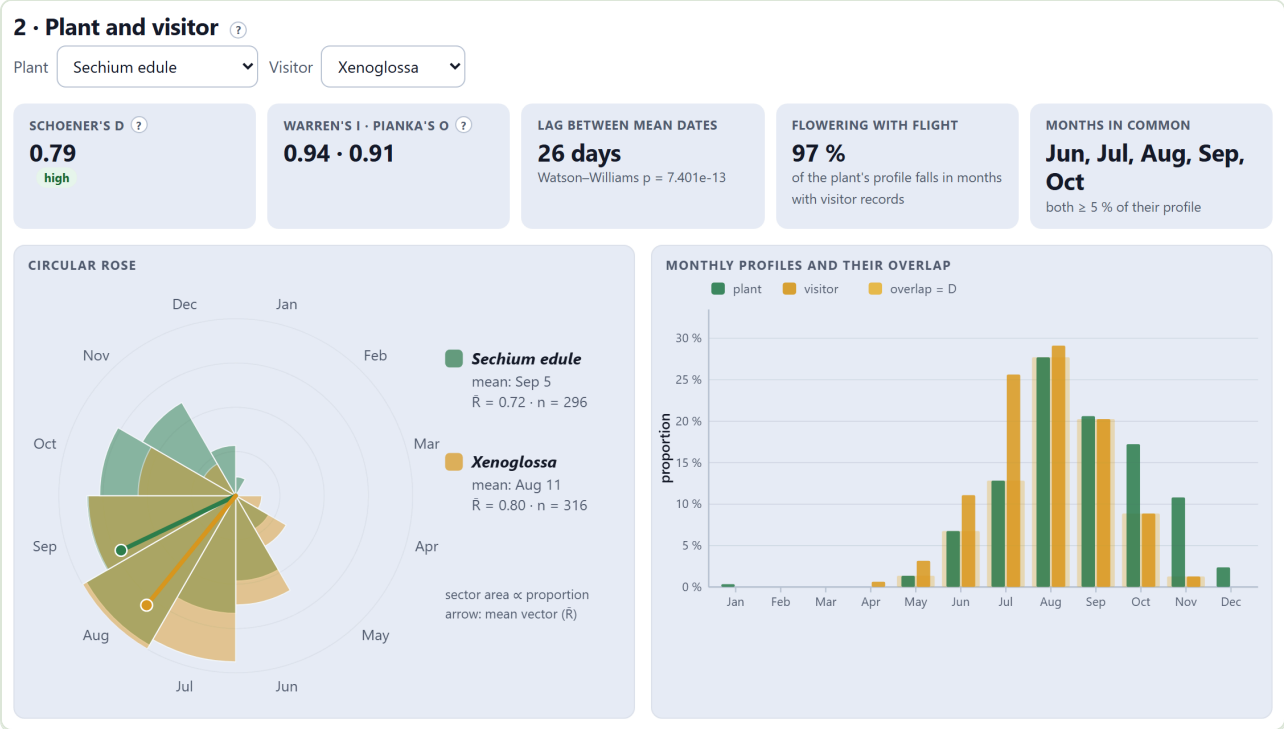


Figure 6.4. Chayote and *Xenoglossa*. In the rose, the area of each sector is proportional to the fraction of the month and the arrow is the mean vector; in the bars, the monthly profiles and, in light color, their overlap (the minimum of the two in each month, whose sum is D).

Tile	What it says	Chayote × *Xenoglossa*
Schoener's D	The fraction of the two calendars that is shared: Σ min(p_i, q_i) over the twelve months, from 0 to 1 (Schoener 1968).	0.79 · *high*
Warren's I · Pianka's O	Two overlap indices that are more lenient with small differences (Warren et al. 2008; Pianka 1973). They are reported along with D.	0.94 · 0.91
Lag between mean dates	The days between the two mean dates, with the Watson–Williams p.	26 days · p ≈ 7×10^{-13}
Flowering with flight	What fraction of the plant's profile falls in months with records of the visitor.	97 %
Months in common	The months in which both have at least 5% of their profile.	Jun to Oct

The reading is that of Section 1.4: the practice chayote flowers on average on September 5 and *Xenoglossa* flies on average on August 11; with hundreds of records, that 26-day lag is highly significant, and yet 97% of the flowering falls in months with bees flying and the two calendars share 79% of their profile. The visitor comes ahead of the plant, but it keeps it company.

▼ Every pair

Plant	Visitor	Schoener's D	I	O	Lag (days)	Watson–Williams p	Flowering with flight
Cucurbita	*Xenoglossa*	0.85 very high	0.97	0.98	1	0.837	98 %
Sechium edule	*Xenoglossa*	0.79 high	0.94	0.91	26	7.401e-13	97 %
Sechium chinantlense	*Xenoglossa*	0.69 high	0.92	0.88	19	0.001	100 %
Sechium hintonii	*Xenoglossa*	0.63 high	0.87	0.78	28	5.800e-6	100 %
Sechium mexicanum	*Xenoglossa*	0.59 moderate	0.74	0.71	44	1.226e-10	100 %
Cucurbita	*Apis mellifera*	0.56 moderate	0.83	0.71	100	6.918e-39 *	100 %
Sechium compositum	*Xenoglossa*	0.56 moderate	0.80	0.72	45	5.228e-16	98 %
Sechium edule	*Apis mellifera*	0.49 moderate	0.74	0.61	127	9.010e-53 *	100 %
Sechium compositum	*Apis mellifera*	0.33 low	0.60	0.47	146	1.048e-41 *	100 %
Sechium chinantlense	*Apis mellifera*	0.32 low	0.57	0.48	120	4.353e-27 *	100 %
Sechium mexicanum	*Apis mellifera*	0.28 low	0.51	0.44	145	5.931e-29 *	100 %
Sechium hintonii	*Apis mellifera*	0.26 low	0.54	0.42	129	4.458e-28 *	100 %

* κ < 1: the Watson–Williams test is unreliable with dates this spread.

Figure 6.5. Every pair, from highest to lowest D. The pairs with the honey bee carry an asterisk on the Watson–Williams p: with dates this spread out (κ < 1), the test is unreliable.

B6

DECISION RULE · SCHOENER'S D

If D < 0.2	**then** no or very limited overlap: flowering and flight hardly coincide.
If D is between 0.2 and 0.4	**then** low.
If D is between 0.4 and 0.6	**then** moderate.
If D is between 0.6 and 0.8	**then** high.
If D > 0.8	**then** very high: the calendars are almost the same.

The scale is that of Rödder and Engler (2011) for the niche, applied to the calendar. A year-round generalist (the honey bee) has a moderate D with almost every plant because its profile is spread out: always read D together with flowering with flight.

↓ Units (.csv) and ↓ Pairs (.csv) download the two tables (`phenology_units.csv` and `phenology_pairs.csv`), with the monthly profiles of each unit.

6.5 Elevation

The third card compares the elevational distribution of the pair with kernel densities (Silverman 1986) and computes its D. It needs records with elevation, which GBIF provides only sometimes: in the practice study only 23% of the records carry it, and none of the visitors'. Block 7 can fill it in from an elevation layer (Section 7.2); Figure 6.6 was made after doing so with the practice elevation layer.



Figure 6.6. The elevation of chayote and *Xenoglossa* with elevation filled in from the practice layer: D = 0.78 (*high*), and 96% of the plant's records fall within the central 90% of the visitor's range. The table gives the minimum, quartiles, median and maximum of each unit.

The sentence below the figure gives the elevational D and what fraction of the plant's records falls within the central 90% of the visitor's range. The table helps to read the wild relatives: in the practice data, *S. chinantlense* is a lowland species (median 317 m) and *S. mexicanum* a highland one (1,687 m).

THE ELEVATION OF A LAYER IS NOT THAT OF THE RECORD

The elevation read from a digital elevation model is that of the cell (at 0.1°, an average over about 11 km), not that of the exact point: in mountains it can differ by hundreds of meters. Records that carried their own elevation keep it; the layer only fills in those that lack it.

6.6 The warnings in this block

Message	What it means and what to do
Only N % of the records carry elevation: the profile may not represent the whole unit	Fill it in from a layer in Block 7, or read elevation with caution.
No record carries elevation…	Load a digital elevation model in Block 7 and fill in elevation from there.
The chosen pair has no elevations	One of the two units has no record with elevation.
* κ < 1 in the pairs table	The dates of one of the units are so spread out that Watson–Williams is unreliable. Use D and flowering with flight.
"possible" under Two peaks?	The unit may have two seasons. Check the phenogram and consider separating populations or subspecies (Block 3).
Few records with flowers	With Plants: only records with flowers turned on, a plant with few flowering records can end up with a noisy profile. Use it to confirm, not as the only analysis.

7

BLOCK 7 · ENVIRONMENTAL NICHE

Environmental niche

Do the plant and the visitor live in the same climates? They are compared in environmental space —the first two principal components of your layers—, correcting for what the region's environment offers, with two randomization tests: whether the niches are identical, and whether they are more alike than the mere similarity of the environments would produce. Then, variable by variable, what sets the two apart.

7.1 Comparing climates, not places

7.2 The environmental layers

7.3 Background and parameters

7.4 The environmental space

7.5 Equivalency and similarity

7.6 Variable by variable

7.7 The warnings in this block

7.1 Comparing climates, not places

Two species may live in different places and share a climate, or live close together and use different environments. For a pollination study, the second matters: if the visitor tolerates a narrower temperature range than the plant, there will be parts of the plant's range where the visitor cannot live, even if today they share a region. And climate change moves each species according to its own niche.

> THEORY IN BRIEF · THE METHOD OF BROENNIMANN ET AL. (2012)
>
> 1) The background cells —the available environment— are summarized with a principal component analysis of the standardized variables. 2) The records of each unit are projected onto the first two axes. 3) On a 100 × 100 grid of that plane, the density of the records and that of the background are estimated with a Gaussian kernel; the density of the records is kept only where there is available environment and, if requested, divided by that of the background. 4) **Schoener's D** and **Warren's I** compare the densities of the two units, from 0 (niches with nothing in common) to 1 (identical). 5) The **equivalency test** shuffles the records at random between the two units; the **similarity test** moves the visitor's niche to a random place in the background; both are repeated many times (Warren et al. 2008).

7.2 The environmental layers

B7

The block works with **your layers**: GeoTIFF (`.tif`) or ASCII grids (`.asc`), in degrees, preferably all at the same resolution. You pick them with Choose files... or, more conveniently, Choose a folder... . Each layer is read only within the window that covers the records plus a **margin** (3° by default): a global layer at 30" has 933 million cells and the study needs a few million, so the reader decompresses only that window. If a layer has a different grid from the first, it is resampled to the grid of the first and the table says so.

The app brings no real climate layers: the global climate databases do not allow redistributing their files, and they are downloaded once from their site (the 19 bioclimatic variables, at 10', 5', 2.5' or 30"). For learning, the drop-down No layers, or just practicing? Fictional layers generated by the program loads, with Load the practice layers , four invented layers for Mexico and Central America (Section I.7).

Part of the card	What it does
The table of layers	The **Use** box decides which variables go in; the name is recognized from the end of the file name (`…bio_1` → BIO1, with its long name) and `elev` , `alt` or `dem` → Elevation. The × removes one layer; Remove all , all of them.
Correlation between the variables in use	Pearson's r between each pair of variables over the cells of the area (a sample of up to 20,000). Those beyond |r| = 0.8 show in red. In the practice layers, none: the largest is −0.77 (BIO4 with BIO12).
Elevation from a layer	With an elevation layer chosen, Fill in the elevation of records that lack it reads it at the point of each record without elevation. Records that carried their own keep it. Block 6 uses it right away (Figure 6.6).

1 · Your environmental layers

Choose the files (or the whole folder) of your layers: GeoTIFF (.tif) or ASCII grid (.asc), in degrees, preferably all at the same resolution. Each layer is cropped to the area of the records plus a margin, so a global layer at 30″ also works. The app brings no real climate layers —the climate databases do not allow redistributing their files, and they are downloaded once from their sites—; for practicing, there are fictional ones below.

Choose files... | Choose a folder... | Margin around the records (°) 3 | Remove all

▼ No layers, or just practicing? Fictional layers generated by the program

Four **invented** layers for Mexico and Central America at 0.1°: a fictional relief and three climate variables derived from it with simple rules (BIO1, BIO4 and BIO12). They look like a climate but are not one: they are for learning Blocks 7 and 8 and must not be cited or used to conclude anything. Elevation is loaded unticked, because it is not a climate variable.

Load the practice layers

✓ 4 layers read and cropped to the study area (±3° around the records).

! These are **fictional** layers: for practicing, not for drawing conclusions or publishing.

Use	Layer	File	Resolution	Cells	Min	Max	
☐	**Elevation**	practica_altitud.asc	6′ (~11.1 km)	29,589	0	2,717	×
☑	**BIO1** Annual mean temperature	practica_bio_1.asc	6′ (~11.1 km)	29,589	11.5	29.2	×
☑	**BIO4** Temperature seasonality	practica_bio_4.asc	6′ (~11.1 km)	29,589	140	545	×
☑	**BIO12** Annual precipitation	practica_bio_12.asc	6′ (~11.1 km)	29,589	70	3,222	×

Correlation between the variables in use

	BIO1	BIO4	BIO12
BIO1		−0.66	0.48
BIO4			−0.77
BIO12			

No pair exceeds |r| = 0.8.

Elevation from a layer

Elevation layer Elevation | Fill in the elevation of records that lack it

The background

Background cells near the study's records | Distance (km) 200 | Density of the records | Test replicates 100

Compute the environmental space

Figure 7.1. The practice layers cropped to the study area: 29,589 cells of 6′ (about 11 km). Elevation is loaded unticked because it is not a climate variable; BIO1 (annual mean temperature), BIO4 (temperature seasonality) and BIO12 (annual precipitation) enter the analysis. The amber message reminds you that they are fictional.

✓ DECISION RULE · WHICH VARIABLES TO USE	
If two variables have |r| > 0.8	**then** keep one: the principal component analysis absorbs the correlation, but two nearly identical variables give double weight to what they measure.
If you do not know which ones to choose	**then** choose a few that make sense for the biology of the pair: mean temperature, seasonality, annual precipitation and dry-season precipitation are usually enough for plants and bees.
If your records come from GBIF with unknown precision	**then** 2.5′ or 5′ is a reasonable resolution; 30″ only makes sense with GPS coordinates.

7.3 Background and parameters

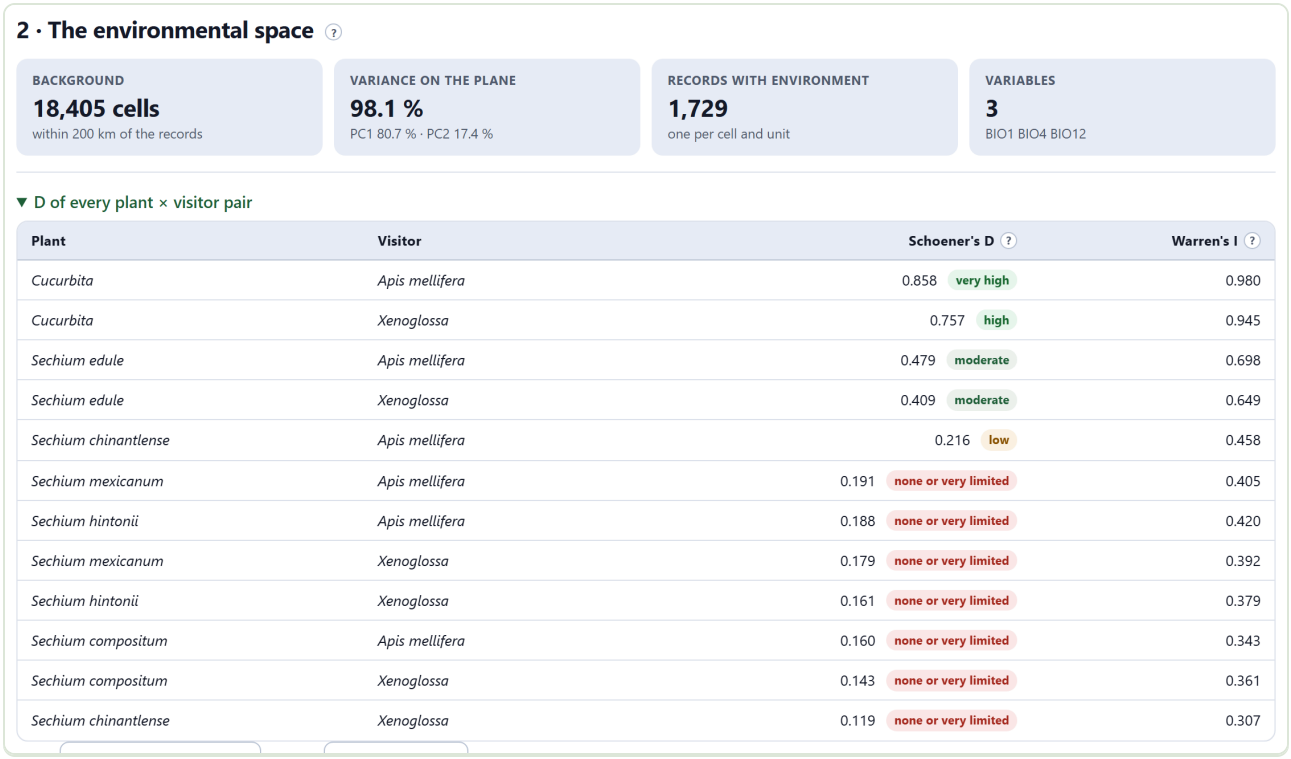
Control	What it does
Background	*Cells near the study's records* (within the chosen distance, 200 km by default) or *the whole area of the layers*. The background is the environment that was within reach of the species: the principal components and the density of the available environment are computed on it.
Density	*Of the records* (the default) or *corrected by the environment*: dividing by the density of the background corrects for a very common climate gathering records just because it is common, but it amplifies the edges of the available space, where there are few cells and a single record weighs a lot.
Test replicates	50, 100, 200 or 500. With 100, the smallest possible p is 1/101 ≈ 0.01; for publication, 500.
Compute the environmental space	Builds the background, the principal component analysis, the densities and the tests of every pair.

B7

> **! THE BACKGROUND CHANGES D**
>
> A background that is too wide includes environments that were never within reach of the species and makes the niches look more alike than they are. State the background in the methods and, if you can, report the result with two backgrounds (for instance, 200 km and the whole area). The same goes for the density: if the two versions give similar D values and the same conclusions, the result is robust.

7.4 The environmental space

The **records with environment** are one per cell and unit: ten records of the same species in the same 11-km cell say the same thing about the climate and must not weigh ten times. The table D of every plant × visitor pair gives the first reading: *Cucurbita* shares much of its niche with both visitors (D = 0.86 and 0.76); the wild relatives with small ranges, almost nothing.



2 · The environmental space

BACKGROUND	VARIANCE ON THE PLANE	RECORDS WITH ENVIRONMENT	VARIABLES
18,405 cells	98.1 %	1,729	3
within 200 km of the records	PC1 80.7 % · PC2 17.4 %	one per cell and unit	BIO1 BIO4 BIO12

▼ D of every plant × visitor pair

Plant	Visitor	Schoener's D	Warren's I
Cucurbita	*Apis mellifera*	0.858 very high	0.980
Cucurbita	*Xenoglossa*	0.757 high	0.945
Sechium edule	*Apis mellifera*	0.479 moderate	0.698
Sechium edule	*Xenoglossa*	0.409 moderate	0.649
Sechium chinantlense	*Apis mellifera*	0.216 low	0.458
Sechium mexicanum	*Apis mellifera*	0.191 none or very limited	0.405
Sechium hintonii	*Apis mellifera*	0.188 none or very limited	0.420
Sechium mexicanum	*Xenoglossa*	0.179 none or very limited	0.392
Sechium hintonii	*Xenoglossa*	0.161 none or very limited	0.379
Sechium compositum	*Apis mellifera*	0.160 none or very limited	0.343
Sechium compositum	*Xenoglossa*	0.143 none or very limited	0.361
Sechium chinantlense	*Xenoglossa*	0.119 none or very limited	0.307

Figure 7.2. The space of the practice study: 18,405 background cells within 200 km of the records; the first two components explain 98.1% of the variance (80.7 and 17.4%); 1,729 records with environment, one per cell and unit. The table sorts the twelve pairs by D.

In Figure 7.3, BIO4 (seasonality) points to the right and BIO12 (rainfall) down to the left: the first component runs from rainy, weakly seasonal climates to dry, seasonal ones. The practice chayote occupies the rainy part and *Xenoglossa* extends toward the dry part: D = 0.409, *moderate*.

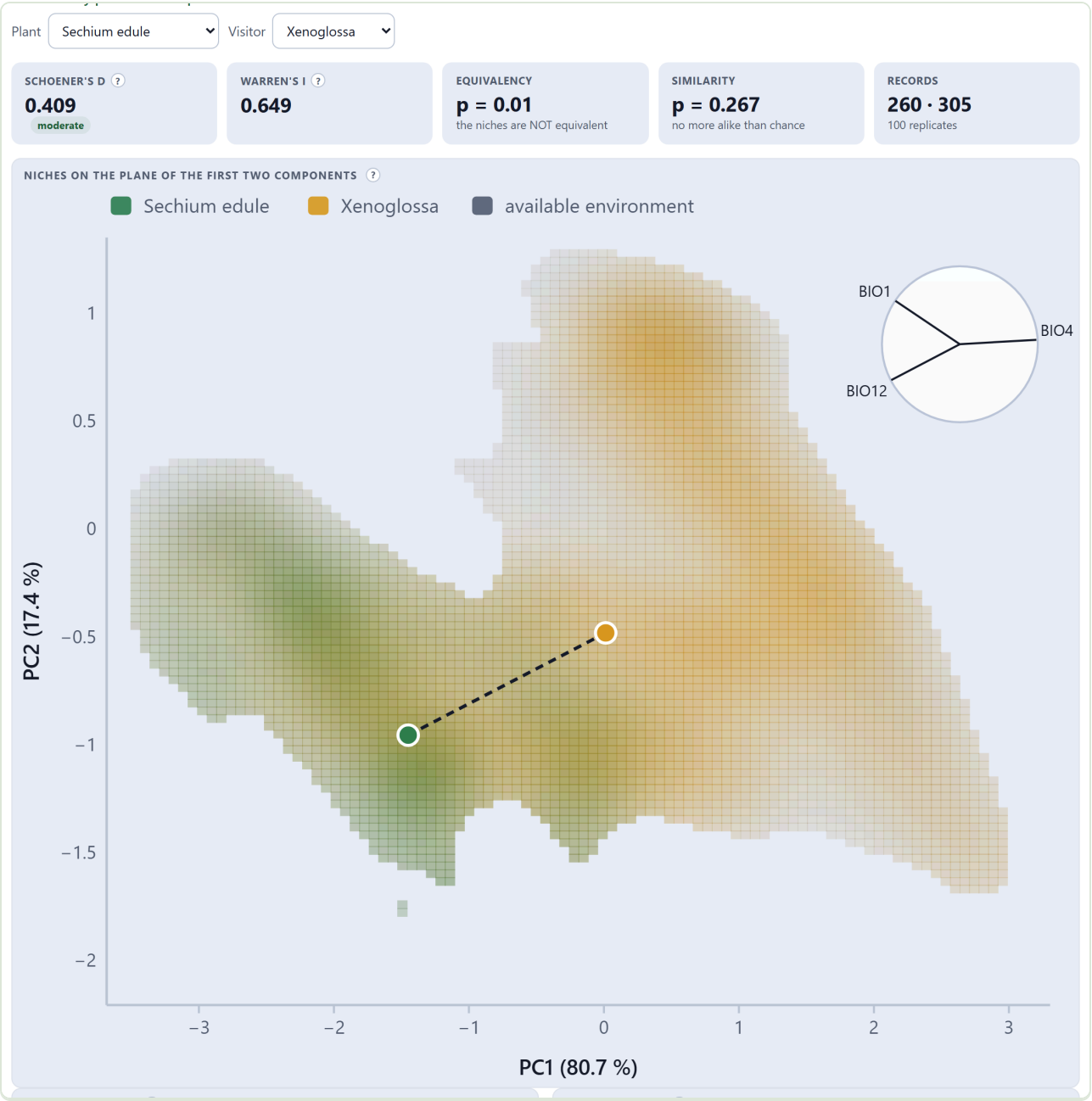


Figure 7.3. Chayote and *Xenoglossa* on the plane of the two components. Green is the density of chayote; amber, that of *Xenoglossa*; where they blend, they share environment. The two circles are the centroids of each niche and the line measures how far apart they are. The circle in the corner tells which way each variable grows on the plane.

B7

7.5 Equivalency and similarity

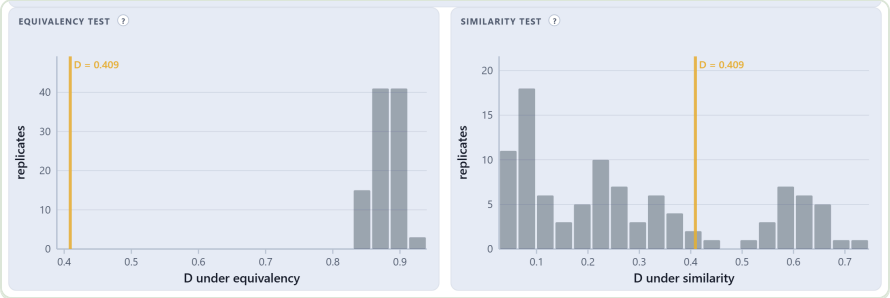


Figure 7.4. The two tests for chayote and *Xenoglossa* with 100 replicates. On the left, the D values obtained by shuffling the records at random between the two units: all above 0.8, far from the observed one (0.409). On the right, the D values obtained by moving the visitor's niche to random places in the background: the observed one falls in the middle.

Test	Question	Chayote × *Xenoglossa*
Equivalency	Are the two niches identical? If the observed D is smaller than almost all those of the random shuffles ($p < 0.05$), they are **not equivalent**.	$p = 0.01$ · the niches are NOT equivalent
Similarity	Are they more alike than the mere similarity of the available environments would produce? If the observed D is larger than almost all those of the randomly moved niches ($p < 0.05$), they are **more alike than chance**: there is niche conservatism.	$p = 0.267$ · no more alike than chance

B7

WORKED EXAMPLE · HOW TO WRITE IT UP

"In the space of the first two principal components of the environment (98.1% of the variance), the niche of Sechium edule overlapped moderately with that of Xenoglossa (D = 0.41; I = 0.65). The equivalency test rejected that the niches were identical (p = 0.01), and the similarity test did not indicate that they were more alike than expected from the available environments (p = 0.27; 100 replicates each; Warren et al. 2008; Broennimann et al. 2012)."

Again: these are the numbers of the practice layers, useful for learning how to write it up.

DECISION RULE · THE TWO TESTS TOGETHER

If equivalency $p < 0.05$ and similarity $p < 0.05$	**then** the niches are not identical, but they are more alike than the environments explain: they share preferences.
If equivalency $p < 0.05$ and similarity $p \geq 0.05$	**then** the niches differ and their resemblance goes no further than that of the environments: each uses the climate in its own way (the case of the example).
If equivalency $p \geq 0.05$	**then** you cannot reject that the niches are identical. With few records the test has little power: say how many there were.

7.6 Variable by variable

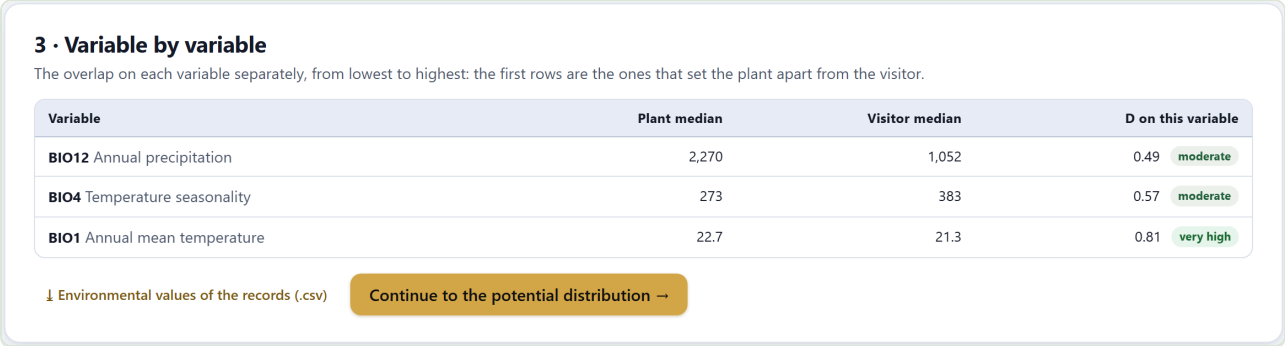


Figure 7.5. D on each variable separately, from lowest to highest. Annual precipitation is the one that most sets chayote apart from *Xenoglossa* in the practice layers (medians of 2,270 and 1,052 mm; D = 0.49); mean temperature, the one that least does (22.7 and 21.3 °C; D = 0.81).

The first rows are the ones that set the plant apart from the visitor: the variable worth thinking about when looking for where the relationship might break. ↓ Environmental values of the records (.csv) downloads, as `environmental_values.csv`, for each record with environment, its unit, its coordinates, its position on the two components and the value of each variable.

7.7 The warnings in this block

Message	What it means and what to do
There are no .tif, .tiff or .asc files in the selection	The folder or the files are not layers. Check the extension.
An error reading a layer (format, compression)	The reader accepts GeoTIFF with or without LZW, Deflate or PackBits compression, classic or BigTIFF, in strips or tiles, and ASCII grids. A layer in meters (projected) will not do: reproject it to degrees.
the layer does not cover the study area	The layer does not overlap the records. Check that it is in degrees and that it covers the area.
N pairs with |r| > 0.8 (in red)...	Two nearly identical variables. Keep one of each pair (rule of Section 7.2).
At least two layers in use and a background with valid cells are needed	Tick at least two variables, or widen the background distance.
These are fictional layers: for practicing, not for drawing conclusions or publishing	The practice layers were loaded. The Block 10 report repeats it in its niche section.

8

BLOCK 8 · POTENTIAL DISTRIBUTION

Potential distribution and mismatch

Where the environment is suitable for the plant and for each visitor, and where the plant is left without any potential visitor. Three suitability models — envelope, Mahalanobis and background logistic— assessed by cross-validation; the map of the plant with and without a visitor; and, with future climate layers, how that mismatch changes.

8.1 Where the plant might be left alone

8.2 The models

8.3 Evaluation and suitability maps

8.4 The mismatch

8.5 With a future climate

8.6 The warnings in this block

8.1 Where the plant might be left alone

Block 7 compared the niches in environmental space; this block takes them back to the map. With a suitability model for the plant and another for each potential visitor, every cell falls into one of four classes: suitable for both, for the plant only, for the visitor only or for neither. The cells **for the plant only** are the **mismatch**: the part of the plant's range that would depend on other visitors, or that would be left without pollination if the relationship were exclusive (Schweiger et al. 2008). Under a future climate, the mismatch may grow or shrink (Kerr et al. 2015).

> **THIS BLOCK USES THE LAYERS OF BLOCK 7**
>
> The variables are the ones ticked in Block 7 (they are listed at the top of the card). If there are not at least two layers loaded, the block asks you to go and get them. In this chapter they are the practice layers —BIO1, BIO4 and BIO12—, and everything said about their maps is about an invented climate.

8.2 The models

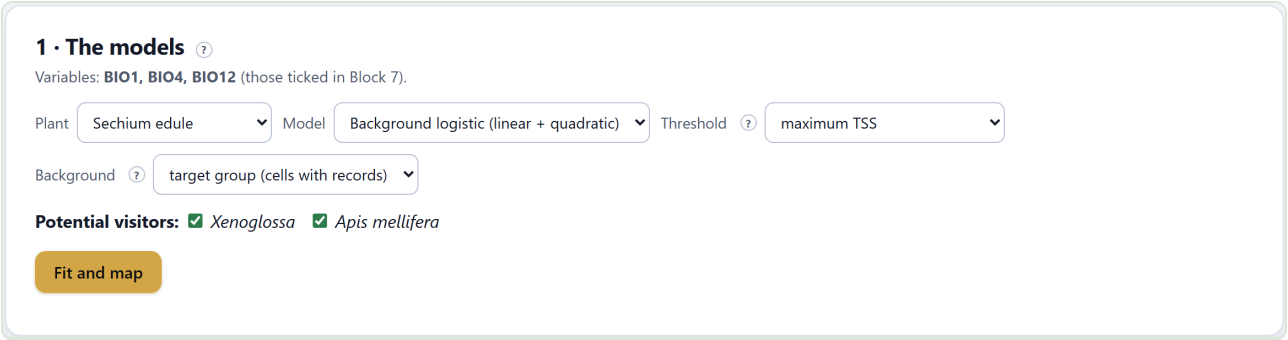


Figure 8.1. The controls: the plant, the model, the threshold rule, the background and the potential visitors, which by default are all the visitors on the list.

Model	How it decides suitability
Background logistic (the default)	Compares the cells with records against background cells with linear and quadratic terms of the standardized variables and a ridge penalty, fitted by iteratively reweighted least squares. It is the most flexible (Phillips et al. 2009; Elith et al. 2011).
Mahalanobis distance	How far a cell is from the climatic center of the records, taking into account how the variables covary; suitability is the upper tail of χ^2 at that distance (Farber and Kadmon 2003).
Bioclimatic envelope	Suitable is whatever falls within the range of the records on every variable, with more weight toward the center; the simplest and the most conservative (Nix 1986).

B8

Control	What it does
Threshold	The value from which a cell is declared suitable: *maximum TSS* (the one that best separates records from background), *10th percentile of presences* (leaves out the 10% least suitable records) or *minimum presence* (includes every record; the widest) (Liu et al. 2005).
Background	*Target group* (the cells with records of any unit of the study: it corrects for sampling bias) or *the whole area*. It only counts in the logistic model and in the evaluation.
Potential visitors	The visitors that count for the mismatch. Unticking one asks what would happen without it.
Fit and map	Fits the model of the plant and of each ticked visitor, evaluates them and draws the maps.

8.3 Evaluation and suitability maps

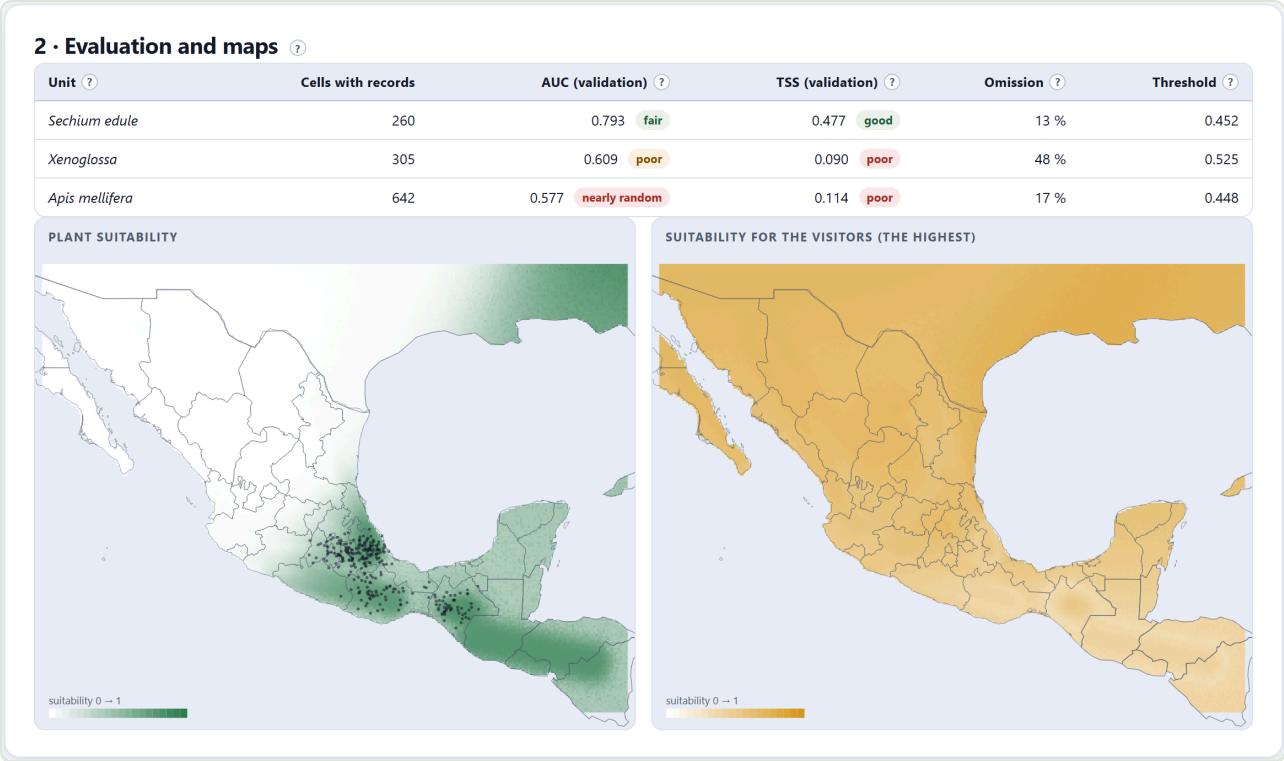


Figure 8.2. The evaluation of the three logistic models with the practice layers, and the two suitability maps: that of chayote (with its records) and the highest among the visitors in each cell.

Each model is evaluated by **four-fold cross-validation**: it is fitted with three quarters of the cells with records and tested with the remaining quarter, four times. That way the numbers say how well the model predicts records it did not see.

B8

Column	What it says	Chayote · *Xenoglossa* · *Apis*
Cells with records	The presences of the model (one per layer cell).	260 · 305 · 642
AUC (validation)	The probability that a cell with a record receives higher suitability than a random background cell (Swets 1988).	0.793 · 0.609 · 0.577
TSS (validation)	Sensitivity + specificity − 1 with the chosen threshold; 0 is chance (Allouche et al. 2006).	0.477 · 0.090 · 0.114
Omission	The fraction of validation records that fall outside the suitable area.	13 % · 48 % · 17 %
Threshold	The suitability value of the cut.	0.452 · 0.525 · 0.448

✓ DECISION RULE · AUC AND TSS

AUC < 0.6 · 0.6–0.7 · 0.7–0.8 · 0.8–0.9 · > 0.9	nearly random · poor · fair · good · excellent.
TSS < 0.2 · 0.2–0.4 · 0.4–0.6 · > 0.6	poor · fair · good · very good.
If a visitor has a nearly random AUC	**then** its map does not discriminate: it is usually a generalist that lives in almost every climate of the area (the honey bee) or a unit with very scattered records. Its high suitability everywhere will erase the mismatch; say that this is due to lack of discrimination, not to a shared niche.

With presence-only data, AUC depends on the size of the background: compare models with the same background, not against numbers from other papers (Lobo et al. 2008).

In Figure 8.2, the chayote model is fair and its map follows the humid southern and Gulf slope of the practice layers; those of the visitors hardly discriminate, and the map on the right —the highest suitability of the two in each cell— comes out high almost everywhere. The plant's map also shows a real phenomenon of these models: high suitability in the northeast, outside the area of the records, where the model **extrapolates** to climates it did not see.

B8

8.4 The mismatch

Tile	What it says	In the practice study
Plant suitable area	The surface of the cells suitable for the plant, in km² on the sphere.	716,078 km²
With a potential visitor	The fraction and the area where it is also suitable for at least one visitor.	40 % · 284,500 km²
Mismatch: plant without visitor	The fraction and the area where it is suitable only for the plant.	60 % · 431,578 km²
Visitor only	The area suitable for some visitor but not for the plant.	2,450,600 km²



Figure 8.3. The mismatch of chayote with the two potential visitors in the practice layers: of 716,078 km² suitable for the plant, 40% have a potential visitor and 60% do not.

Below the map, the app says what fraction of the cells lies outside the environmental range of the plant's records in at least one variable: *"35 % of the map cells lie outside the environmental range of the plant's records in at least one variable: there the model extrapolates and its suitability must be read with caution."* ⤓ Classes by cell (.csv) downloads, as `mismatch_cells.csv`, each cell with its coordinates, the suitability of the plant and its current and future class.

> **A POTENTIAL MISMATCH**
>
> The map rests only on environmental suitability: it does not consider dispersal, barriers, land use, the behavior of the visitors or other visitors that are not on the list. A large mismatch tells you where it is worth looking in the field for what visits the plant; it does not say that the plant is not pollinated there.

DECISION RULE · HOW TO DEFEND THE MISMATCH	
If the mismatch changes a lot with the threshold	**then** show it with two rules (maximum TSS and 10th percentile of presences) and use the same one for the plant and the visitors.
If the visitor models have a poor TSS	**then** draw no conclusion about the mismatch: the visitors' map is not reliable.
If a large part of the map is extrapolated	**then** restrict the interpretation to the area where there are records and say so in the methods.

8.5 With a future climate

The fourth card applies the models already fitted, without refitting them, to the same variables computed for a future period under a climate model and an emissions pathway. Choose the future layers... takes the files, which are recognized by the end of their name ("bio_1", "bio_12"...) and must be the same variables in use. For practice, Use the practice future (fictional) loads an invented future, warmer and drier, which only works with the practice layers.



Figure 8.4. The mismatch under the practice future. The suitable area of chayote falls by 62%, the area with a potential visitor by 45% and the mismatch by 73%: in this invented future, the plant loses above all the zones where it was already alone.

The table compares, in km², the suitable area of the plant, the area with a potential visitor and the mismatch, today and in the future, with its change in percent. The Block 10 report takes it up again with the names of the future files.

> **SEVERAL CLIMATE MODELS, NOT ONE**
>
> A single climate model is one of many possible projections. For a paper, repeat the projection with three or more climate models and two emissions pathways, and report the range of the change. The app projects one set of layers at a time: choose the future layers again for each combination and write down the result.

8.6 The warnings in this block

Message	What it means and what to do
Environmental layers are needed	Load at least two layers in Block 7.
First fit the models with the current climate	The future is projected with the models already fitted: press Fit and map first.
Future layers are missing for: …	The future layers must be the same variables in use and have the same ending in their name ("bio_1"…).
The practice future only goes with the practice layers of Block 7	With your real layers, choose your own future layers.
N % of the map cells lie outside the environmental range…	There the model extrapolates. Read it with caution and say so in the methods.
A unit with very few cells with records	Cross-validation needs records in each fold; with very few, the model and its evaluation are fragile. Merge species into their genus or widen the area.

9

BLOCK 9 · INTERACTION NETWORK

Interaction network

Here we no longer speak of coincidences but of recorded visits: your field observations and what others have published. The plant–visitor network says how specialized the relationship is —connectance, nestedness, specialization and modules, against null models—, which visitors are key for each plant and how many remain to be recorded.

9.1 From coincidence to visit

9.2 Which interactions go in

9.3 The network and its null models

9.4 Plants and visitors, one by one

9.5 One plant and its visitors

9.6 The warnings in this block

B9

9.1 From coincidence to visit

Blocks 5 to 8 measured the opportunity for interaction. This block works with the fourth link of the chain of evidence (Section 1.7): the visit to the flower. A visit network is a matrix of plants (rows) by visitors (columns) with the number of visits observed in each cell, or with a 1 when all that is known is that the link exists.

Pollination networks have a characteristic structure: they are **nested** —specialists visit subsets of the plants that generalists visit (Bascompte et al. 2003)—, more or less **modular** —groups of plants and visitors that interact more among themselves— and they have a measurable **specialization**. Each index is compared with what a random network with the same totals would give: a number without a null model does not say whether the structure is remarkable.

9.2 Which interactions go in

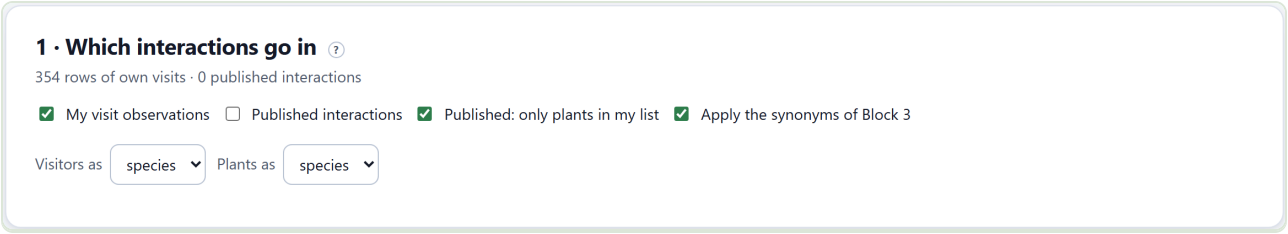


Figure 9.1. The sources of the network of the practice study: 354 rows of own visits and no published interactions (the practice data do not query the internet).

Control	What it does
My visit observations	The visits imported in Block 2, with their counts.
Published interactions	The pairs brought from the interaction database; each published record counts as one visit.
Published: only plants in my list	Leaves out the published plants that do not belong to an entry of your list (the database also returns the other plants that your visitors visit).
Apply the synonyms of Block 3	Rewrites the names with the synonym rules: *Peponapis crassidentata* becomes *Xenoglossa crassidentata*.
Visitors as · Plants as	Species or genera. Grouping by genus helps when the sources identify at different levels.

> **MIXING OWN AND PUBLISHED VISITS CHANGES WHAT CAN BE COMPUTED**
>
> With your visits only, the network is **quantitative**: each cell is a comparable number of visits. When you add the published ones, each published record counts 1 and the numbers stop being comparable: the app then computes the structure (connectance, NODF, modules) but not H2' or d', which assume counts, and says so in its tile.

9.3 The network and its null models

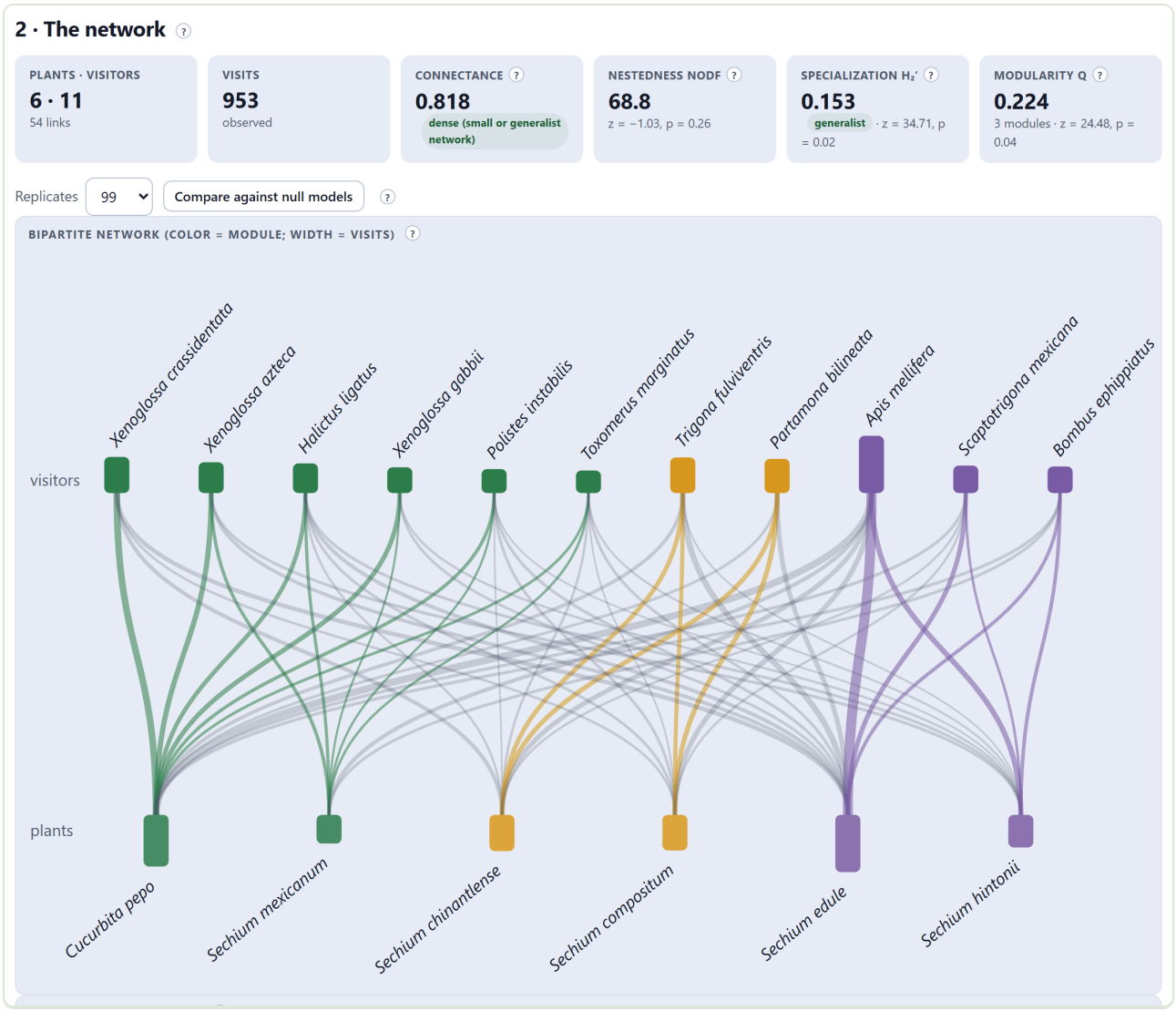


Figure 9.2. The network of the practice study after Compare against null models with 99 replicates: 6 plants, 11 visitors, 54 links and 953 visits. In the drawing, the visitors are at the top and the plants at the bottom; the width of each line is the number of visits and the color, the module.

Index	What it measures	In the practice study
Connectance	The observed links among the possible ones (Jordano 1987). Large networks usually have between 0.15 and 0.4.	0.818 · *dense (small or generalist network)*
NODF	Nestedness, from 0 to 100, by overlap and decreasing fill of the binary matrix (Almeida-Neto et al. 2008).	68.8 · z = −1.03, p = 0.26

B9

Index	What it measures	In the practice study
H_2'	The specialization of the whole network, from 0 (each visitor spreads its visits like the supply) to 1 (each one visits exclusive plants) (Blüthgen et al. 2006).	0.153 · *generalist* · z = 34.71, p = 0.02
Modularity Q	How divided the network is into groups (Barber 2007), searched for with label propagation and restarts (Liu and Murata 2010; Beckett 2016).	0.224 · 3 modules · z = 24.48, p = 0.04

The **null models** keep the totals of each plant and each visitor and distribute the links at random: the *curveball* algorithm for the binary matrix (Strona et al. 2014) and the random allocation of individual visits for the quantitative one (equivalent to that of Patefield 1981). The tile gives **z** —how many standard deviations the observed value lies from the null mean— and a two-tailed p; the one for modularity uses up to 49 replicates because each one requires searching for modules.

WORKED EXAMPLE · READING THE PRACTICE NETWORK

The practice network is small and highly connected (0.818): almost every visitor reaches almost every plant. Its nestedness (68.8) is no different from that of a random network with the same totals (p = 0.26). Its specialization is low in absolute terms (H2' = 0.153, *generalist*), but much higher than that of chance (z = 34.7): the visitors distribute their visits with preferences, albeit weak ones. And it has three modules: the squash bees, with *Halictus*, *Polistes* and *Toxomerus*, together with *Cucurbita pepo* and *S. mexicanum*; the stingless bees *Trigona* and *Partamona* with *S. chinantlense* and *S. compositum*; and the honey bee, *Scaptotrigona* and the bumblebee with chayote and *S. hintonii*.

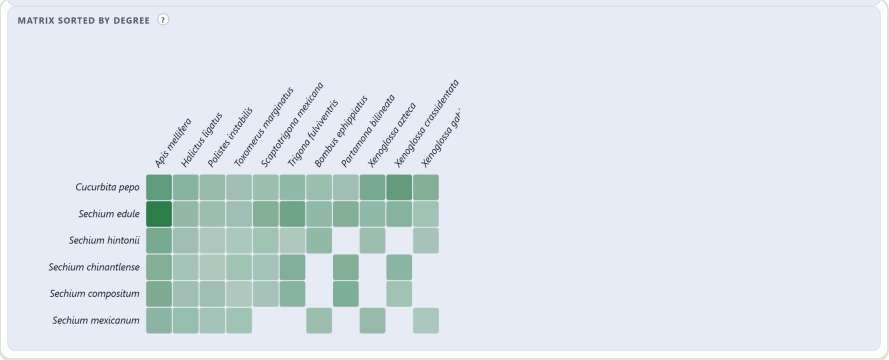


Figure 9.3. The matrix sorted by degree: the visitors and plants with the most links at the top and on the left. In a nested network, the filled cells crowd into the upper left corner. ↓ Matrix (.csv) downloads it as `interaction_matrix.csv`.

B9

DECISION RULE · WHAT TO REPORT ABOUT THE NETWORK	
If the network is small (fewer than 10 species per side)	**then** connectance will be high and nestedness hardly informative; report H_2', the modules and the sampling completeness before NODF.
If an index does not differ from the null model	**then** the structure it measures is not remarkable given the totals: do not discuss it as a trait of the network.
H_2' < 0.3 · 0.3–0.6 · > 0.6	generalist · intermediate · specialized.

9.4 Plants and visitors, one by one

3 · Species

Plants

Species	Visits	Degree	Normalized degree	Strength		d'
Sechium edule	348	11	1.00	3.38	0.07	generalis
Cucurbita pepo	271	11	1.00	3.64	0.16	generalis
Sechium chinantlense	105	8	0.73	1.09	0.16	generalis
Sechium compositum	100	8	0.73	1.05	0.16	generalis
Sechium hintonii	77	9	0.82	0.98	0.17	generalis
Sechium mexicanum	52	7	0.64	0.87	0.22	generalis

Visitors

Species	Visits	Degree	Normalized degree	Strength		d'
Apis mellifera	354	6	1.00	2.13	0.04	genera
Partamona bilineata	90	4	0.67	0.65	0.22	genera
Trigona fulviventris	102	5	0.83	0.61	0.11	genera
Xenoglossa crassidentata	106	4	0.67	0.49	0.16	genera
Xenoglossa azteca	66	4	0.67	0.47	0.17	genera
Halictus ligatus	59	6	1.00	0.46	0.04	genera
Bombus ephippiatus	41	4	0.67	0.38	0.17	genera
Scaptotrigona mexicana	46	5	0.83	0.23	0.05	genera
Polistes instabilis	29	6	1.00	0.20	0.04	genera
Xenoglossa gabbii	37	4	0.67	0.19	0.17	genera
Toxomerus marginatus	23	6	1.00	0.19	0.04	genera

⤓ Species (.csv)

Figure 9.4. The species of the network. The honey bee has the highest strength (2.13) and the lowest specialization (d' = 0.04): it is the visitor on which the plants of the practice network depend most.

Column	What it says
Visits · Degree	How many visits, and with how many species of the other side it interacts.
Normalized degree	The degree divided by the number of species of the other side: 1 if it interacts with all of them.

Column	What it says
Strength	The sum of its partners' dependences on it (Bascompte et al. 2006): a visitor with high strength is one on which many plants depend for a large fraction of their visits. It is the measure of a "key visitor".
d'	The specialization of each species with respect to the availability of its partners, from 0 to 1 (Blüthgen et al. 2006): *generalist* < 0.3, *intermediate* 0.3–0.6, *specialist* > 0.6.

↓ Species (.csv) downloads both tables as `network_species.csv`. In the practice network no species exceeds d' = 0.22; the most specialized are *Partamona bilineata* and *S. mexicanum* (0.22).

9.5 One plant and its visitors

The last card looks at one plant at a time: who visits it, in what proportion, and whether sampling has already found all its visitors. It works even if the study has a single plant —in that case it is the only possible reading, because with one plant there is no network to describe.

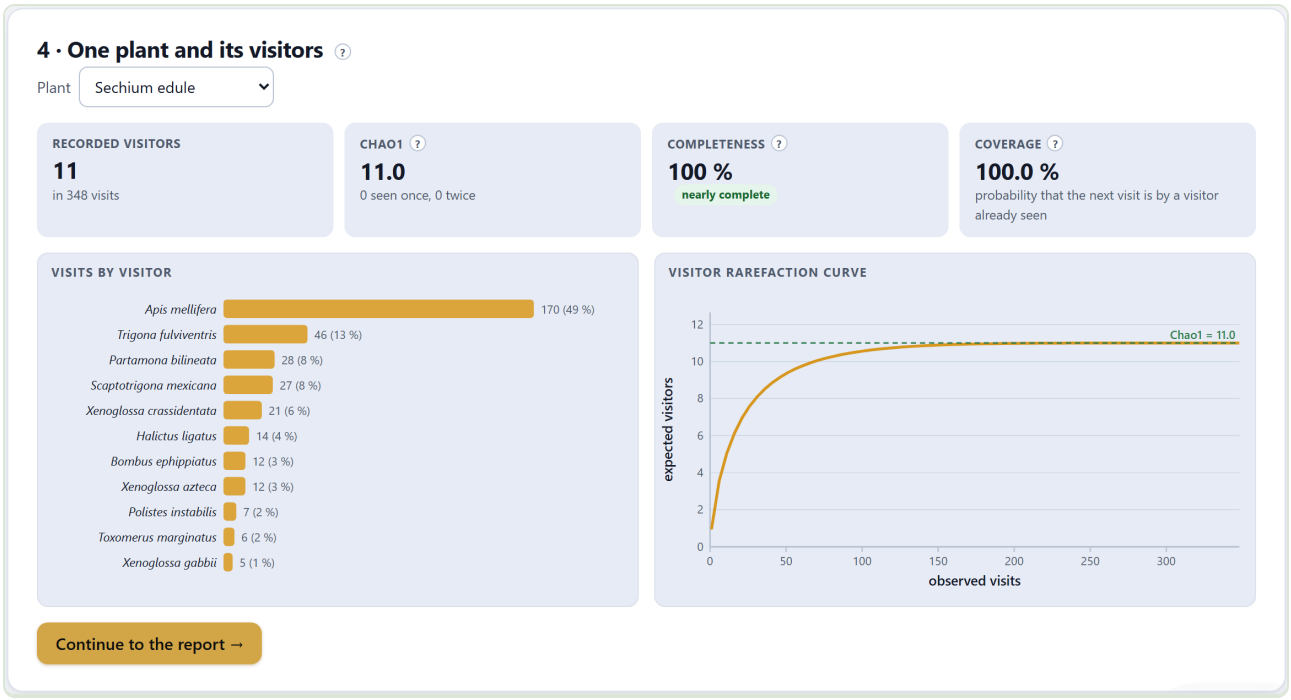


Figure 9.5. The practice chayote: 11 visitors in 348 visits, half of them by the honey bee. The rarefaction curve flattens well before 348 visits and reaches the Chao1 line: sampling is nearly complete.

Tile	What it says
Recorded visitors	How many different visitors in how many visits.
Chao1	The estimated minimum number of visitors, from how many were seen once and how many twice (Chao 1987).

B9

Tile	What it says
Completeness	Recorded divided by Chao1: what fraction of the visitors has already been found (Chacoff et al. 2012). *Very incomplete* < 50%, *incomplete* 50–80%, *good* 80–95%, *nearly complete* > 95%.
Coverage	The probability that the next visit is by a visitor already seen.

DECISION RULE · DO YOU NEED TO GO BACK TO THE FIELD?

If completeness is below 80%	**then** visitors remain to be recorded: do not claim that the plant has few visitors or that it depends on one; go back to the field or say so in the discussion.
If there are many visitors seen only once	**then** Chao1 is unstable: it is a minimum of the total, not the total.

WITH PUBLISHED INTERACTIONS ONLY, THERE IS NO COMPLETENESS

Completeness needs visit counts. With published interactions, all that is known is whether the link exists, and the app says so instead of drawing the curve.

9.6 The warnings in this block

Message	What it means and what to do
Interactions are needed	Import your visit observations or search the published interactions in Block 2.
There are no interactions with these sources...	The chosen boxes leave the network empty. Tick another source or remove the filter of plants in the list.
With a single plant (or a single visitor) there is no network to describe...	The visitors, their shares and the completeness are shown; the network indices do not apply.
Sampling completeness needs visit counts...	There are only published interactions for that plant.
H_2': needs visit counts	The network mixes sources or has only published ones (Section 9.2).
no null model yet	Press Compare against null models.

10

BLOCK 10 · REPORT

Figures, report and package

Everything you computed, ready to leave the app: every figure of the study in PNG, JPG or SVG up to 900 dpi in the style you choose —that of all of them at once from the figure studio, or that of each one with its ✎ editor—; a report that opens in any browser and prints to PDF, with the methods written from what you actually ran; and a .zip with the report, the figures, the tables and the project that reproduces it.

10.1 What comes out of the app

10.2 The state of the study

10.3 The catalog and the two figure editors

10.4 The report

10.5 The reproducible package

10.6 Before you hand it in

10.1 What comes out of the app

A study is delivered with three things: the figures, the text that says what was done and the data that allow it to be redone. Block 10 produces all three from whatever exists at that moment in the computed blocks. It recomputes nothing: if you change something in a block, come back here and generate again.

10.2 The state of the study

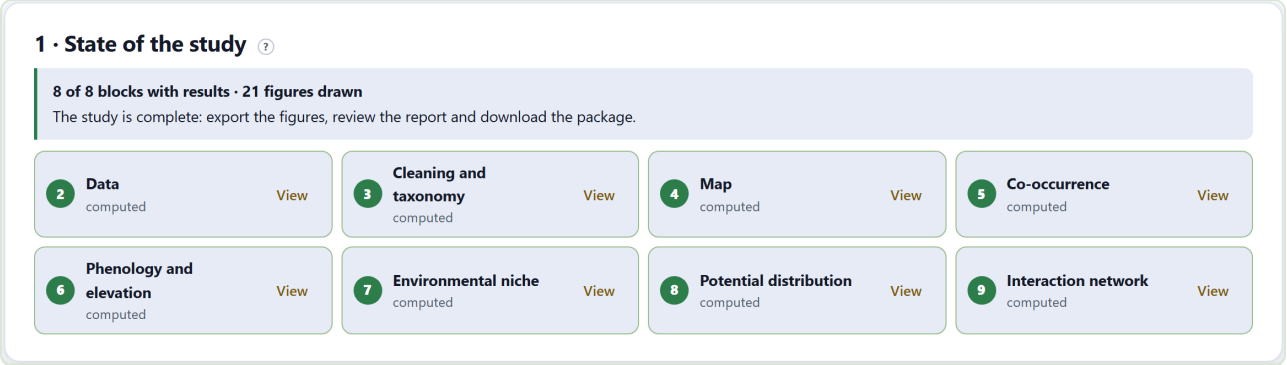


Figure 10.1. The practice study with the eight working blocks computed and 21 figures drawn.

Each block appears as *computed* (with the View button) or *not computed* (with Go). A block is computed when you open it; the ones that were not opened do not appear in the report or in the catalog, and the message at the top reminds you of it. Blocks 7 and 8 also need layers, and Block 9, interactions.

10.3 The catalog and the two figure editors

Control	What it does
Format	PNG, JPG or SVG (vector and editable).
Resolution	150 dpi (screen), 300 (print), 600 (journal) or 900 (poster).
Theme · **Background**	Light, dark or the one on screen; background as styled, white, that of the card or transparent.
With the title inside the figure	Writes the title of the figure into the file. For a journal, which puts the title in the caption, untick it.
↓ Export the selected ones · All the selected ones in a .zip	Downloads the ticked ones; with more than six they come in a .zip (`<study>_figures.zip`), because the browser blocks many downloads in a row.

There are two ways to change the look of the figures, and they combine.

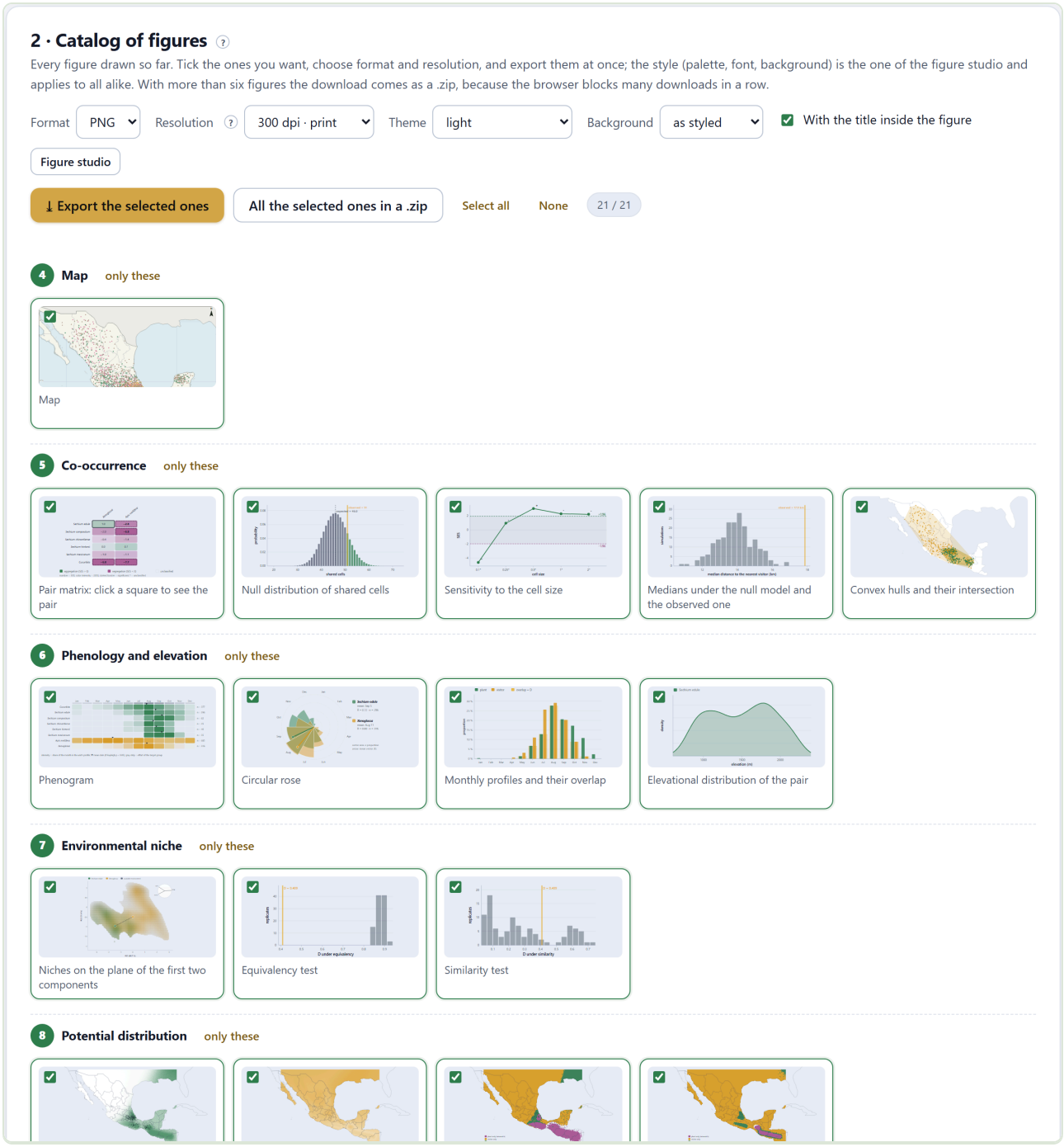


Figure 10.2. The catalog gathers all the figures drawn, grouped by block, with a box to tick each one and an only these button per block.

The figure studio: all at once

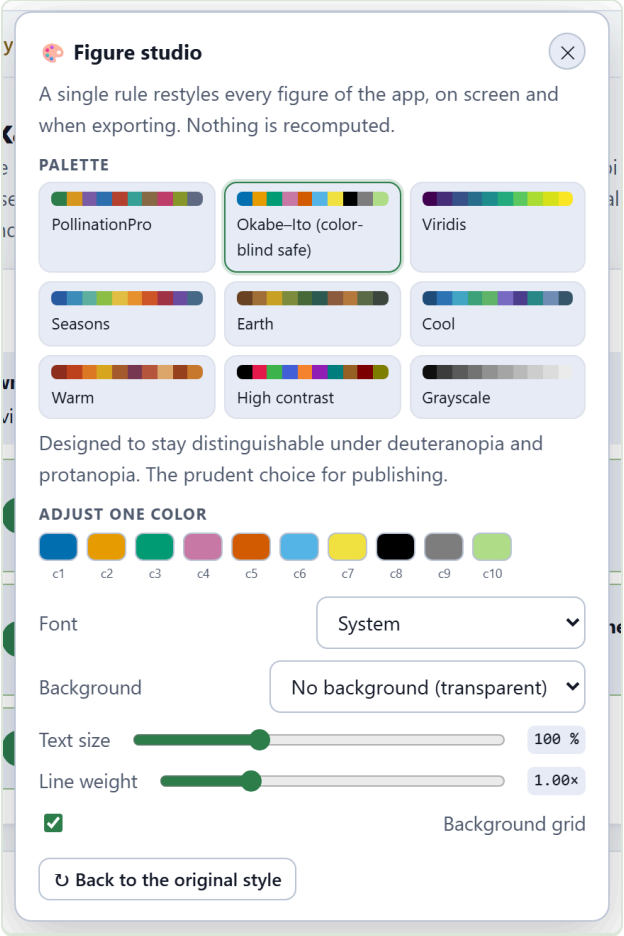


Figure 10.3. The figure studio, with the Okabe–Ito palette.

It opens with the palette button of the top bar, from any block. A single rule restyles every figure of the app, on screen and when exporting, without recomputing anything.

- **Nine palettes:** PollinationPro, Okabe–Ito (color-blind safe), Viridis, Seasons, Earth, Cool, Warm, High contrast and Grayscale.
- **Adjust one color**: any of the ten of the palette.
- **Font** (seven families that exist on any computer), **background**, **text size**, **line weight** and **background grid**.
- ↻ Back to the original style.

The ✎ editor: one figure at a time

Every figure of the app has, next to the ↓ button, a ✎ button that opens its own editor. Whatever you change there applies only to that figure, is kept even if the figure is recomputed and comes out in its export, in the catalog, in the report and in the package. The editor has six tabs — General, Axes & grid, Series, Legend, Texts and Annotations —; the three you will use most are these.

Tab	What is edited
General	The title inside the figure, with its subtitle, footnote, size and alignment; the font family, the size of every text (from 0.6 to 2.2 times), the color of every text, the line weight and the point size (from 0.3 to 3 times), the background and a border around the whole figure. The colors of the axes and the grid lines are on the Axes & grid tab.
Texts	Each text of the figure, by its original wording: it can be rewritten, resized, set in bold (B) or italic (I), recolored or hidden. It serves to translate a label, shorten a name or remove a label the journal does not want.
Series	Each color the figure uses, named after its legend entry, to replace it with another, change its weight, stroke or opacity, or hide the series.

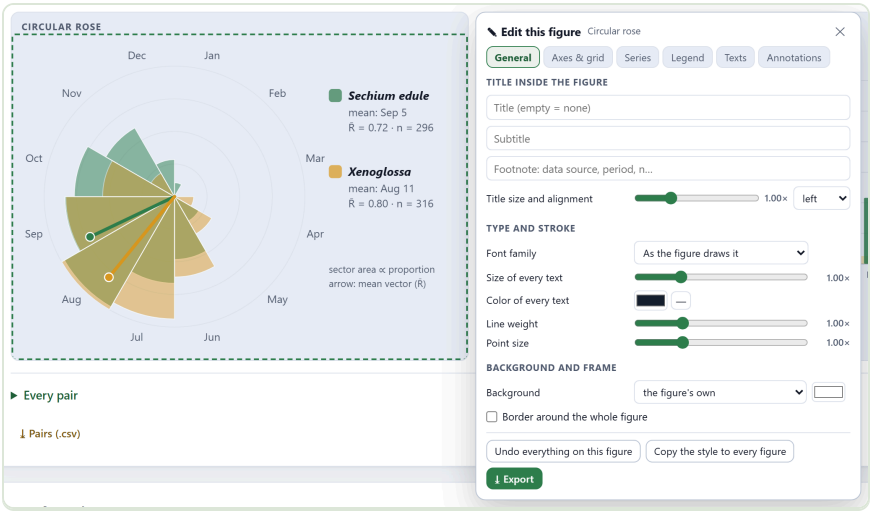


Figure 10.4. The General tab of the editor, opened on the circular rose of Block 6.

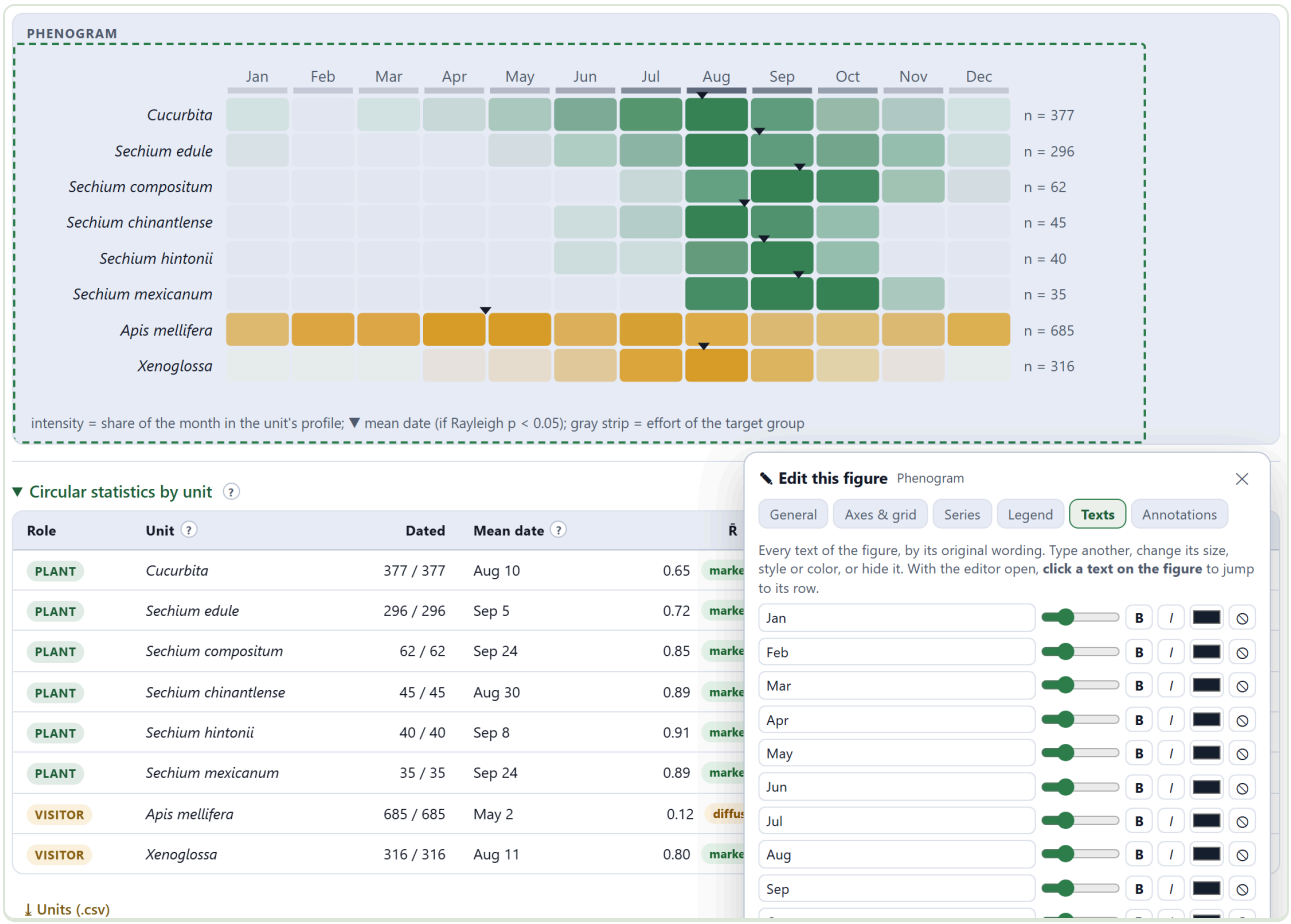


Figure 10.5. The Texts tab on the phenogram: each month, each unit and each "n =" are edited separately. At the foot of the editor (Figure 10.4), Undo everything on this figure and Copy the style to every figure.

A TEXT IS RECOGNIZED BY ITS ORIGINAL WORDING

The editor remembers each text change by what the figure said. If you change the language of the app, the texts are different and are edited separately. And if a figure changes its content —another pair, another cell —, the texts that no longer exist are simply not applied.

10.4 The report

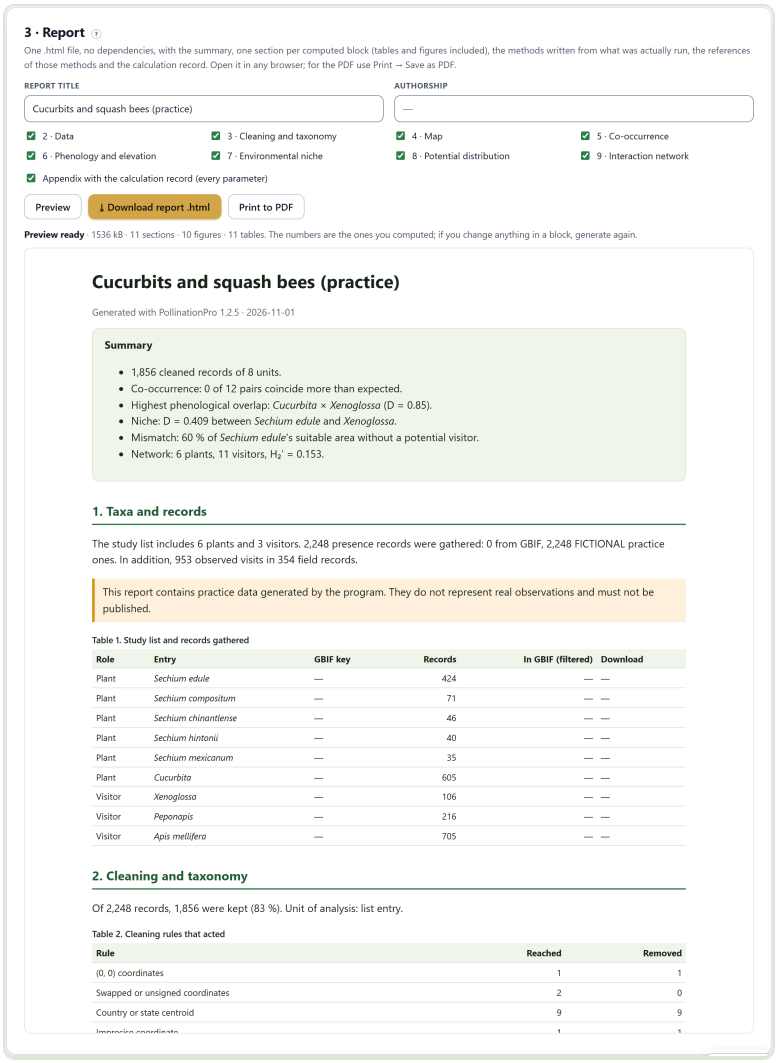


Figure 10.6. The preview of the report of the practice study: 11 sections, 10 figures and 11 tables. The amber box warns that it contains practice data.

The report is a single `.html` file, without dependencies, that opens in any browser. You give it a **title** and an **authorship**, tick the **sections** —one per computed block— and the **appendix with the calculation record**. Preview shows it in the card; ↓ Download report .html saves it as `<study>_report.html`; Print to PDF opens the browser's print dialog (Save as PDF).

Part of the report	What it contains
Summary	One line per block: the cleaned records, the pairs that coincide more than expected, the largest phenological overlap, the D of the niche, the mismatch and the specialization of the network.
One section per block	What the block computed, in prose, with its tables and its figures (with the changes of the ✎ editor and the style of the figure studio).
Methods	Written from what was actually run: the cleaning rules with their parameters, the universe and the cell of the co-occurrence, the effort correction, the background and the density of the niche, the model and the threshold, the null model of the network. Only what was used appears.
References	Only the works of the methods used.
GBIF datasets	The datasets that contributed cleaned records, because GBIF asks for them to be credited, and the recommendation to repeat the download on its portal to obtain the DOI.
How to cite and **appendix**	The citation of the program and, if ticked, the calculation record with every parameter of every block.

> **! THE REPORT SAYS WHEN THE DATA ARE FOR PRACTICE**
>
> If the study uses the practice data, the report warns about it at the start; if the niche or the distribution were computed with the practice layers, their sections warn about it too. Such a report is for learning to read it, not for handing in.

10.5 The reproducible package

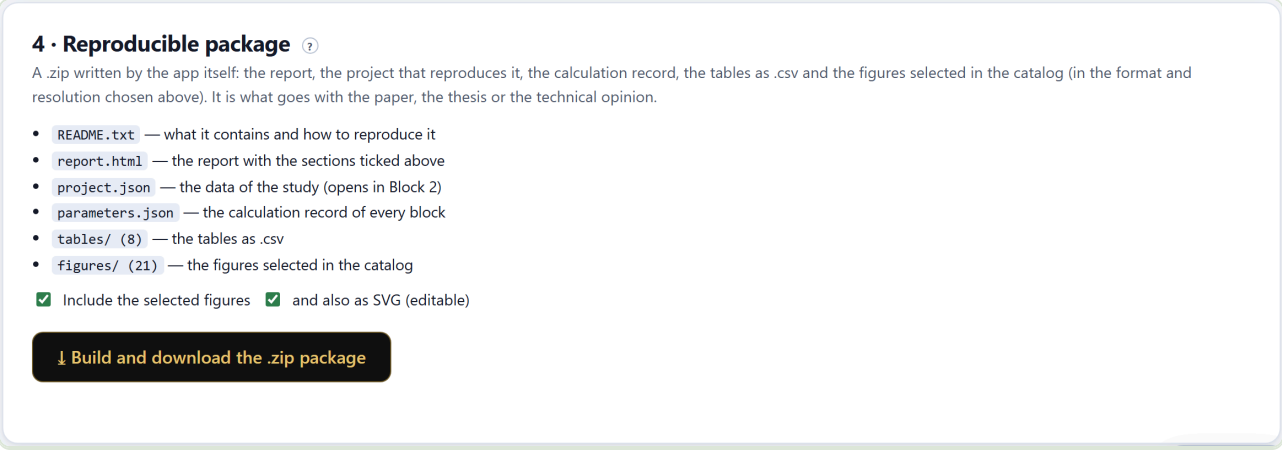


Figure 10.7. The package of the practice study: the report, the project, the calculation record, 8 tables and 21 figures.

File	Content
`README.txt`	What the package contains and how to reproduce the study.
`report.html`	The report with the ticked sections.

File	Content
`project.json`	The data of the study: it opens in Block 2 with Open a project.
`parameters.json`	The calculation record of every block: rules, cells, universes, backgrounds, models, thresholds, replicates.
`tables/`	The tables as .csv: cleaned and removed records (`clean_records`, `removed_records`), co-occurrence pairs (`cooccurrence_pairs`), phenological units and pairs (`phenology_units`, `phenology_pairs`), niche pairs (`niche_pairs`), evaluation of the distribution (`distribution_evaluation`), species of the network (`network_species`) and GBIF datasets (`gbif_datasets`).
`figures/`	The figures ticked in the catalog, in the chosen format and resolution, and also as SVG if and also as SVG (editable) is ticked.

The .zip, `<study>_package.zip`, is written by the app itself, byte by byte, without any library. It is what goes with the paper, the thesis or the technical opinion: whoever receives it can open the project, read the parameters and redo every number (Sandve et al. 2013; Wilkinson et al. 2016).

10.6 Before you hand it in

1. **Check the state of the study.** If a block the paper uses is missing, open it.
2. **Reread the methods of the report.** They must match what you say in the paper: cell, universe, background, model, threshold, replicates.
3. **Cite.** The program, the works of the methods you used and each GBIF download with its DOI, repeating it on its portal with the same filters.
4. **Check the wording.** Blocks 5 to 8 give *potential pollinators* and *opportunity for interaction*; only Block 9 gives interactions.
5. **Export the figures at the journal's size** and check in the ✎ editor that no text is cut off or overlaps.
6. **Keep the package** with the manuscript. It is the supplementary material and the guarantee that the study can be redone.

A

APPENDICES

Appendices

What you need to have at hand and did not fit in any chapter: which files the app reads and writes, the decision rules of the blocks gathered in a single list, the vocabulary of pollination ecology with occurrence data, what to do when something does not work, the references behind each calculation and the license of the tool.

A The files the app reads and writes

B The decision rules, gathered

C Glossary

D When something does not work

E References

F The tool: license and how to cite it

A The files the app reads and writes

PollinationPro runs in the browser and does not send your data to any server: every file comes in from and goes out to a folder on your computer. Only name resolution, the GBIF download and the search for published interactions query the internet, and only when you ask for them.

A.1 · What the app reads

Format	Where and what for
.csv, .txt, .tsv	Block 2, occurrence records and visit observations. Any separator, UTF-8 or Windows-1252; the columns are recognized by their heading in Spanish or English, or by the Darwin Core names.
.xlsx	Block 2, the same, from a spreadsheet; if the workbook has several sheets, you choose which one. The reader is written by the program itself from the public specification of the format.
pasted	Block 2, ...or paste the table: the cells copied from a spreadsheet with their row of headings.
project **.json**	Block 2, Open a project: the list, the records, the visits, the interactions and the filters saved earlier by the app.
.tif, .tiff (GeoTIFF)	Blocks 7 and 8, current and future environmental layers: uncompressed, LZW, Deflate or PackBits, with predictors; in strips or tiles; classic or BigTIFF; in degrees.
.asc (ASCII grid)	Blocks 7 and 8, the same in the text format that any geographic information system writes.
style **.json**	Block 4, Load a style...: a map style saved earlier.

A.2 · What the app writes

Block	Button	Content
2	⤓ Download an empty template	The columns the app recognizes, for occurrences or for visits (`records_template.csv`, `visits_template.csv`).
2	⤓ Interactions (.csv)	The published records: plant, visitor, type, source and coordinates when there are any (`published_interactions.csv`).
2	⤓ Raw records (.csv)	All the records as they arrived (`raw_records.csv`).
2	Save the project (.json)	The whole study, to reopen it on another computer (`pollinationpro_<date>.json`).
3	⤓ Cleaned records (.csv) · ⤓ Removed records, with the reason (.csv)	The ones that enter the analysis, with their unit and the original and harmonized name (`clean_records.csv`); the rest, with the rule that removed them (`removed_records.csv`).
4	⤓ Download the map · ⤓ Save this style (.json) · ⤓ Points shown (.csv)	The map in PNG, JPEG, WebP, SVG, TIFF or GeoTIFF; the map style (`map_style.json`); the visible points (`map_points.csv`).
5	⤓ Pairs (.csv)	Each pair with cells, expected, SES, both tails of p, the adjusted p, the verdict and the Jaccard, Sørensen, Simpson and cover indices (`cooccurrence_pairs.csv`).
6	⤓ Units (.csv) · ⤓ Pairs (.csv)	The circular statistics and the monthly profiles (`phenology_units.csv`); the phenological pairs (`phenology_pairs.csv`).
7	⤓ Environmental values of the records (.csv)	Each record with environment, its components and its variables (`environmental_values.csv`).
8	⤓ Classes by cell (.csv)	Each cell with the suitability of the plant and its current and future class (`mismatch_cells.csv`).

Block	Button	Content
9	⭳ Matrix (.csv) · ⭳ Species (.csv)	The network matrix (`interaction_matrix.csv`); the tables of plants and visitors (`network_species.csv`).
all	⭳ of each figure	The figure in PNG, JPG or SVG, 150 to 900 dpi, with the changes of the ✎ editor.
10	⭳ Export the selected ones · ⭳ Download report .html · ⭳ Build and download the .zip package	The figures of the catalog; the report (`<study>_report.html`); the package (`<study>_package.zip`) with the report, the project, the parameters, the tables and the figures.

B The decision rules, gathered

The reading scales of the chapters, in a single table. They are conventions for reading a number, not laws; the ? sheet of each index in the app says where each one comes from.

Index or decision	Reading	Ch.
Co-occurrence universe	GBIF or collection records → target group; systematic sampling → the cells surveyed.	1, 5
Records per name	A generalist capped at 20,000 is enough; the plant or visitor of interest, uncapped; fewer than 20 in the area, weak tests.	2
Remove or flag	Remove what does not represent where the organism was; flag and review what is doubtful but possible.	3
Records kept	> 80% usual · 50–80% check which rule removes most · < 50% review the parameters.	3
Map format	Journal: TIFF or PNG at 600 dpi at column width · retouching: SVG · GIS: GeoTIFF without title.	4
SES	< −1.96 segregation · −1.96 to 1.96 chance · ≥ 1.96 aggregation.	5
Cell size	0.1–0.25° for the foraging scale · 0.5–1° for the region · if the verdict changes, report the sensitivity.	5
R^-	< 0.3 diffuse · 0.3–0.6 moderate · > 0.6 marked.	6
Schoener's D (calendar, elevation, niche)	< 0.2 none or very limited · 0.2–0.4 low · 0.4–0.6 moderate · 0.6–0.8 high · > 0.8 very high.	6, 7
Correlation between layers	|r| > 0.8 → keep one of the two.	7
Equivalency and similarity	Equivalency p < 0.05: not identical · similarity p < 0.05: more alike than the environments.	7
AUC	< 0.6 nearly random · 0.6–0.7 poor · 0.7–0.8 fair · 0.8–0.9 good · > 0.9 excellent.	8
TSS	< 0.2 poor · 0.2–0.4 fair · 0.4–0.6 good · > 0.6 very good.	8
Mismatch	Show it with two thresholds; draw no conclusion with poor visitor models; restrict it to the zone without extrapolation.	8
Connectance	< 0.15 sparse · 0.15–0.4 usual · > 0.4 dense (small or generalist network).	9

Index or decision	Reading	Ch.
H_2' and d'	< 0.3 generalist · 0.3–0.6 intermediate · > 0.6 specialized (specialist).	9
Completeness	< 50% very incomplete · 50–80% incomplete · 80–95% good · > 95% nearly complete.	9

C Glossary

Each term is followed, in italics, by its name in the Spanish interface when it differs.

Term	Meaning
AUC	Area under the ROC curve: the probability that a cell with a record receives higher suitability than a random background cell.
Background (*fondo*)	The cells of available environment against which what the species use is compared.
Centroid (*centroide*)	The point assigned when only the country or the state is known: the area centroid or the center of the bounding box.
Chao1	Estimator of the minimum number of species (here, of visitors) from those seen once and twice.
Circular statistics (*estadística circular*)	Statistics that treat dates as angles, because the year is a circle.
Clip (*recorte*)	Drawing the map only inside the outline of Mexico or of the chosen states.
Co-occurrence (*coocurrencia*)	Two taxa having records at the same site (cell). It is an opportunity for interaction, not an interaction.
Completeness (*completitud*)	Recorded visitors divided by estimated visitors: how much of the sampling has already been done.
Connectance (*conectancia*)	Observed links divided by possible links of a network.
Convex hull (*envolvente convexa*)	The smallest polygon without indentations that encloses all the points of a unit.
Cross-validation (*validación cruzada*)	Evaluating a model with records that were not used to fit it.
d'	Specialization of a species with respect to the availability of its partners, from 0 to 1.
Environmental space (*espacio ambiental*)	The plane of the first two principal components of the climate variables of the background.
GBIF	Global Biodiversity Information Facility: the portal that gathers occurrence records from collections and observations around the world.
H_2'	Specialization of a whole quantitative network, from 0 to 1.
List entry (*entrada de la lista*)	Each name of the study list (species, genus or family), with its role.
Living collection (*colección viva*)	A specimen from a botanical garden or a field genebank; it does not say where the wild population lives.
Mismatch (*desajuste*)	Area suitable for the plant and for no potential visitor.
Modularity Q (*modularidad Q*)	How much a network divides into groups that interact more among themselves.

Term	Meaning
NODF	Measure of the nestedness of a matrix, from 0 to 100.
Null model (*modelo nulo*)	The random version of the data that keeps what is not in question (the totals, the geography of sampling) and breaks only what is being tested.
Omission (*omisión*)	Fraction of validation records that fall outside the suitable area.
Opportunity for interaction (*oportunidad de interacción*)	Coincidence in space, time and environment: a necessary, not sufficient, condition for pollination.
Potential pollinator (*polinizador potencial*)	A visitor that coincides with the plant in space, time or environment, but whose visits and pollen transport have not been documented.
Published interaction (*interacción publicada*)	A plant–visitor pair recorded in the literature or in collections and gathered by a database of biotic interactions.
R^-	Length of the mean vector of the dates: their concentration, from 0 to 1.
Rayleigh test (*prueba de Rayleigh*)	Test that a set of dates is seasonal and not uniform over the year.
Role (*papel*)	Plant or visitor.
Sampling bias (*sesgo de muestreo*)	The uneven distribution of recording effort: near roads, cities and stations.
Schoener's D (*D de Schoener*)	Overlap of two probability distributions (monthly, elevational or in environmental space), from 0 to 1.
SES (*EEF*)	Standardized effect size: (observed – expected) / null standard deviation.
Suitability (*idoneidad*)	How much the environment of a cell resembles that of the records of a unit, according to a model.
Synonym (*sinónimo*)	An earlier name of the same taxon; the app gathers them under the current name and keeps the original.
Target group (*grupo objetivo*)	The sites (or months) where any taxon of the study was recorded: the universe that corrects for sampling bias.
Threshold (*umbral*)	The suitability value from which a cell is declared suitable.
TSS	Sensitivity + specificity – 1 of a binary map, from –1 to 1.
Unit of analysis (*unidad de análisis*)	What Blocks 4 to 8 compare: a list entry or a species.
Universe (*universo*)	The set of sites among which chance is distributed in a co-occurrence test.
Veech test (*prueba de Veech*)	Exact hypergeometric test of the number of sites shared by two taxa.
Watson–Williams test (*prueba de Watson–Williams*)	Test that two sets of dates have the same mean date.

D When something does not work

What happens	What to do
The app opens blank or without styles	A folder is missing (js, css, data). Copy the whole program folder.
Names cannot be resolved or downloaded	No internet, or GBIF does not respond. Try later; meanwhile, use the practice data or your own records.
A name was resolved as another taxon (a fossil, another kingdom)	Choose the right one in the selector of that name (Section 2.2).
The download was cut off halfway	Repeat Download the records: only the GBIF records of each name are replaced.
My columns are not recognized	Choose each field in the column assignment, or download the template and use its headings.
My dates come out with the day and month swapped	Change **Dates written as** to day/month/year or month/day/year.
There are points in the sea or in another country	Check the coordinate rules in Block 3; "In the sea or outside the outlines" and "Outside the stated country" are set to *flag only* by default.
Almost all pairs come out "unclassified"	Too few records for the cell: use larger cells or merge species into their genus.
Block 7 does not read a layer	It must be in degrees (latitude and longitude). Reproject layers in meters.
The layers look shifted or are marked "resampled"	They have different grids. Use layers from the same source and resolution.
Block 8 says that future layers are missing	Name the future files with the same ending as the current ones ("bio_1"...).
The network has no H_2' or d'	You mixed your own visits with published interactions: use only your visits for those indices.
A figure looks different from how I left it	Check the ✎ editor of that figure and the figure studio: the changes are kept in the browser. Undo everything on this figure returns it to how the app draws it.
The report lacks a block	That block is not computed: open it (it computes itself on opening) and go back to Block 10.
The browser blocks many downloads	Export the figures as a single .zip from the catalog of Block 10.
I want to start from scratch	Empty the list in Block 2 clears the study; Reset everything, on the Looks tab of the map studio, returns the map to its starting values.

E References

The sixty references of the app's home page, by family. The Block 10 report cites only those of the methods the study used.

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F The tool: license and how to cite it

F.1 · What the app carries inside

PollinationPro is written in HTML, CSS and JavaScript without any third-party library: not for the charts, not for the maps, not for reading GeoTIFF files or spreadsheets, not for writing the .zip. The methods are those of the original publications, cited in the code, on the home page and in the report. The outlines of the countries and of the states of Mexico come from Natural Earth (public domain). The practice data and the practice layers are fictional and are generated by the program itself. The file PROCEDENCIA.md in the program folder details the origin of everything, tests/index.html checks the engine with 191 tests, and the validation/ folder compares it number by number with R packages written by other people (34,299 comparisons, none in disagreement).

F.2 · License

The program is distributed under the **GNU General Public License, version 3 or later** (GPL-3.0-or-later): it can be freely used, copied, studied and modified, on condition that modified versions are distributed under the same license and keep the attribution. The full text is in the LICENSE file. This manual is © 2026 Luis Ángel Barrera-Guzmán, all rights reserved.

F.3 · How to cite it

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The DOI is the concept DOI in Zenodo: it always resolves to the latest version. The citation of the program is also in CITATION.cff and at the end of every Block 10 report. Also cite the methods you used to their original works, and each GBIF download with its DOI.